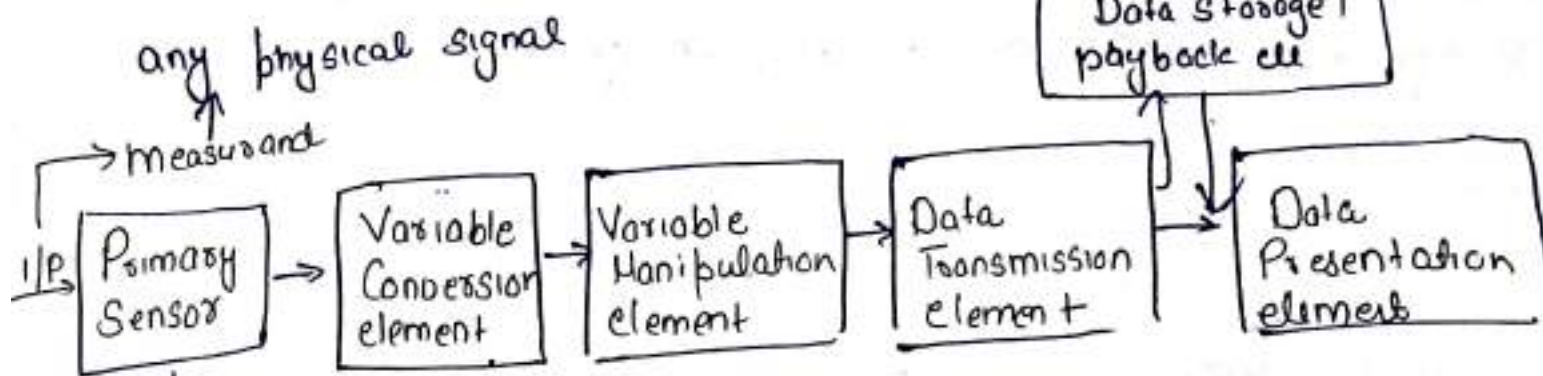


T.S.C Basics

Block diagram of a generalised measurement system



Convert input to a suitable output signal

RTD → Resistance Temperature Detector

e.g. →



$$\Delta R = f(T)$$

$$R(T) = R_0 (1 + \alpha(T - T_0))$$

R_0 → nominal resistance
 T_0 → nominal temperature
 α → Temperature coefficient.

① Variable Conversion → convert the input variable (ele/non-ele) to suitable output (ele/non-ele)

e.g. in case of RTD we get $\Delta R \rightarrow$ voltage

② Manipulation → amplification
 → modification → (preserving the original nature) $V \rightarrow V$

③ Transmission → Coaxial Wires, Wavelets, long dist (Repeater)

2) Storage media - to store the data for future uses.

3) Data - presentation → To present the data

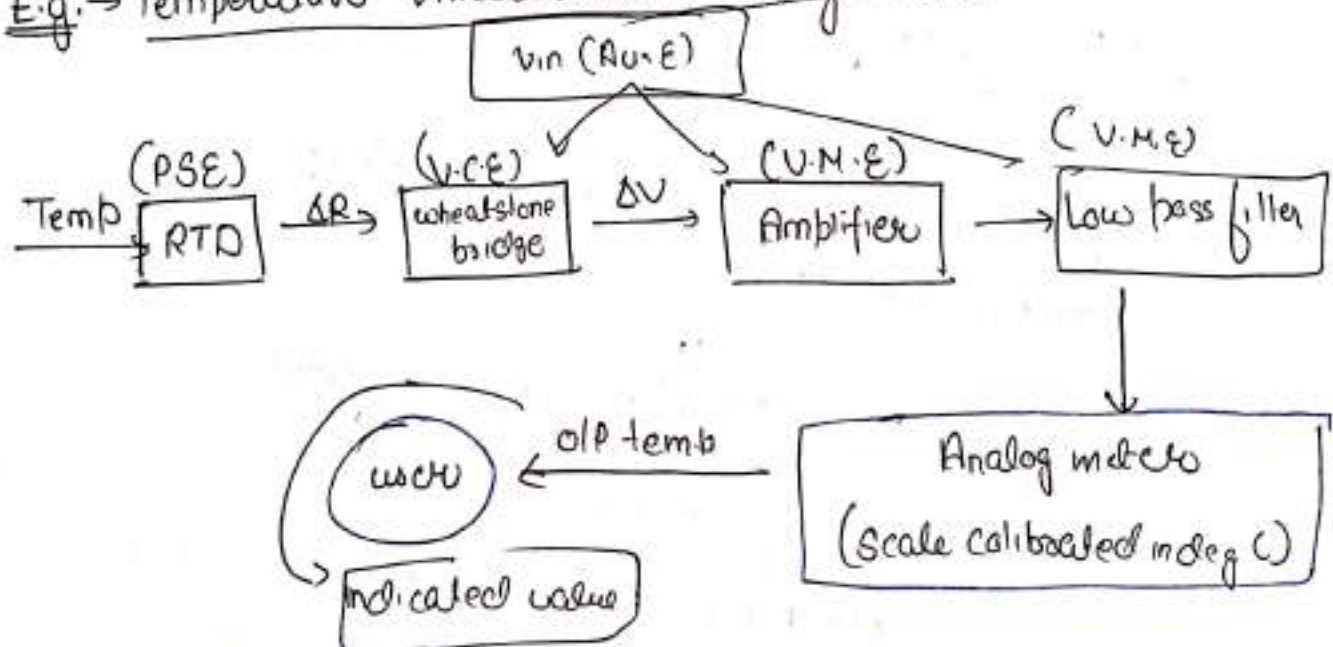
(Scale should be calibrated to read the output)

7 Segment, L.C.D, oscilloscope, analog-meter

Stages

- i) Signal transducer stage → primary sensor
- ii) Signal conditioning and processing → variable conversion + manipulation
- iii) Data transmission stage → Data transmission + storage element
- iv) Data visualisation

E.g. → Temperature measurement using RTD

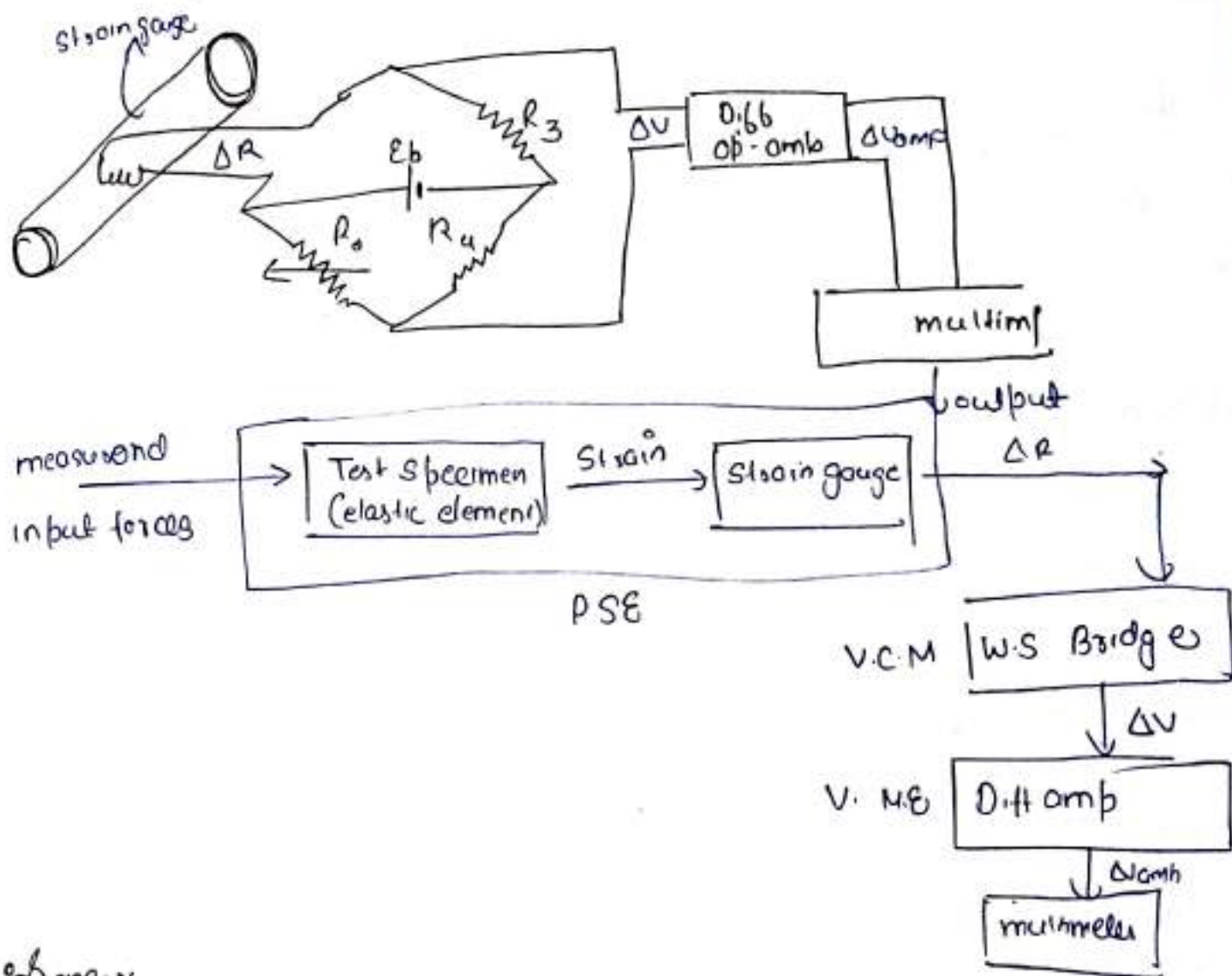


$$R_T = R_0 (1 + \alpha (T - T_0))$$

where R_0 and T_0 - nominal value of RTD (100Ω at 0deg)

* Low pass filter → reject higher frequency noise

Strain gauge

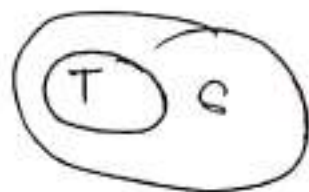


Sensor

Output provide (ele/non ele) in response to when a quantity being measured (ele/non ele)

• Transducers → provide electrical output response to a specific measured

• Transducer → Transform energy into electrical form



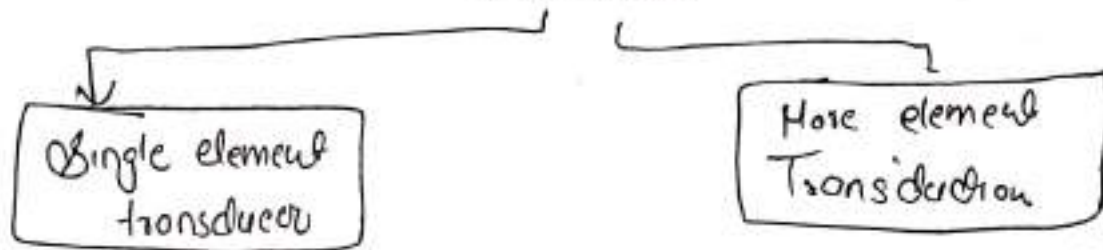
Why electrical
↓

easy trans, processing, modification, can be interfaced with electrical and non-electronic devices

* Inverse transducer I/P → electrical
O/P → non electrical

→ actuators, relays, servos

Transducer



RTD → (Temp directly to electrical quantity)

Borden tube Pressure gauge (Bellows + LVDT)

→ Primary transduction element (PSE), called a mechanical transducer if the output is mechanical

→ Secondary transducer element (VSE), called electrical transducer if the o/p is ele.

Sensor Classification

① Working Principle

electro-mechanical, electro-chemical, R, C, I

② Signal domain

Mechanical, Thermal, Radiation, Optical etc.

③ Variable

Temp, colour, force, light intensity, sound pressure, speed, acc.

④ Power require.

i) Active

Requires an excitation signal (ext. power) to provide o/p for an i/p

→ Resistive, Capacitive, Inductive.

ii) Passive

→ Self generation

→ Value will be very low

→ Piezoelectric, thermocouple, photodiode.

① Nature of output

Non electrical, analog, digital, pulse-generating.

② Physical contact

Contact type, non-contact

③ Material

Metal-Alloys, Si-based

④ Method of fabrication

Non-IC conventional, IC-based smart sensor.

⑤ Application

Industrial, biomedical, automobile, aerospace.

Sensor-performance characteristic

* Performance characteristic. Static, Dynamic, Environment

Mechanical-electrical, reliability → lifetime
Certain specification that tell about sensor perform

Static characteristic → When the I/P conditions are static

Dynamic → frequency based I/P

When signal changes w.r.t time

Slow varying signal

→ We compare sensors based on performance characteristic

* Static

When I/P is constant or changing very slowly.

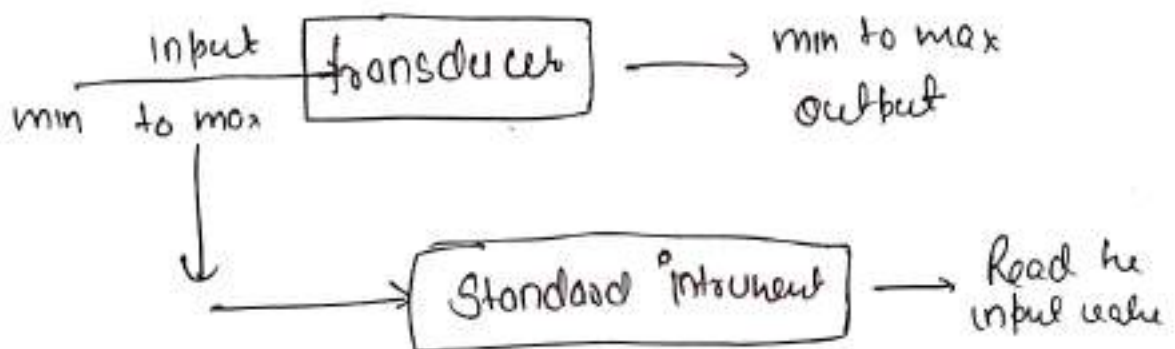
Static characteristic → help in determine accuracy

reliability and suitability.

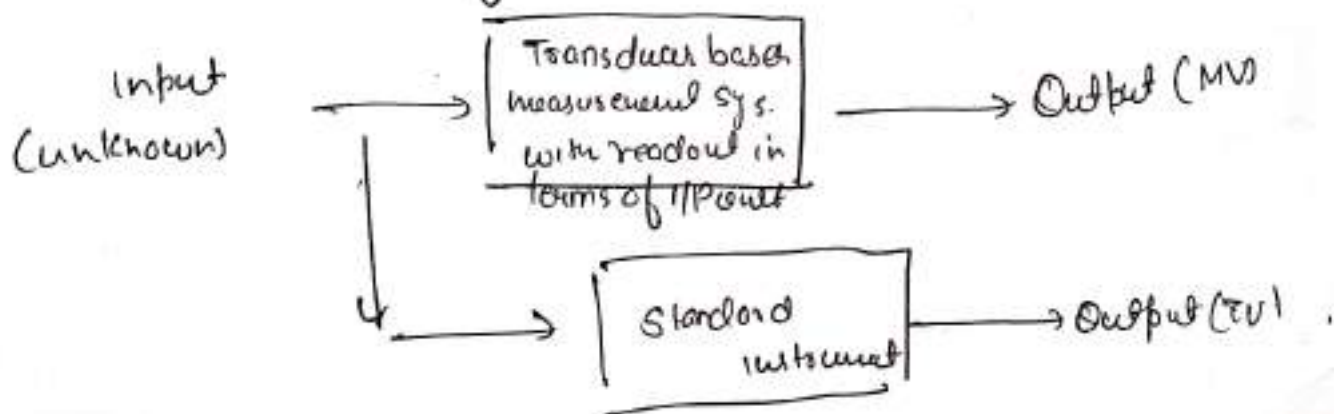
→ these characteristics are established through static calibration
↓
performed in industry

Static calibration

A process that indicates the relationship between O/P and I/P of the sensor, input is changed over the full scale range under (standard operating / test condition). The I/P is measured by a secondary / standard instrument and output of the sensor (voltage) is measured by an indicating instrument to provide output-input characteristics (calibration curve) of the device.



static calibration of Transducer instrument

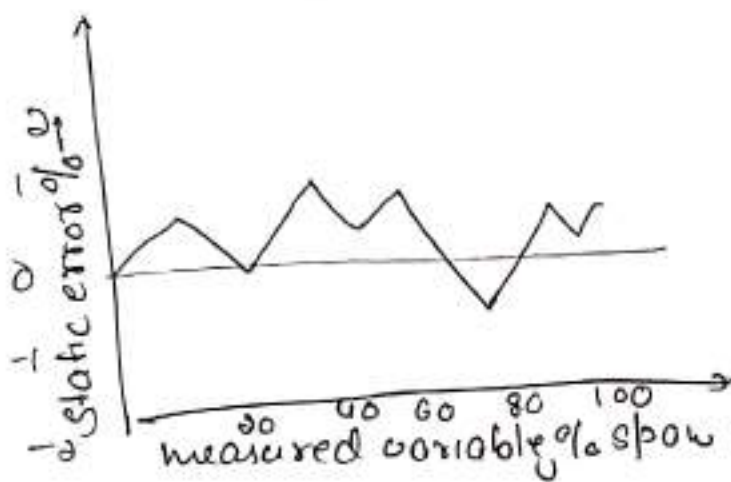


Calibration procedure involves comparison of a particular instrument with a sec. standard of higher accuracy to determine the accuracy of instrument

$$\boxed{\text{Abs Error} = MV - TV}$$

Error calibration curve

$$\% \text{ static error} = \frac{MV - TV}{\underbrace{X_{\max} - X_{\min}}_{\text{maximum and min value of the sensor}}} = \frac{\text{absolute error}}{\text{span}}$$



→ Relationship between % error and input variable in terms of span.

1] Accuracy

Refers to the closeness of the transducer's reading to the true or accepted value of measured quantity.

- indicate the degree of correctness
- specified in terms of static or relative error (specified in terms of $TV / FSD / \text{span}$)

±] Percentage of true value

$$= \frac{M.V - T.V}{T.V} \times 100$$

±] Percentage of Full scale deflection (max. reading)

$$= \frac{MV - T.V}{F.S.D} \times 100$$

3] Percentage of span

$$= \frac{MV - TV}{X_{max} - X_{min}} \times 100$$

→ if nothing is given (taken to be span)

Q] A thermometer is calibrated 150° to 200°C . The accuracy is ± 0.25 percent, what is max/min static error

$$\pm 0.25\% = \frac{\text{Static error}}{50} \times 100$$

$$\pm \frac{0.25 \times 50}{100} = \text{Static error}$$

$$\boxed{S.E = \pm 0.125^{\circ}\text{C}}$$

Indicate how 200°C will be gauged by the thermometer

→

$$\boxed{200^{\circ}\text{C} \pm 0.125^{\circ}\text{C} = 199.875^{\circ}\text{C} \text{ to } 200.125^{\circ}\text{C}}$$

Q] A transducer is measuring relative humidity in range 0 to 100%, accuracy = $\pm 1\%$ of T.V find the limits in which value of 80% relative humidity

$$\pm 1\% = \frac{\text{Static error}}{80} \times 100$$

$$\pm \frac{80}{100} = S.E$$

$$S.E = \pm 0.8$$

$$80\% \pm 0.8\%$$

Gauged value $\Rightarrow 79.2\% \text{ to } 80.8\%$

└ T.V

So F.S.D Error will be more because it is in all the reading but w.r.t T.V error will be different and small in each value.

Note: Specification of relative error in reading ref to T.V is more accurate than ref to F.S.D

2] True value

└ measured by secondary standard instrument
or
└ mean of M.V

3] Absolute static error and relative static error

$$\text{Absolute S.E} = M.V - T.V$$

↓
does not precisely indicate the accuracy of measurement

→ Better way of specifying accuracy is in term of relative static error

$$\rightarrow \epsilon_r = \frac{\epsilon_o}{T.V} = \frac{M.V - T.V}{T.V}$$

$$\downarrow \quad \epsilon_r = \frac{M.V}{T.V} - 1$$

$$\Rightarrow T.V = \frac{M.V (1 - \epsilon_r)}{(1 - \epsilon_o)}$$

if ϵ_r is very small $\epsilon_o \ll 1$

$$T.V = M.V (1 - \epsilon_r)$$

$$\text{or } TV = MV (1 - \epsilon_s)$$

$$\boxed{MV = TV + \epsilon_r} \rightarrow \boxed{MV = TV + \epsilon_o}$$

(The measured measures the TV with a constant error)

4] Static correction

→ It is the difference between the true value and the measured value

$$\bullet \Delta C = TV - MV$$

$$\bullet \Delta C = -\epsilon_o$$

$$\bullet \boxed{TV = MV + \Delta C}$$

Q] A thermometer reads 94.45°C static correction -0.08°C , determine TV, absolute static error

$$\Delta C = -0.08^\circ\text{C}$$

$$\boxed{\epsilon_A = 0.08^\circ\text{C}}$$

$$\Delta C = TV - MV$$

$$-0.08^\circ\text{C} = TV - 94.45$$

$$TV = 94.45 + 0.08^\circ\text{C} \rightarrow \boxed{TV = 94.53^\circ\text{C}}$$

$$\epsilon_s = \frac{0.08}{94.53} \times 100 = 0.000845$$

Q] $TV = 1.50\text{V}$, $MV = 1.46\text{V}$ range 0 to 2.5V ,

what is ϵ_A and ΔC , ϵ_r (in terms of TV)

$$= \epsilon_A = 1.46 - 1.50$$

$$= -0.04\text{V}$$

$$\boxed{\Delta C = -0.04\text{V}}$$

$$E_b (F.S.D) \% = \frac{-0.04}{1.5} \times 100 = -2.66\%$$

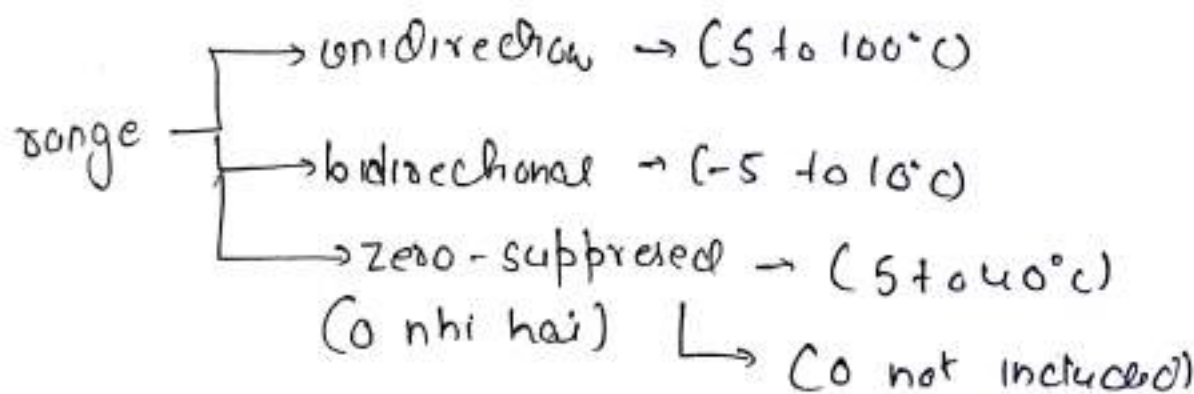
$$E_b (F.S.D) \% = \frac{-0.04}{2.8} \times 100 = -1.6\%$$

5 Range

↳ Range represent the minimum and maximum value of the quantity transducer can measure accurately

6] span

↳ difference between minimum and maximum value



7 Precision

Q] A temp. transducer has range from 20°C to 250°C
a measurement results in the value of 55°C.
Specify the error within range if it is specified
with an accuracy of

i) $\pm 0.5\%$ FSD

ii) $\pm 0.75\%$ of span

iii) $\pm 0.8\%$ of T.V

$$\pm 0.5 = \frac{E_A}{250} \times 100$$

$$\frac{0.5 \times 250}{100} \cdot E_A = \frac{5}{4} = \pm 1.25^\circ\text{C}$$

$$\text{reading} = 55 \pm 1.25^\circ\text{C}$$

$$\text{range} = 53.75^\circ\text{C} - 56.25^\circ\text{C}$$

ii)

$$0.25 = \frac{E_A}{230} \times 100$$

$$E_A = \frac{0.25 \times 230}{100} = \frac{0.75 \times 230}{100} = \pm 0.1725^\circ\text{C}$$

$$\text{reading} = 55 \pm 0.1725^\circ\text{C} = 54.8275^\circ\text{C} - 55.1725^\circ\text{C}$$

iii)

$$0.8 = \frac{E_A}{55} \times 100$$

$$E_A = \frac{0.8 \times 55}{100} = \pm 0.44^\circ\text{C}$$

$$\text{reading} = 55 \pm 0.44^\circ\text{C}$$

$$\left. \begin{array}{l} \\ \\ \end{array} \right\} (54.56 - 55.44)^\circ\text{C}$$

7* Precision

→ Closeness among the group of reading taken for an input under same condition by same instrument by the same user.

→ Precision defines repeatability } how many times input makes the same output

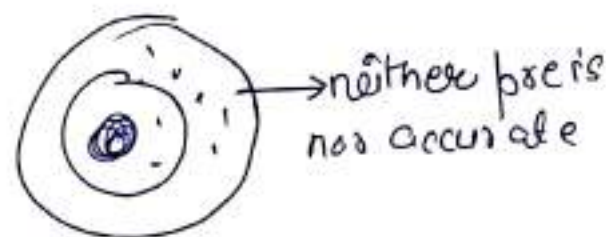
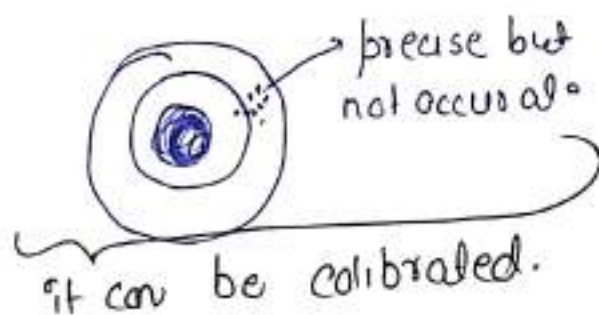
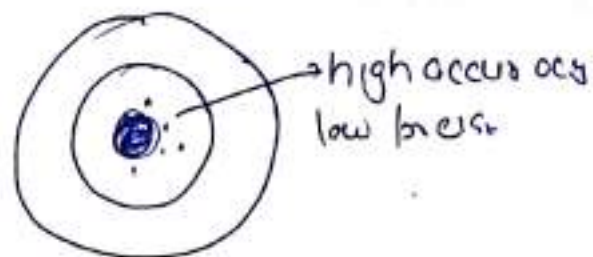
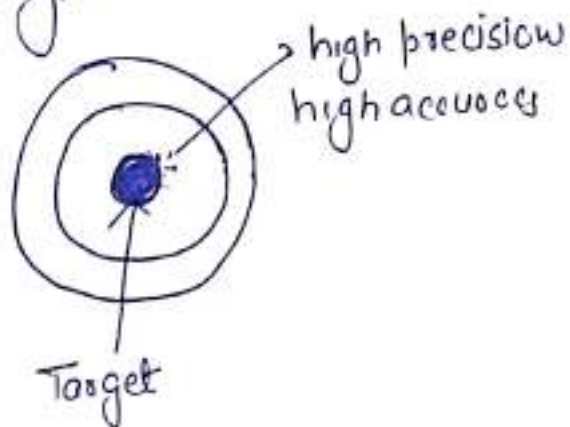
Q] Classify different type of errors and how to minimize them.

high precision means random errors in instrument are less

* Assignment

→ Precision is determined by standard deviation
→ Standard deviation

8) Reproducibility → if group of the readings are taken for same input through different instruments of different types



9) Sensitivity
= $\frac{\text{change in output}}{\text{change in input}}$

Static sensitivity → small change of input provide large (desirable) change of output.

→ but very high is not reliable because noise can change the output.

→ for linear characteristic → static sensitivity remains constant.

→ for non-linear characteristic → sensitivity → non constant

Thermistor $\rightarrow$ non linear

e.g. $y = a_0 + a_1x + a_2x^2$ } non linear

10] Inverse sensitivity

$$= \frac{\Delta I/P}{\Delta O/P}$$

for linear sensor without offset ($c=0$)
 $y = mx$
o/p $\downarrow$ Sensitivity $\downarrow$ I/P

$\rightarrow$ $x = \frac{y}{m}$ $\rightarrow$ giving the output we are finding I/P
 $\frac{1}{m} = \text{reverse sensitivity}$

Imp $\rightarrow$ because we wanna find input

Q] Assume that transducer is having sensitivity of $10 \text{ mV}/^\circ\text{C}$
If the transducer provide o/p 0.35 V find the

i) Inverse sensitivity $0.1^\circ\text{C}/\text{mV}$

ii) Temp $\rightarrow 0.35 \text{ V}$

iii) Temp range if the output voltage has range of $0-1 \text{ V}$

i) $10 = \frac{0.35}{T}$

$$T = \frac{0.35 \times 10^3}{10} = 35^\circ\text{C}$$

iii) $10 = \frac{(0-1)}{T}$

$$T = \frac{(0-1) \times 10^3}{-10} = \underline{\underline{T = (0-100)^\circ\text{C}}}$$

Thermistor $\rightarrow$ non-linear characteristic

e.g. $y = a_0 + a_1x + a_2x^2$ } non linear

10] Inverse sensitivity

$$= \frac{\Delta I/P}{\Delta O/P}$$

for linear sensor without offset ($c=0$)
 $y = mx$
↓
O/P ↓ I/P
 Sensitivity

$\rightarrow$ $x = \frac{y}{m}$ $\rightarrow$ giving the output we are finding I/P
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$$T = \frac{0.35 \times 10^3}{10} = 35^\circ\text{C}$$

iii) $10 = \frac{(0-1)}{T}$

$$T = \frac{(0-1) \times 10^3}{-10} = \underline{\underline{T = (0-100)^\circ\text{C}}}$$

o/p $1/P = (100 - 500)^{\circ}\text{C}$

Q] $y = mx + c \rightarrow$ where $m = 0.1 \text{ V/kN/m}^2 = 0.1 \text{ V/Pa}$
 and offset is 0.5 kN/m^2 if the output voltage is
 0V find the input P

$$0 = 0.1 \times x + 0.5$$

$$1.5 = 0.1 x$$

$$x = 15 \text{ Pa}$$

II Resolution

→ Smallest increment which can easily be detected
 → minimum division that can be read with
 certainty

eg. → if voltmeter range 0-100V scale contains 100 divisions
 then resolution

$$= \frac{100}{100} = 1 \text{ V}$$

Q] Consider a meter that can read 0-200A there are 10
 major divisions and 1/10 of the major division can be
 read with certainty

$$= \frac{200}{10} = 20 \text{ A}$$

$$\text{resolution} = \frac{20}{10} = 2 \text{ A}$$

Resolution of Sensor

→ the minimum increment of the input which can be
 detected by sensor.

Resolution → also depends upon scale

* Cross-sensitivity

When sensor also have interfering input as signals

$$CS = \frac{\Delta O/P}{\Delta \text{Interfering I/P signals}}$$

Cross sensitivity of the sensor to that interfering input
Eg → A pressure sensor may be sensitive to temp irrespective of pressure variation

$$0.01 \text{ kN/m}^2 / ^\circ\text{C}$$

Q] Consider a sensor which has desired sensitivity as 0.01V per unit change in the desired I/P the sensor is calibrated at 25°C when the sensor is used the environment temp is 50°C CS → 0.01V per unit change in temp find the error in the reading when the ~~val~~ value of I/P = 20 units

$$S = 0.01\text{V}$$

$$G.T = 25^\circ$$

$$I/P = 20$$

$$CS = 0.01^\circ\text{C}$$

$$T = 50$$

$$O/P = 0.01 \times 20$$

$$d = 0.2\text{V}$$

output

$$\text{Output} = 0.2 + 0.01 \times 25$$

$$= 0.45\text{V}$$

$$\text{Measured Output} = 0.2 + 25 \times 0.01$$

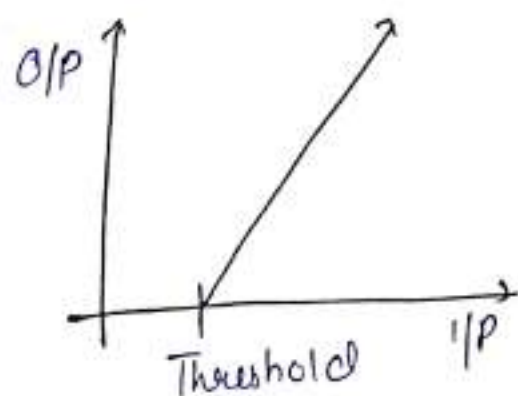
$$= 0.45$$

$$\Delta E = |0.45 - 0.2| = 0.25$$

$$\text{Output} = \text{Sensitivity} \times \Delta \text{I/P} + \text{C.S.} \times \Delta \text{Interf.}$$

* 13 Threshold

Minimum I/P required to give the output i.e. change it from 0.



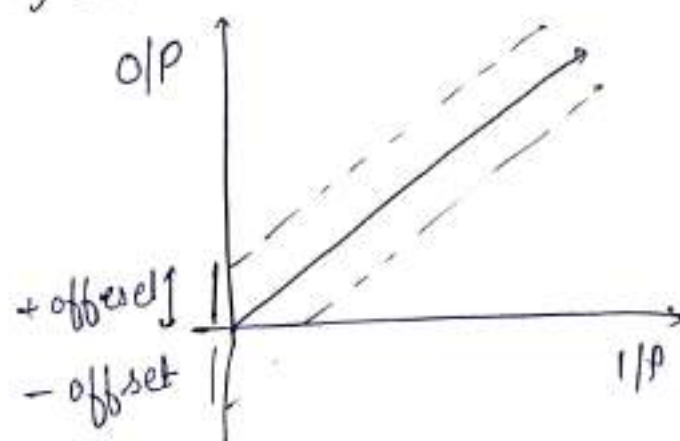
→ It is a limiting case of resolution

* 14 drift

Slow changes of characteristic of instrument w.r.t time due to which errors can result into the measurement

i) zero drift → I/P is 0 and O/P is not zero

↓
zero offset



$$\text{Corrected value} = \text{Measured value} - \text{offset}$$

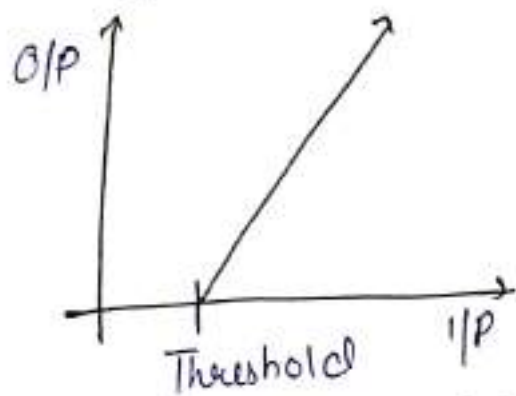
ii) Sensitivity drift

→ Slope of curve change i.e. not coinciding with where it have to be

$$\text{Output} = \text{Sensitivity} \times \Delta \text{I/P} + \text{C.S.} \times \Delta \text{Interf. Output Input}$$

* 13 Threshold

Minimum I/P required to give the output i.e. change it from 0.

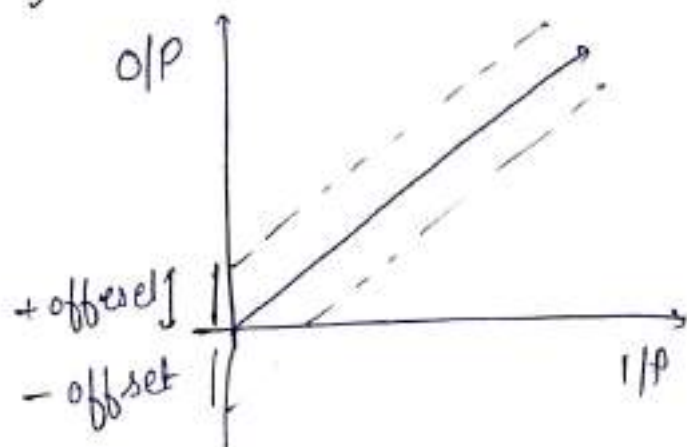


→ It is a limiting case of resolution

* 14 drift

Slow changes of characteristic of instrument w.r.t time due to which errors can result into the measurement

i) zero drift → I/P is 0 and O/P is not zero
↓
zero offset



$$\text{Corrected value} = \text{Measured value} - \text{offset}$$

ii) Sensitivity drift

→ Slope of curve change i.e. not coinciding with where it have to be



→ slope change can be +ve and can be -ve too

iii) Zero drift and sensitivity drift



Q] Consider a load cell which is calibrated at 20°C and it has following calibration chart

load	deflection (mm)	40°	50°
0 - 0.4	0	3	0
0.4 - 0.8	10	14	11
0.8 - 1.2	20	25	22
1.2 - 1.6	30	36	33
1.6 - 2.0	40	47	44
	50	58	55

In environment of 40°

→ Zero offset = 3

→ Sensitivity drift → calculated on full scale deflection = 8

$$\text{Slope (20)} \quad \text{Sensitivity at } 20^\circ\text{C} = \frac{10}{0.04}$$

$$= \frac{10 \times 100}{4}$$

$$= 250 \text{ mm/unit of load (kN)}$$

Sensitivity at $40^\circ\text{C} = \frac{11}{0.04}$
 $= 11 \times 25$
 $= 275 \text{ mm/cm of load (kN)}$

Sensitivity drift = 25

15] Linearity → Output is proportional to I/P and it's a straight line relationship

$y = mx + c$

→ many transducers has to have linear design why?

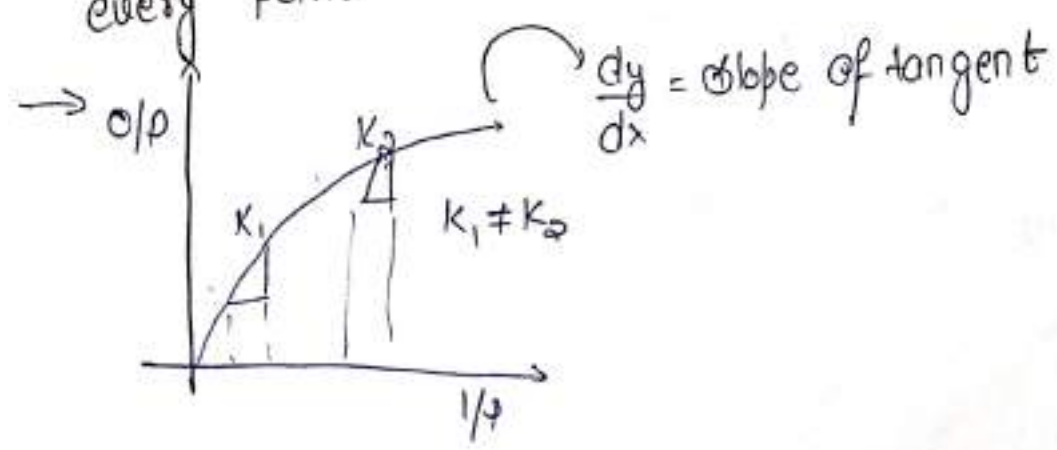
→ correct reading after removing zero offset

relationship is governed by scale factor

→ deviation of the sensor characteristic from ideal linear char. is known as linearity non-linear error

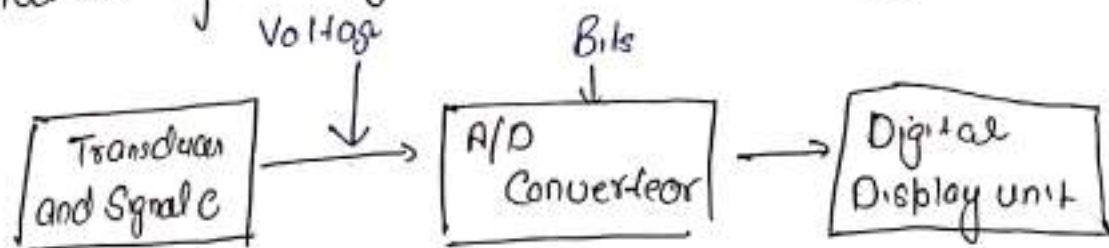
% non linearity → $\frac{\text{maximum deviation of reading from ideal characteristic}}{\text{F.S.D of the instrument}}$

→ Density of a non-linear sensor is calculated at every point



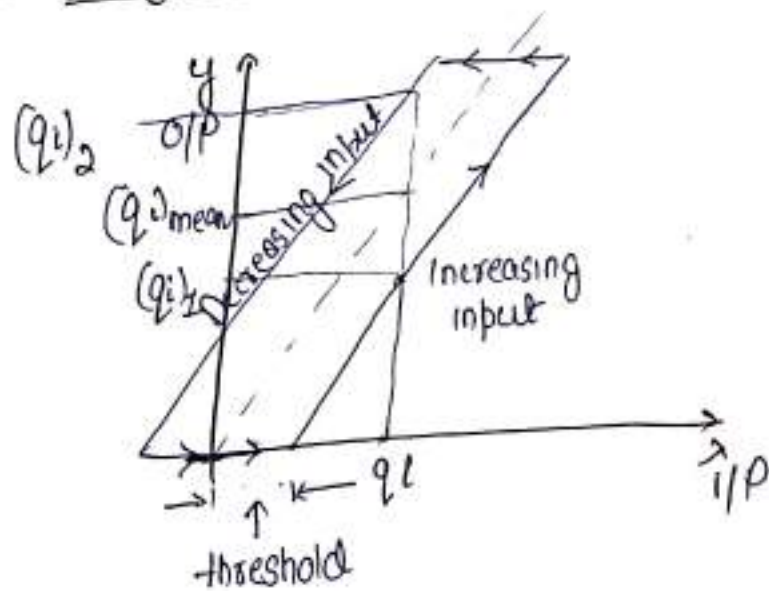
• 16 Hysteresis

16) Resolution of a digital transducer meter



$$\text{Resolution} = \frac{\text{Range of measurement (Span)}}{\text{Number of bits}}$$

17] Hysteresis



$$\text{hysteresis error} = (q_i)_2 - (q_i)_1$$

hysteresis error can be minimised by taking mean of two readings of the sensor

for some input, mean value $(q_i)_{\text{mean}} = \frac{(q_i)_1 + (q_i)_2}{2}$

• Caused due to frictional effects, Magnetic characteristics, elastic deformation, backlash

→ Try hysteresis effect in the sensor provides a different output for the same input when the sensor is subjected to the input in two decreasing and two increasing direction

18 deadband

It is defined as the largest change of the measurement to which the instrument does not respond

19 backlash

Q] The dead zone in a certain pyrometer is 0.125% of span of $^{\circ}\text{C}$. The cal. is 400° to 1000°C . What temperature might occur before it is detected

$$\text{Span} = 1000 - 400 = 600^{\circ}\text{C}$$

$$\text{Dead zone} = \frac{0.125}{100} \times 600 = 0.75^{\circ}\text{C}$$

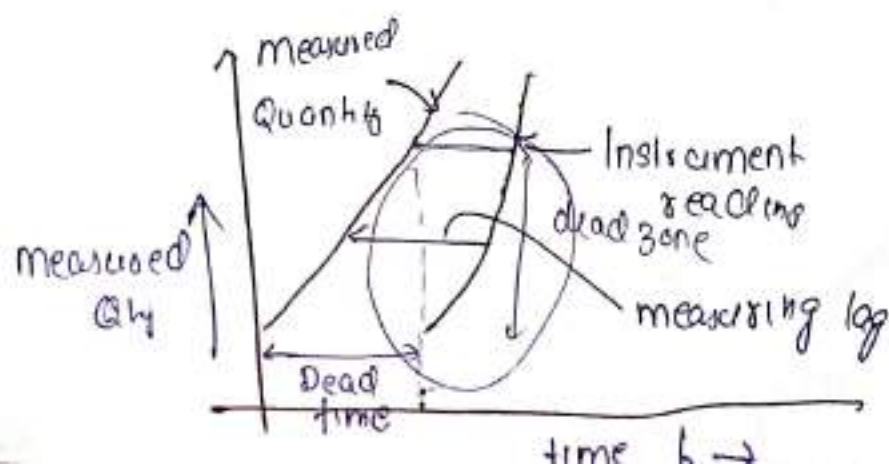
A change of 0.75°C must be occurred to be detected

20] Dead time and measurement lag

→ Deadtime is the time required by a sensor / or transducer instrument to begin to respond to the changes in

I/P

→ Time taken by a sensor to provide an output when the I/P starts changing. Measurement lag is the time lag in measuring the output corresponds to input

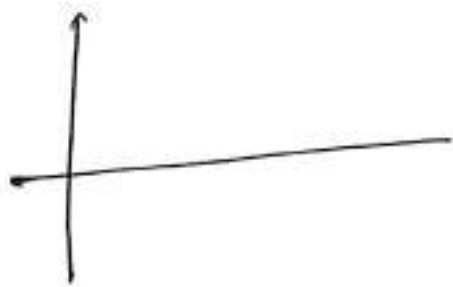


measurement lag is the time lag between I/P and O/P readings

② measured value $\rightarrow$ error due to zero drift + error due to

* Dynamic characteristic of device

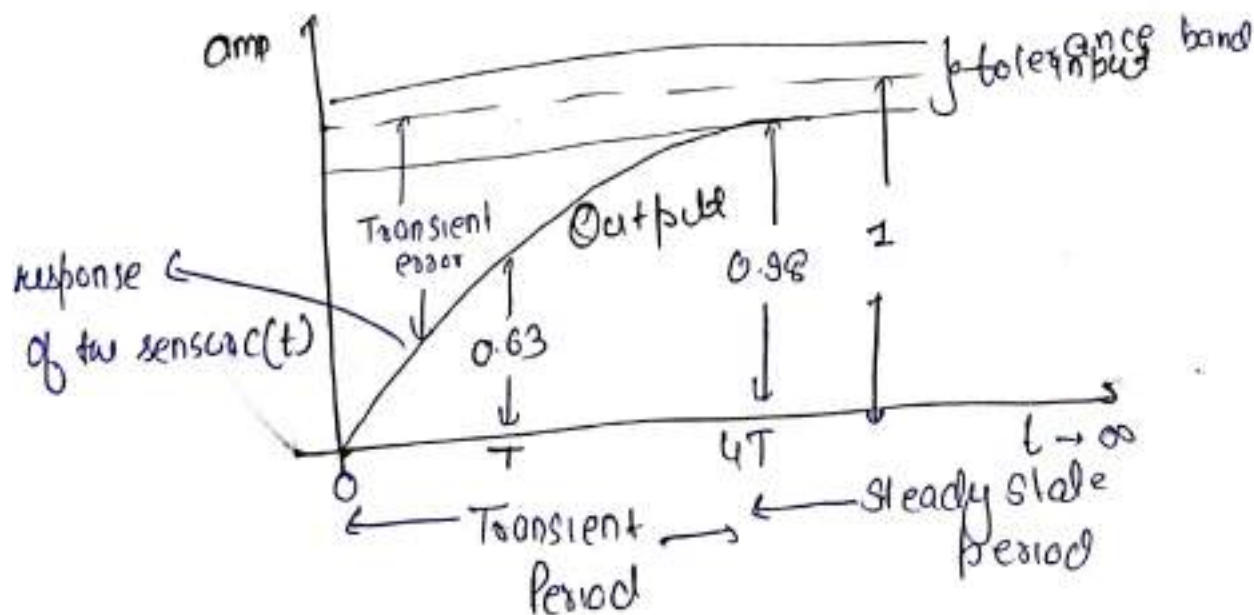
$\rightarrow$ when the I/P is changing or varying over time



I/P $\rightarrow$ sine
 $\rightarrow$ 三角
 $\rightarrow$ square
 $\rightarrow$ step

$\rightarrow$ transiently random signal

③ Transient response and steady state period



1) Dynamic error

① Algebraic difference between I/P and O/P when the I/P is function of time

dynamic error

- transient error $\rightarrow$ error during transient response, before the SS output
- steady state error $\rightarrow$ dynamic error at SS output

2] Speed of response

The rapidity with which the sensor responds to the change in the dynamic input.

3] Response time (O/P reaches steady state)
Time taken for a sensor to reach a certain percentage (often 90%) of its final output value after a step change in the I/P is applied

4] Time constant

Time constant denotes the speed of response. It is the time taken for a sensor to reach 63% of I/P value

$$R.T = G \times T.C$$

① Steady state always have some tolerance

Q) Assume a sensor with the T.C of 1s connected whether this sensor is suited for the temperature variation of 10 Hz

Temp variation $\Rightarrow$ 10 times per second

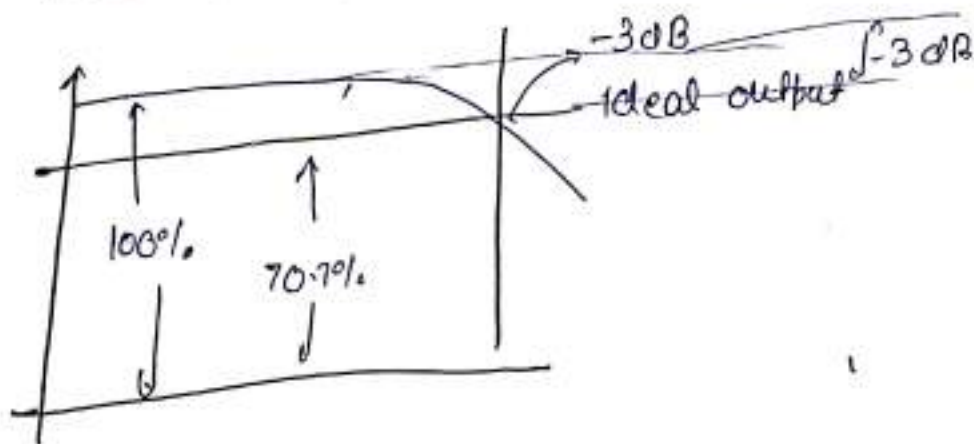
$\rightarrow$ not suitable

5] Frequency Response

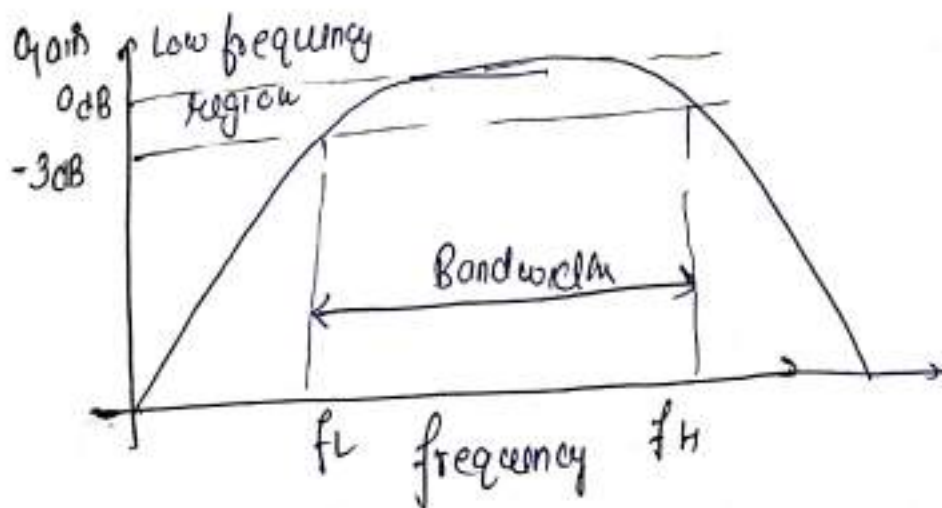
V/P is periodic and the frequency of signal carrier

→ is the ability

BW is the frequency range over which the sensor's output is considered acceptable. Typically defined as the range where the output is within $-3dB$ of the ideal O/P



$$dB = 20 \log \left(\frac{V_o}{V_{in}} \right)$$



Q7 Consider a sensor with B.W 10KHz maximum frequency change is i) 5KHz, ii) 30KHz, iii) 100KHz in which case sensor can use effectively

$$\begin{aligned} & BW = 10 \pm 7.07 = 2.97 - 17.07 \text{ KHz} \\ & \text{Answer: } 10 \pm 7.07 = 2.97 - 17.07 \text{ KHz} \end{aligned}$$

6] Loading effect

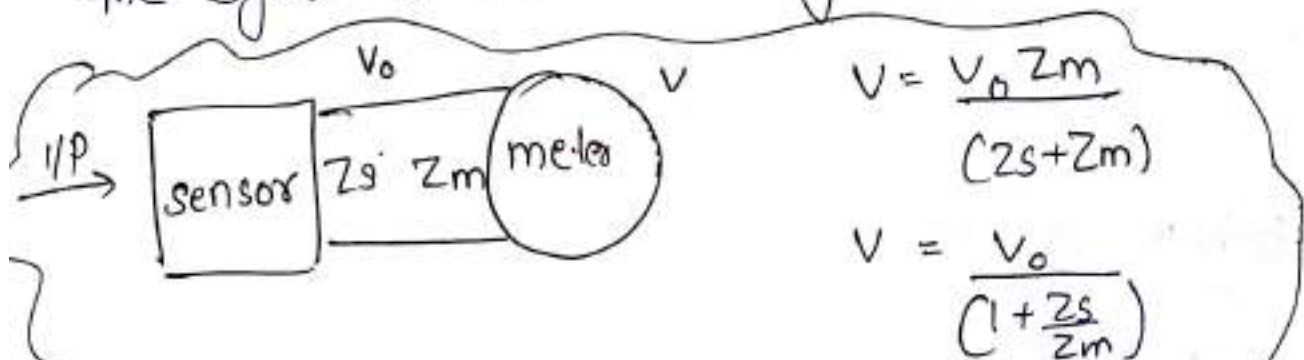
- The output is not measured as it is because of the impedance of the devices
- Input sensor can load the input quantity or output of the sensor can load the meter
- Loading → not measuring the actual true value

→ The loading effect in sensors refers how the sensor's presence and characteristics (like resistance, c) alter the input signal.

→ To minimise output loading effect of a sensor, it's important to choose sensors with impedance that are well-matched with the circuit

→ High input impedance is generally desired to avoid loading down the signal source

→ Low I/P Z is often preferred to efficiently deliver the signal to the next stage.



Loading error = $V - V_0$
To minimise loading error $V = V_0$
 $Z_m \gg Z_s$ or input/output imped of ins should be low

Q] A transducer has a source impedance that varies from $2-5\text{ k}\Omega$. During its operation,

i) Having an input impedance of $1.0\text{ M}\Omega$

ii) Having an input impedance of 100Ω

Which instrument is going to give a more accurate

i) $\frac{Z_s}{Z_m} = \frac{2 \times 10^3}{10^6} = 2 \times 10^{-3}, 5 \times 10^{-3}$

ii) (i) instrument will give more accurate results

Assume, consider Z_{op} of
sensor is maximum

$$= \frac{V_o - \frac{V_o}{(1 + 5 \times 10^{-3})} \times 100}{V_o} = \frac{100}{1 + 5 \times 10^{-3}} = \left(\frac{1}{1 + 5 \times 10^{-3}} - 1 \right) \times 100 = 0.1\%$$

$$V = 0.99V_o$$

ii) $V = 0.19V_o$

$$\frac{(0.19V_o - V_o) \times 100}{V_o}$$

$$\frac{(1 - 0.19) \times 100}{(0.81) \times 100} = 81\%$$

* Selection criteria for a transducer

- Mech suitability $\rightarrow$ size, weight, shape, mounting arrangement, cable-connectors

ii) Measurement suitability → Desired measurement range, accuracy, resolution, overload protection

iii) Performance suitability → linear relation, high sensitivity, good repeatability

iv) Electrical suitability → loading effect, power requirement, excitation signal range

v) Environment suitability → less sensitive to interfering inputs or external loads - load error, self heating effect, temp, humidity, vibration, dust, magnetic field, supply freq, voltage

vi) Maintenance suitability

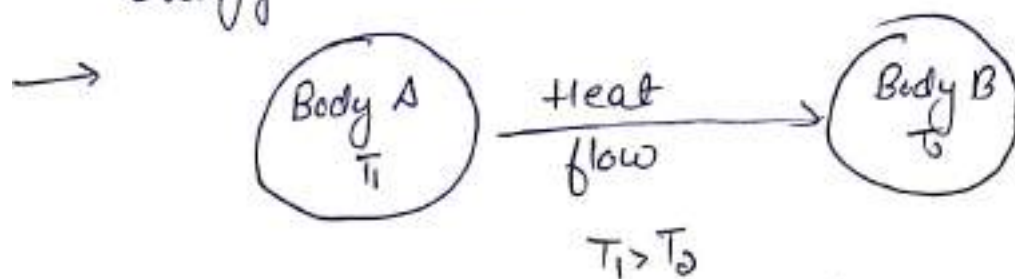
vii) Robustness suitability

2. Temperature Measurement

→ The measure of the degree of hotness or coldness of an object.

→ Measure of the average kinetic energy of the particles

→ Measure of the degree of (amount of) heat energy contained within the object



(At equilibrium both bodies at same temp T)

of Temperature measurement

transducer and the object → equilibrium

→ Temp is measured indirectly by measuring the effect of heat on the sensing element & that changes its physical or electrical properties. The change in any properties response to temp.

→ ΔR

→ ΔVol

→ Δcolor $\Delta \text{intensity of radiation}$

→ Calibration of sensor o/p - i/p characteristic is done through comparison with established standard.

$\rightarrow 0^\circ\text{F} = \left(\frac{9}{5}^\circ\text{C} + 32 \right) \rightarrow \text{Ice point} = 0^\circ$
 $\text{Steam point} = 100^\circ\text{C}$
 $\text{divided into 100 parts}$
 $\rightarrow \text{Ice point} = 32^\circ\text{F}$
 $\text{S.P.} = 212^\circ\text{F} \quad (180 \text{ parts})$

$$\rightarrow {}^\circ\text{K} = {}^\circ\text{C} + 273.2$$

$\rightarrow$ Absolute zero (no activity):-

* Temperature self points for calibration of temp scale

Temp ($^\circ\text{C}$) Ref points
 -182.97°C Fund.

1) Oxygen point

$\rightarrow$ Temp of equilibrium between
 Oxygen and its vapour

2) Ice point

$\rightarrow$ equilibrium between ice
 & water

0°C

"

3) Steam point: Temp of

equilibrium between liquid
 water and its vapour

100°C

"

4) Sulphur point

eq. between liquid sulphur
 and its vapour

444.6°C

"

5) Silver point

-960.8°C

"

6) Gold point

1063°C

7) Tin melting point

231.9

Secondary

8) Lead melting point

327.3

"

10) Pt/Platinum point

eq. between solid and liquid

→ Mercury filled thermometer use to calibrate
 other temp transducer (RTD, thermistors) use same
 range (-30-350°C)

1) RTD

(Oxygen point to ice point)

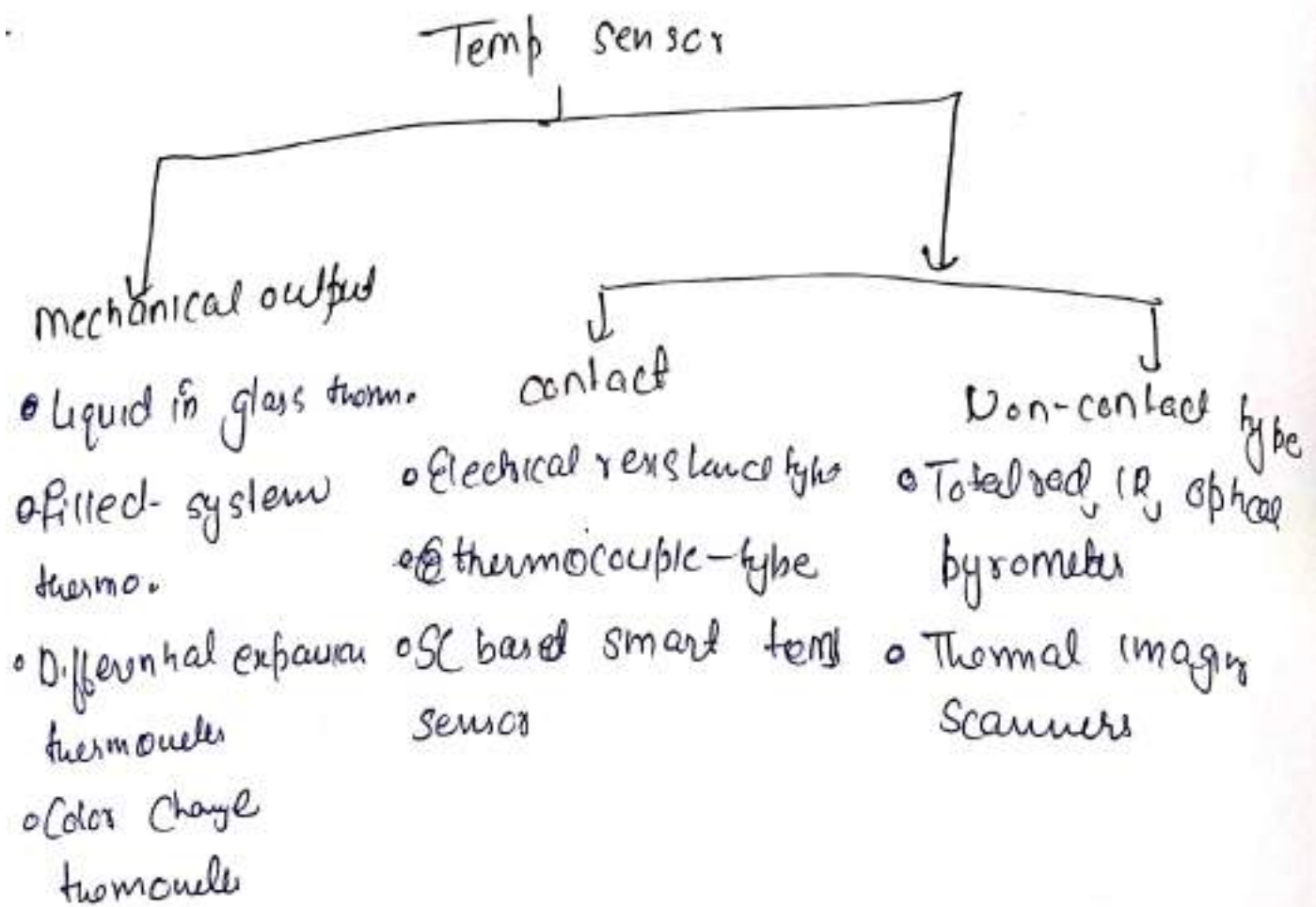
2) RTD

(ice point - 630.5°)

3) Pt-Pt Rhodium alloy thermocouple (630.5 to 1063°C)

4) Above 1063°C

Optical pyrometers



Q2) A diaphragm strain gauge connected with wheatstone bridge.

Ex 20V

R Ex 12V

P.R 200 kPa

O/P 0.1 mV/V/kPa

Accuracy $\pm 0.5\%$ FSD

Repeatability $\pm 0.2\%$ FSD

Linearity and hysteresis error = $\pm 0.4\%$ FSD

Thermal sensitivity drift = $0.02\%/^{\circ}\text{C}$

The resistance of bridge 350 Ω

meter resistance = 2 k Ω

Overloading capacity 250% FSD

We are applying very high I/P that needed for short time

2i) Find out static sensitivity at recommended voltage

$$\begin{aligned}\text{Static sensitivity} &= 0.1 \text{ mV} \times 12 / \text{kPa} \\ &= 1.2 \text{ mV/kPa}\end{aligned}$$

ii) Voltage corresponding pressure of 100 kPa at recommended voltage

$$V = 1.2 \times 100 \text{ kPa} \text{ mV}$$

$$V = 120 \text{ mV}$$

iii) Find the maximum output voltage of the device at 12 at 20V

$$\rightarrow \text{at 12V O/P} = 1.2 \times 200$$

$$= 240 \text{ mV}$$

$$\rightarrow \text{at 20V O/P} = 20 \times 0.1 \times 200 = 400 \text{ mV}$$

iv) Accuracy in kPa at recommended voltage

$$= \pm \frac{0.5}{100} \times 200$$

$$= \pm 1 \text{ kPa}$$

v) Calculate hysteresis and linearity error at F.S.O
Output

$$\rightarrow \pm \frac{0.4}{100} \times 240$$

$$\pm 0.96 \text{ mV}$$

→ hysteresis + linearity always at output

vi) Overload capacity in terms of kPa

i) I/P

$$\frac{250}{100} \times 200 = 500 \text{ kPa}$$

ii) O/P

$$\frac{250}{100} \times 240 = 600 \text{ mV}$$

Our instrument at max can handle it for short time

vii) Loading error

$$V = \frac{V_o Z_m}{(Z_m + Z_s)}$$

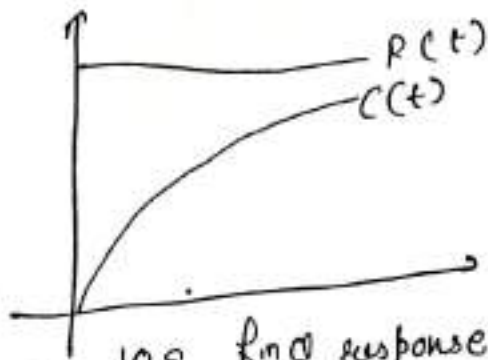
$$= \frac{V_o \cdot 2000}{(2000 + 350)}$$

$$V = 204.25 \text{ mV}$$

$$\% \text{ error} = \left(\frac{240 - 204.25}{240} \right) \times 100 = 14.89\%$$

Q) Assume a sensor is a

$$C(t) = R(t) (1 - e^{-t/\tau})$$



at a 10s find response of the given sensor given $\tau = 0.1$

$$C(t) = R(t) (1 - e^{-10/0.1})$$

$$= R(t) (1 - e^{-100})$$

$$= 0.999999 \dots R(t) \approx R(t)$$

→ because steady state time $\approx 4\tau$

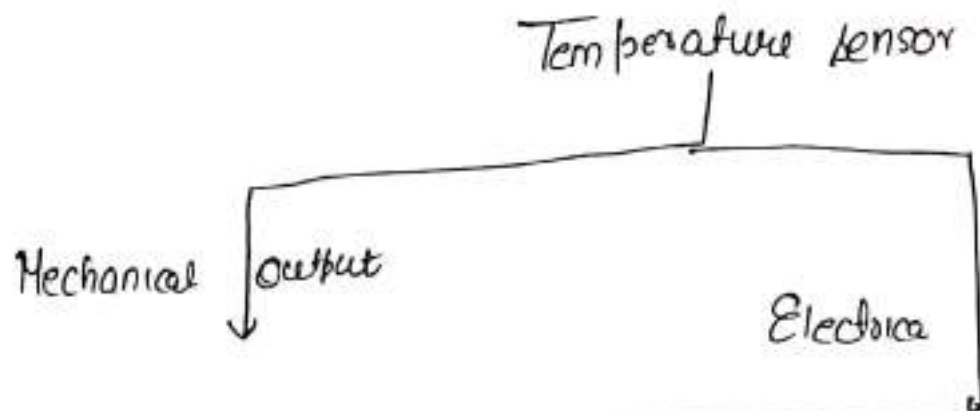
1) at 0.05s if $K = 10\text{mV}$ gain

$$C(t) = K R(t) (1 - e^{-10/0.1})$$

$$\boxed{3.934\text{mV}}$$

* Classification of Temperature sensor

11



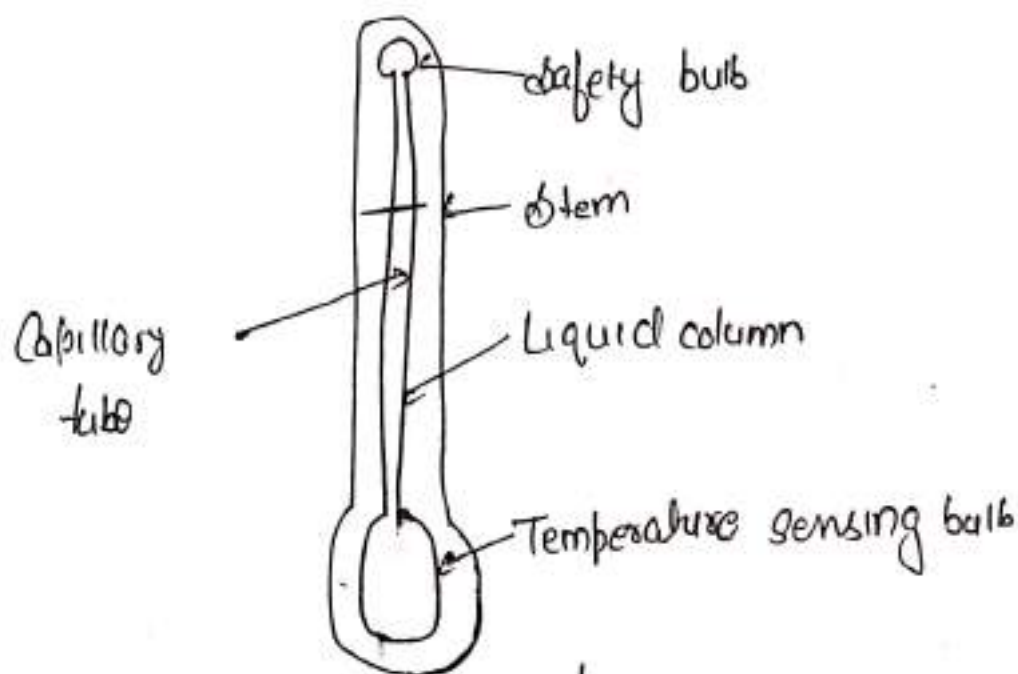
1) Liquid in glass thermometer

→ Liquid in a glass column is calibrated to read temperature

→ glass envelope → liquid column + safety bulb

→ responsive liquid is contained in temperature sensing bulb

→ scale



Liquid
Mercury

Range
-35 to 510

features

Common material, broad temp span, linear coefficient of exp.

Toxic

Alcohol	-80 to 70	Non-toxic, non-linear expansion coefficients lower span
Toluene	-80 to 100	
Pentane	-200 to 30	

Working principle

→ Volumetric expansion of a liquid with temperature

$$V(T) = V_0 (1 + at + bt^2 + ct^3); \quad (\text{non-linear}) \dots i)$$

(a, b, c are coefficients of volumetric expansion).

$$V(T) = V_0 (1 + \beta t) \quad (\text{linear})$$

(β is linear coefficient of volumetric expansion, $0.18 \times 10^{-3}/^\circ\text{C}$ for Hg)

$$\text{Change in volume } \Delta V = V(T) - V_0 = V_0 \beta T = k + T$$

(approx as linear relation)

$$\beta = \frac{\Delta V}{V_0 T} \quad \left. \vphantom{\frac{\Delta V}{V_0 T}} \right\} \begin{array}{l} \text{change in volume per volume per } ^\circ\text{C rise} \\ \text{temperature} \end{array}$$

→ change in volume cause the liquid to expand and rise within capillary

→ Volume of liquid occupied in the capillary ΔV_{cap} depends upon the area 'a' and height 'h' of the capillary tube

$$\Delta V_{\text{cap}} = ah$$

→ change in volume = volume in capillary

$$\Delta V = \Delta V_{\text{cap}}$$

$$K \propto T = a \cdot h$$

$$h = \left(\frac{K}{a}\right) T$$

$$S = \frac{K}{a} = \frac{V_0 B}{a}$$

$$h = S T$$

S = sensitivity of thermometer

a very small $\left(\rightarrow S \rightarrow \text{high} \right)$

$$h \propto T$$

linear relation

Features

- Simple, low cost, secondary temp std, no need of primary instrument, no excitation, resolution: 0.5°C , 1°C
- fragile construction, behave as a first order system, large time constant, there is a time lag between the change in temp and the thermometer steady state response due to the relatively high heat capacity

② Resistance Thermometers / Temperature Transducers

→ Resistance of the sensor element changes with temperature and has an established $R-T$ relationship over the sensor range

→ Passive, analog, resistive type, contact type.

RTD

(Resistance temperature dependent)
Metal, +ve $R-T$ coefficient
linear

Thermistor

Semiconductor,
-ve/+ve, exponential

1) RTD

→ RTD measures temp by detecting changes in R

→ Temp $\uparrow$, $K.E \uparrow$, increases in collision of particles
↓
obstruction to $e^- \rightarrow R \uparrow$

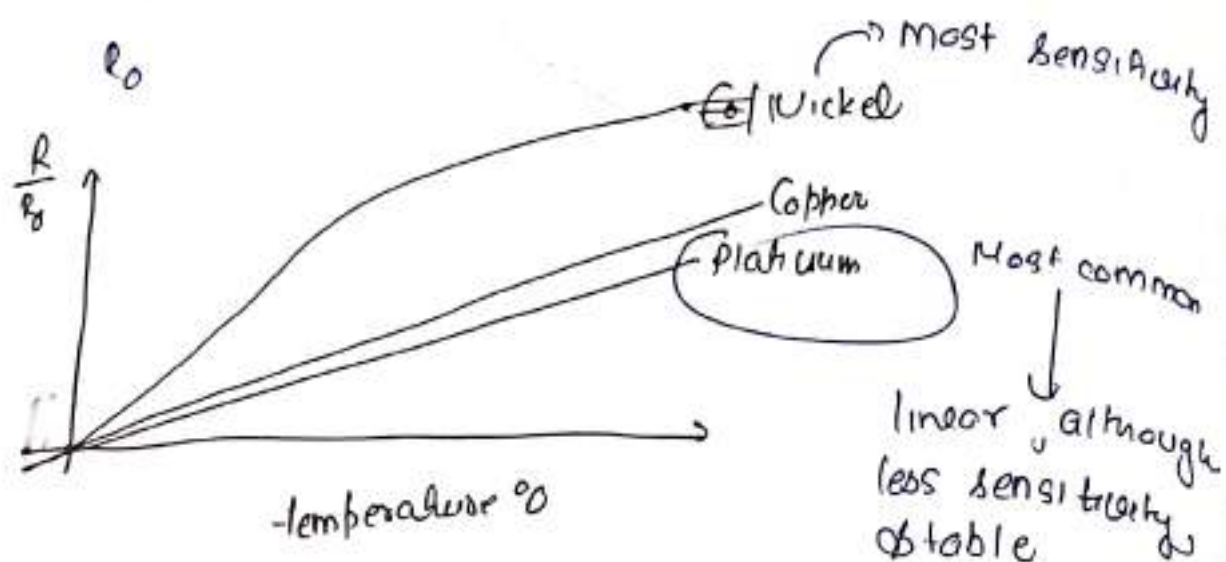
→ RTD (metal) → nickel, copper, Platinum linear non
and predictable. This change in resistance is then
converted into temp. reading.

$R-T$ equation of RTD

1) Quadratic Equation → over wide range, more accurate,
non linear

$$R(T) = R_0 \{ 1 + \alpha (T - T_0) + \alpha^2 (T - T_0)^2 \}$$

where, $R(T)$ is resistance at T
 R_0 is resistance at T_0



ii) Linear approximate relation

(Small span taken) $\rightarrow (T_1 \text{ to } T_2)$

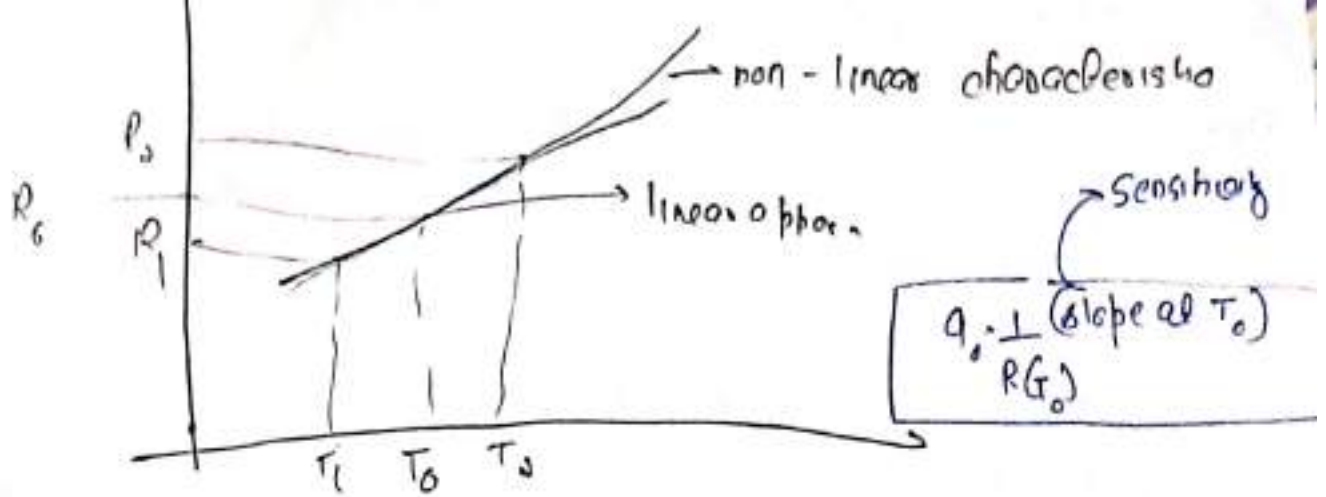
$$R(T) = R_{T_0} (1 + \alpha_{T_0} (T - T_0)) \quad \text{where } T_1 < T_0 < T_2$$

R_{T_0} is resistance at T_0

T_0 : mid way between the range or as specified
 6°C or 25°C $= (T_1 + T_2)$

And α_{T_0} is the fractional slope of curve between two end points denoting linear temperature coefficient of resistance within span,

$$\alpha_{T_0} = \frac{(R_{T_2} - R_{T_1})}{R_{T_0} (T_2 - T_1)}$$



Q) Find a linear approximation between 30°C and 60°C

$$R(30^\circ\text{C}) = 4.8\Omega, R(45) = 5.5\Omega \text{ and } R(60^\circ\text{C}) = 6.2$$

$$\alpha_{T_0} = \frac{1}{5.5} \left(\frac{30}{1.4} \right)$$

$$R_T = 5.5(1 + 0.008484(T - 45))$$

$$\alpha_{T_0} = 3.896$$

$$R_T = R_{T_0}(1 + 3.896(T - 45))$$

$$R_T = 5.5(1 + 3.896(T - 45))$$

$$\alpha = \frac{\Delta R}{R_0 \Delta T}$$

Linear approx $\rightarrow R_T = R_{T_0}(1 + \alpha_{T_0}(T - T_0))$

$$R(50) = ?$$

$$= 5.5(1 + 0.0085 \times 5)$$

$$R_{T_{50}} = 5.73\Omega$$

Practically for any RTD specified as Pt-100, Pt-1000
 $\downarrow \qquad \qquad \qquad \downarrow$
 $100\Omega \text{ at } 0^\circ\text{C} \qquad 1000\Omega \text{ at } 0^\circ\text{C}$

Linear approx
 (small span around 0°)

$$R_T = R_0(1 + \alpha_0(T - T_0))$$

$$T_0 = 0 \quad R_0 = 100\Omega$$

$$\alpha_{0^\circ\text{C}} = 0.00385/^\circ\text{C}$$

$$\Delta R = \alpha_0 R_0 (\Delta T)$$

Change in resistance of RTD for 1°C

$$\Delta R = 0.00385 \times 100 \times 1 = 0.385 \Omega$$

---> change in $R \propto$ temperature

---> S depends upon R_0 and α

---> very low sensitivity

* Materials

Ni, Ni alloys

Cu

Tungsten

Pt(100) - most desirable but low temperature coefficient

	Materials	Temp Range	α	Features
1)	Pt	-182 to 630 C	100 Ω at 0C 0.00385/C	stable ch linear
2)	Ni	-100 to 300 C	0.0067/C	
3)	Cu	120 C	0.004/C	
4)	Tg (W)	-450 to 4000 F	0.0045/10	

$$\textcircled{Q}) \quad \alpha = \frac{1}{599.48} \left(\frac{573.40 - 605.50}{-30} \right)$$

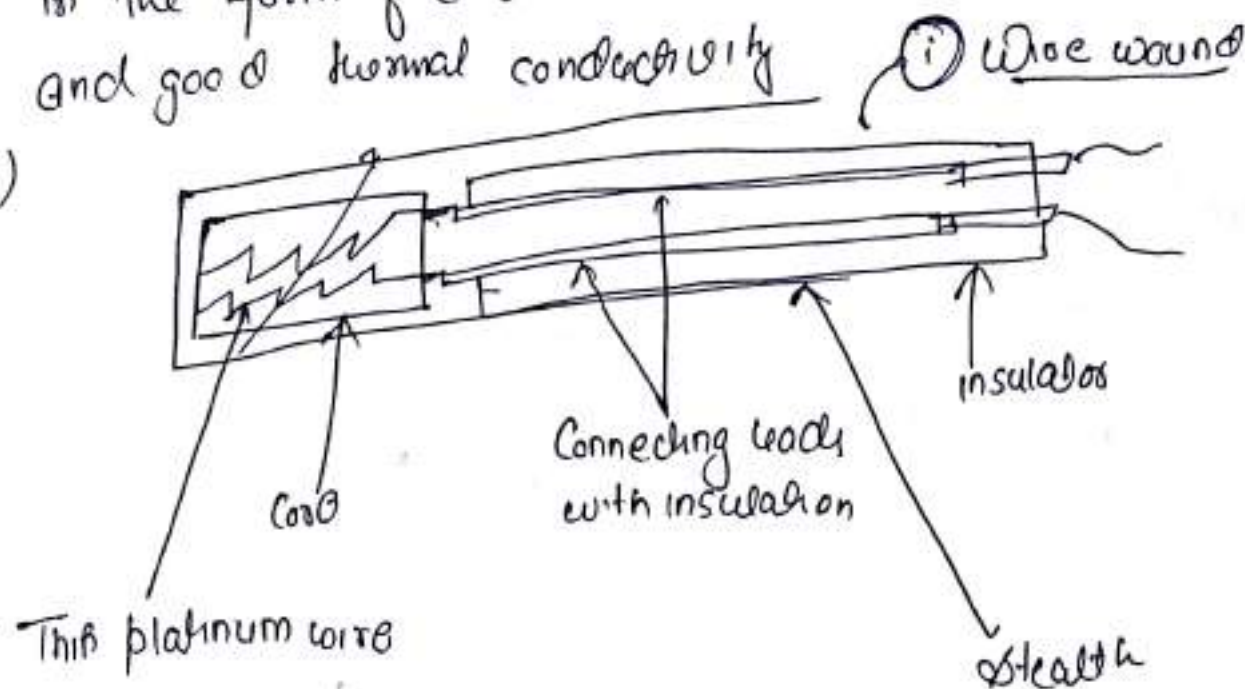
$$\alpha = 0.00182 / ^\circ\text{C}$$

$$\text{Linear Approximation} = 599.48(1 + 0.00182(T - 115^\circ\text{C}))$$

* Construction

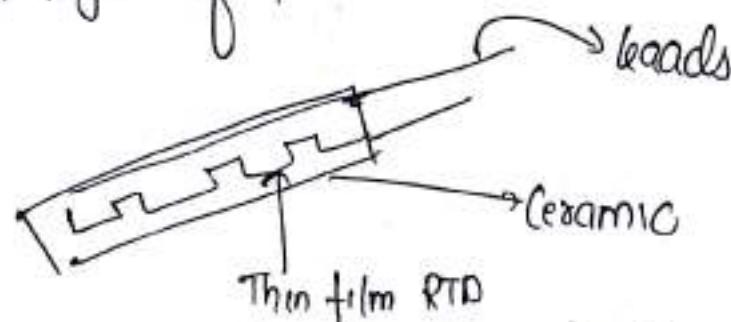
- i) Winding a fine wire usually Pt, around ceramic or glass core
- ii) It is a simple length of wire of Pt that would in the form of coil to achieve a small size and good thermal conductivity

iii)

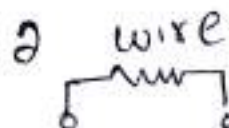


* Thin film RTD

→ Thin layer of Pt deposited onto a ceramic substrate

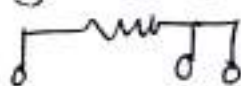


→ They are small in size and can



→ suffer inaccuracy due to resistance of connecting wires
→ short distance

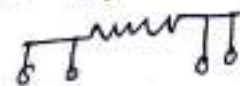
3 wire



→ improved accuracy
→ bit complicated
→ long distance

be less expensive

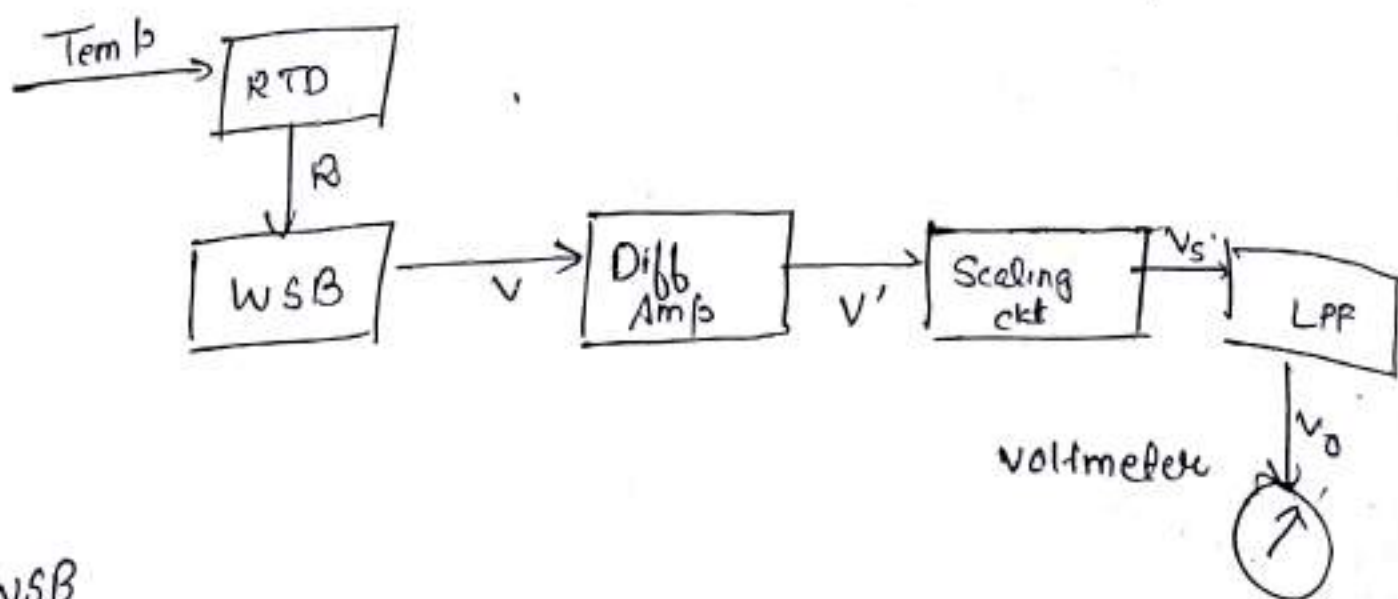
4 wire



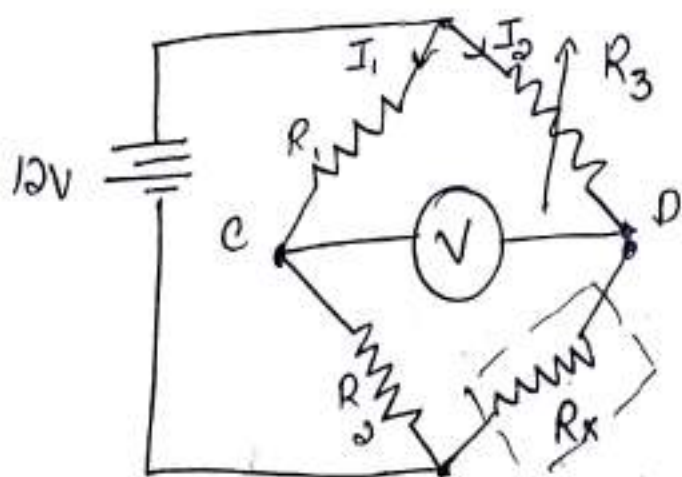
→ highest accuracy by completely eliminating the effect of lead wire
→ Acquires more processing

② Signal conditioning of RTD

- i) Resistance measurement
→ Wheatstone bridge
- ii) Amplification: Differential amplifier
- iii) Scaling circuit / Signal conditioning converter
- iv) Low pass filter: to minimize noise due to high freq components such as surging voltages



WSB



Eqⁿ of WSB at balance condition

$$V_{out} = (V_c - V_d)$$

$$R_c = R_d$$

$$V_c = V_d$$

$$\frac{R_1}{R_2} = \frac{R_3}{R_x}$$

$$R_x = \frac{R_3 \times R_2}{R_1}$$

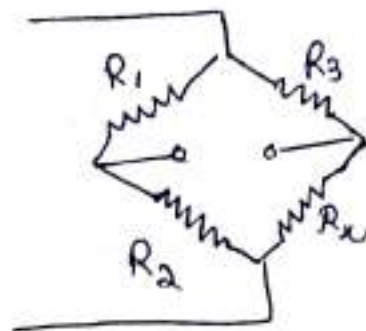
$$\frac{E_x \times R_2}{R_1 + R_2} = \frac{E_x \times R_x}{R_3 + R_x} \quad \text{Proof} \rightarrow$$

$$\frac{R_2}{R_1 + R_2} = \frac{R_x}{R_3 + R_x}$$

$$R_2 R_3 + R_2 R_x = R_x R_1 + R_3 R_x$$

$$R_x = \frac{R_2 \cdot R_3}{R_1}$$

RTD $\leftarrow$



→ At balance condition resistance of RTD is calculated in terms of other resistors, (R_3 is changed to obtain balance, $R_1 = R_2$ generally)

① Under unbalance condition

Consider a WSB in which

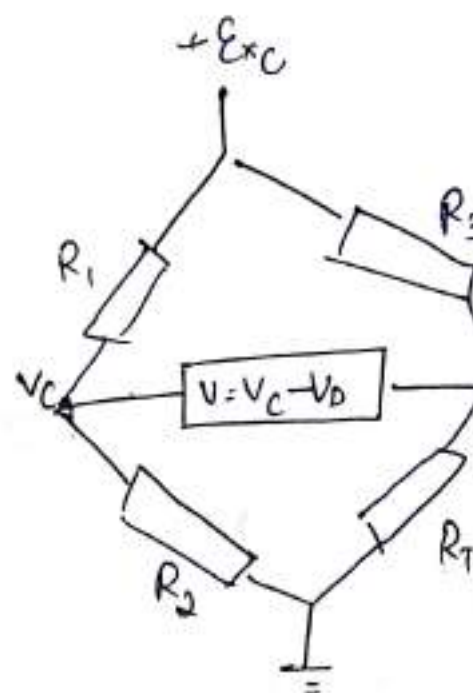
$$R_1 = R_2 = R_3 = R_{TD} = R_0$$

nominal R_0 under condition under which bridge is balanced

→ balanced condⁿ

$$R_{TD} R_1 = R_2 R_3 \quad \dots (1)$$

o/p voltage $V = V_{cd} - V_{db} = 0$



→ Consider that at any time t there is temperature variation from initial value T_0 to T

$$R(T) = R_0 (1 + \alpha (\Delta T))$$

$$\Delta R = R_0 \alpha \Delta T$$

where $\Delta T = T - T_0$

$$\Delta V = V_c - V_d$$

$$= \frac{E_x R_2}{R_3 + R_1} - \frac{E_x R_D}{R_3 + R_{TD}}$$

$$= \frac{E_x}{2} - \frac{E_x (R_0 + \Delta R)}{(R_0 + R_0 + \Delta R)}$$

$$(R_2 = R_1 = R_0)$$

$$= \frac{E_x}{2} - \frac{E_x (R_0 + \Delta R)}{(2R_0 + \Delta R)}$$

$$= E_x \left(\frac{1}{2} - \frac{(R_0 + \Delta R)}{(2R_0 + \Delta R)} \right)$$

$$= E_x \left(\frac{2R_0 + \Delta R - R_0 - \Delta R}{4R_0 + 2\Delta R} \right)$$

$$= \frac{E_x R_0}{4R_0 + 2\Delta R}$$

$$V_0 = \frac{E_{ex} \Delta R}{4R_0}$$

$$\left(\frac{\Delta R}{2R_0} \right)$$

is very small

$$V_0 = \frac{E_{ex} d \Delta T}{4}$$

$$V_0 = K \Delta T$$

$$K = \frac{E_{ex} d}{4}$$

$$S = \frac{E_{ex} d}{4}$$

→ Only valid equation when all resistances are equal initially

* About the output voltage of WSB

i) o/p is linear fn of temp

ii) bridge sensitivity is low depends on $\frac{E_{ex} d}{4}$

ii) Output is very low, requires for amplification

$$\Delta R \text{ for } 1^\circ T = \Delta R = R_0 \Delta T$$

$$= 100 \times 0.00387 \Omega$$

$$= 0.387 \text{ very very low}$$

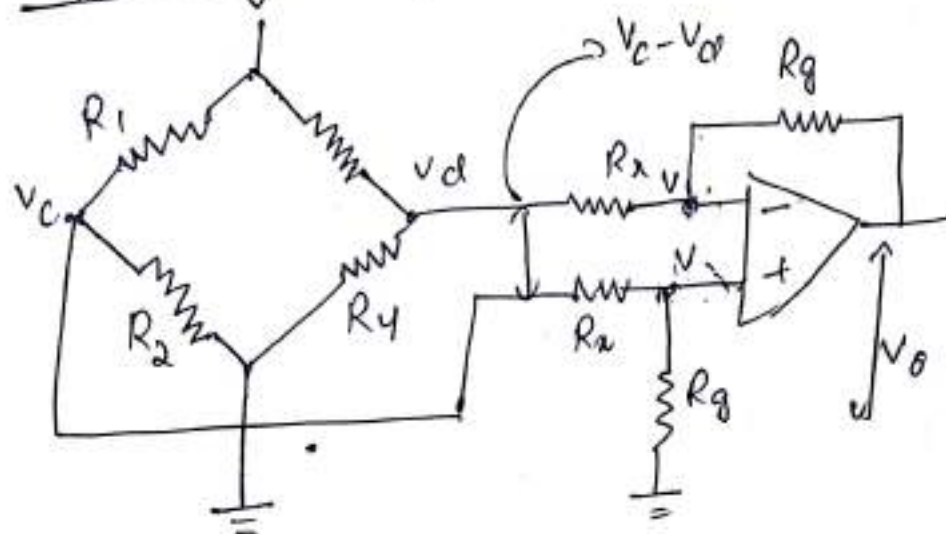
Comparable with lead resistance hence cause error
as total R_{TD} resistance = $R_{TD} + R_e$

$$(R_0 + \Delta R) \quad \downarrow \quad R_{lead}$$

$$V_{op} = \frac{V_{ex} \Delta T}{4}$$

e.g. $\rightarrow V_{ex} = 5 \rightarrow V_{op} = \frac{5 \times 0.0039 \times 1}{4}$
 $= 4.875 \text{ mV}$

* Differential amplifier



$$V_0 = A_d (V_A - V_B)$$

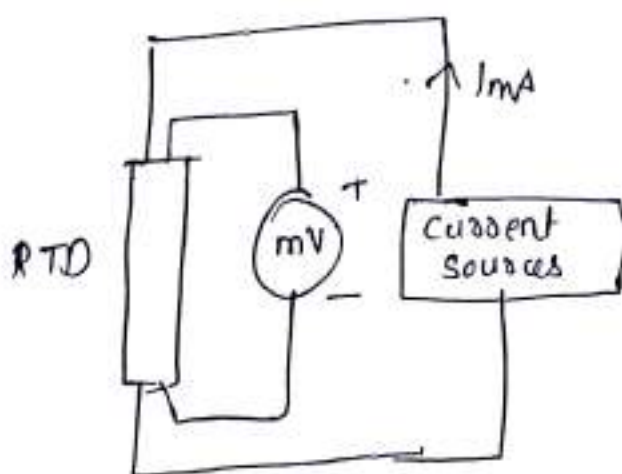
$$A_d = \frac{R_g}{R_x}$$

$$\frac{V - V_c}{R_2} = \frac{V}{R_g}$$

→ sources of errors and compensation

① Lead wire compensation

→ 4 wire RTD

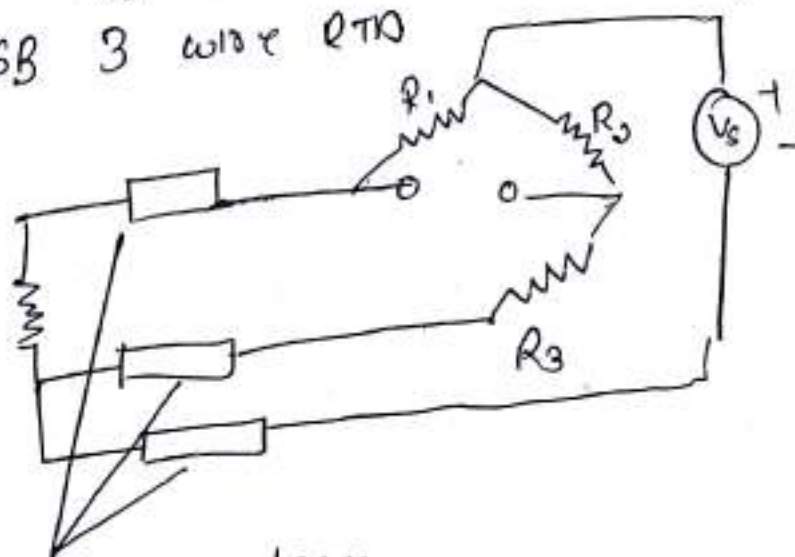


$$V_{(RTD)} = I (R_{TD}) \cdot R(T)$$

$$R(T) = \frac{V_{(RTD)}}{I_{(RTD)}}$$

2 wire configuration (Prone to lead effect)

→ Connect WSB 3 wire RTD



lead resistance

lead resistance is same in all arms so for output
lead resistance (R_l) is cancelled using 3 wire-connection
lead

$$P(RTD) = I^2 R_{TD}$$

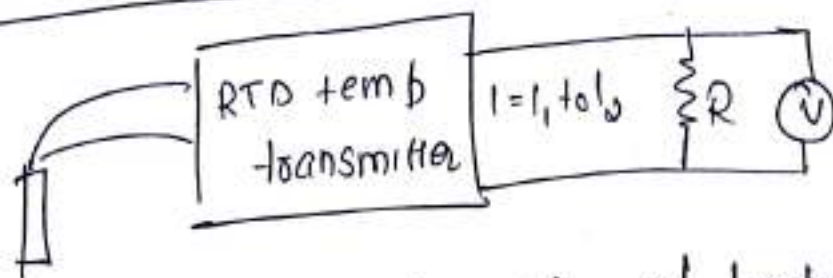
Where I is the current flowing in RTD in μA

$$= \frac{E_x}{(R_{TD} + R)}$$

$$\Delta T = \frac{P_{(RTD)}}{P_0}$$

→ if power dissipation in RTD goes beyond power dissip. constant of RTD P_0 , then there is temp. rise of RTD due to self-heating. Because of self-heating, RTD will sense more temperature

② RTD with Transmitter $\rightarrow T = T_1 \text{ to } T_2 \text{ range}$



→ Current I carries the value of temperature T

$R \rightarrow$ Very high precise

$$I = \frac{V}{R}$$

Corresponding to current I temp

T is indicated

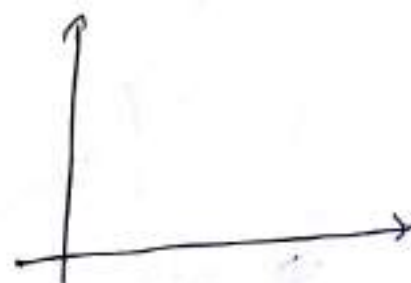
$$T = T_1 + \left(\frac{(T_2 - T_1)}{(I_2 - I_1)} \right) (I - I_1)$$

$$y = mx + c$$

$$y \rightarrow I$$

$$x \rightarrow T$$

$$c =$$



Features of RTD

- Established linear relationship
- Low sensitivity ($0.00385/^{\circ}\text{C}$), requires amplification
- Require lead wire compensation. (3 wire RTD or Pt-1000)
- Time lag → RTD enclosed in protective sheath has more time lag.
- Prone to self-heating effect and undue rise in temp.
careful design of bridge circuit to minimize error
requirement of low current source

Q.2] A copper RTD at 20°C with $\alpha = 0.00393/^{\circ}\text{C}$ is used to indicate the temp of the bearing of the machine. What resistance should not be exceeded if max bearing temp is not to exceed 150°C .

$$R_{\text{Cu}}(20^{\circ}\text{C}) = 100\Omega$$

$$R = 100(1 + 0.00393(150 - 20))$$

Q.3] $I = \frac{V}{R} \rightarrow I = \frac{4}{500} \quad 8\text{mA}$

$$T = T_1 + \left(\frac{T_2 - T_1}{I_2 - I_1} \right) (I - I_1)$$

$$T = 0 + \left(\frac{500}{16} \right) (8 - 4)$$

$$T = \frac{500}{4} = 125^{\circ}\text{C}$$

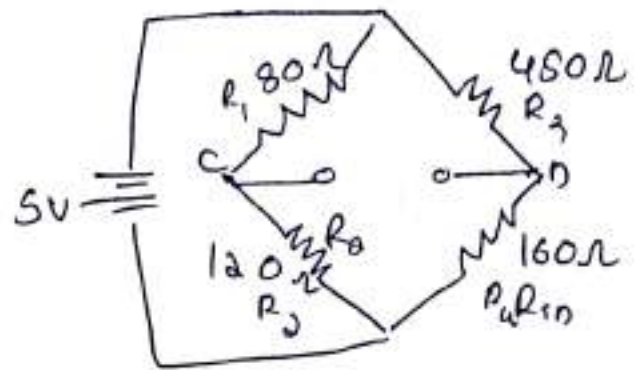
Q-4) $V_0 = \frac{\epsilon_{ex} \alpha \Delta T}{4}$

$V_0 = 5 \times (0.00385) \times$

$V_c = \left(\frac{5 \times 120}{120 + 80} \right) = 3V$

$V_d = \left(\frac{5 \times 160}{160 + 480} \right) = 1.25$

$V_0 = V_c - V_d = 1.75V$



Calculate V_0 under unbalance condition.

ii) Find value of R_3 to put bridge under balance

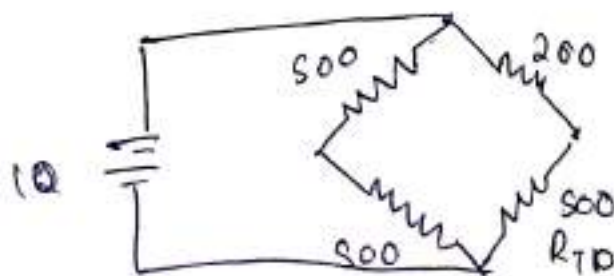
$\frac{80}{120} = \frac{R_3}{160}$

$R_3 = \frac{80 \times 160}{120} = 106.66$

Q-5) An RTD has $\alpha = 0.005/^\circ C$, $R = 500\Omega$ at $20^\circ C$ and dissipation constant of $30mW/^\circ C$. RTD is in bridge

circuit with $R_1 = R_2 = 500\Omega$, R_3 a variable resistor of 200Ω

If $V_s = 10V$, self heating find temperature rise due to



$I_{RTD} = \frac{10}{200 + 500}$

$= 0.0143mA$

$P = I^2 R$
 $= 102.2mW$

$$\Delta T = \frac{P_{em}}{R_c} = \frac{100 \cdot 2}{30} = 6.67^\circ \text{C}$$

ii) if RTD is placed at 0°C , find R_{T0}

$$R = 500(1 - \alpha \Delta T)$$

$$R = 500(1 - 0.005 \times 20) = 450 \Omega$$

Thermistor

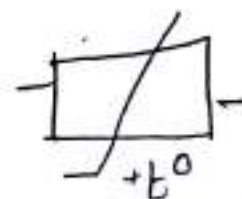
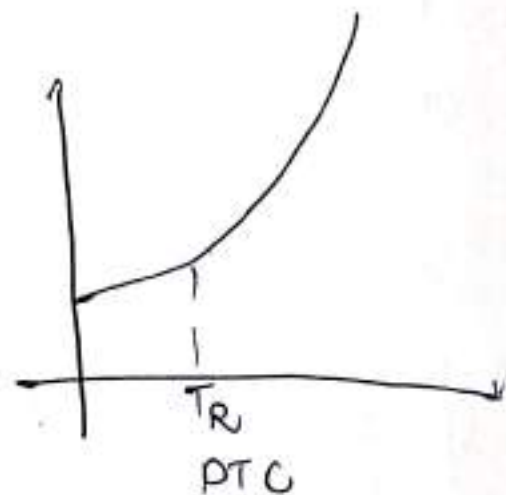
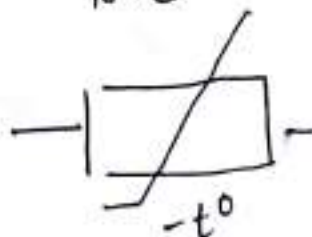
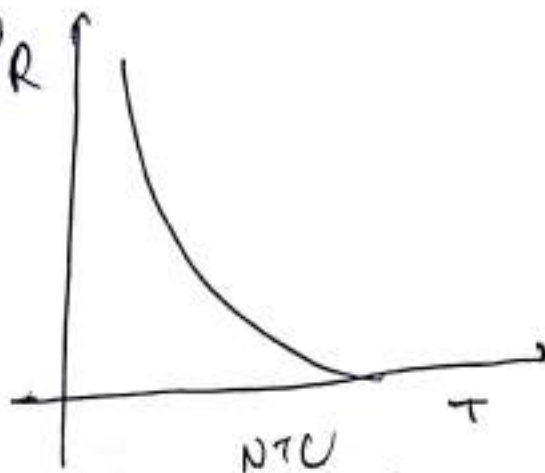
- resistance change non linear with the temperature
- high insensitivity
- They are used for controlling temperature in sensitive environment

S.C → Temp ↑ → e^- and hole pair generate → $R \downarrow$

NTC thermistor: Resistance decreases with increase in

PTC thermistor: $R \uparrow \rightarrow T \uparrow$

① NTC





R-T equation

i) Non-linear characteristics

$$R = R_0 e^{\beta (1/T - 1/T_0)}$$

$T \rightarrow$ in Kelvin

around 3500-4000

R = resistance at T

R_0 = resistance at T_0

β = constant, characteristic of material

T_0 = absolute temp in K

T_0 is the ref temperature 298K, 25°C

ii) Comparing linear temp coefficient of Thermistor (α) with β

$$\frac{dR}{dT R_0} = -\frac{\beta}{T_0^2}$$

Alpha is negative -ve ($= -\frac{4000}{298^2} = -0.04/C$)

(Alpha for Pt = 0.00385/C, Thermistor = -0.04/C)

(10 times more, -ve) $\rightarrow$ 10 times more sensitive

iii) To calculate resistance of the thermistor, using two points

$$R_{T_1} = R_{T_2} e^{B \left(\frac{1}{T_1} - \frac{1}{T_2} \right)}$$

iv] Use of linear-inverse relationship in small span
 $R(T) = R_0 (1 - \alpha(T - T_0))$

Base value of Thermistor ($10k\Omega, 1000k\Omega$) at $25^\circ C$

v] Approx. log relationship to find thermistor resistance at Temp T using resistance at 100 points

$$(R_T = a R_0 e^{\frac{b}{T}})$$

where a and b are some constant $T \rightarrow (K)$

vi] Sternhart - hart equation

$$\frac{1}{T} = A + B(\ln R) + C(\ln R)^3$$

to calculate unknown T at a given value of R

vii] Look-up method

$$Y = Y_1 + \left(\frac{X - X_1}{X_2 - X_1} \right) (Y_2 - Y_1)$$

where (X_1, Y_1) and (X_2, Y_2) is the extreme points on the table.

Q.1) A $10k\Omega$ NTC thermistor has a value of "13" $\Delta R/R$ between the temperature range of $25^\circ C$ and $100^\circ C$. Calculate the resistive value at $25^\circ C$ and again at $100^\circ C$

$$(R_0 = 10k\Omega)$$

$$R = R_0 e^{b/T - b/T_0}$$

$$R = 935 \Omega$$

Q2) A thermistor has a resistance of 3980Ω at ice point (0°C) and 794Ω at 50°C . The resistance relation is $R_T = aR_0 e^{b/T}$ calculate a and b . Calculate the range of resistance to be measured in case temp $\sim 40^\circ\text{C}$ to 100°C

$$\rightarrow 3980 = a \cdot 3980 e^{b/273}$$

$$1 = a e^{b/273}$$

$$\rightarrow 794 = a \cdot 3980 e^{b/323}$$

$$a = 30 \times 10^{-6}$$

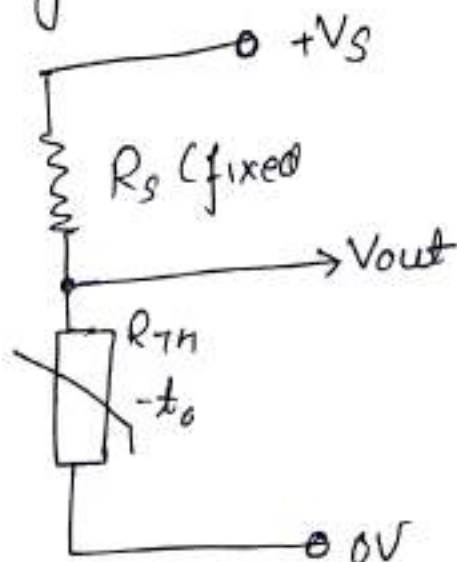
$$b = 2845$$

$$R_{(40^\circ\text{C})} = 1060 \Omega \quad R_{(100^\circ\text{C})} = 245 \Omega$$

$$(1060 - 245) \Omega$$

Measurement circuit (to convert $R \rightarrow V$)

1) Voltage divider circuit



$$V_{out} = V_s \left(\frac{R_{TH}}{R_{TH} + R_s} \right)$$

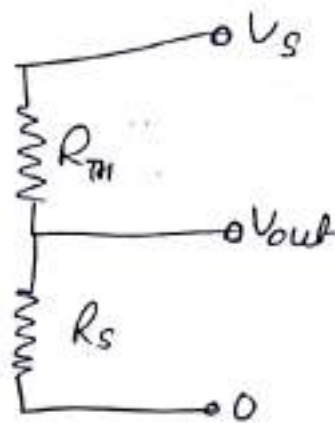
(At nominal temp, d/p voltage = $0.5 V_s$) when $(R_{TH} = R_s)$

as temp increases, R_{TH} decreases, V_o decreases

↳ so to avoid this

$$\rightarrow V_{out} = V_s \left(\frac{R_s}{R_s + R_{TH}} \right)$$

$T \uparrow \rightarrow R_{TH} \downarrow \rightarrow V_{out} \uparrow$



Find Value of R_{TH} in terms of V_{out}

$$V_{out} = \frac{V_s R_{TH}}{R_{TH} + R_s}$$

$$V_{out} = \frac{V_s}{\left(1 + \frac{R_s}{R_{TH}}\right)}$$

$$1 + \frac{R_s}{R_{TH}} = \frac{V_s}{V_{out}}$$

$$\frac{R_s}{R_{TH}} = \frac{V_s - V_{out}}{V_{out}}$$

$$R_{TH} = \frac{R_s \times V_{out}}{V_s - V_{out}}$$

Now we can use Steinhart-Hart equation to find temperature

$$\frac{1}{T} = A + B(\ln R) + C(\ln R)^3$$

① Choice of R_s and Excitation current for the thermistor in VD and WSB circuit.

→ Requirement

$$V = \frac{V_o}{1 + \frac{Z_{in}}{Z_m}}$$

1) Current need stable and precise excitation current.

Excitation should be carefully controlled to prevent self-heating

Power diss $P_{TH} = I^2 R_{TH}$
 Temp Rise, $\Delta T = \frac{P_{TH}}{P_D}$ } $P_{TH} < P_D$

→ The highest temperature should not cause self heating

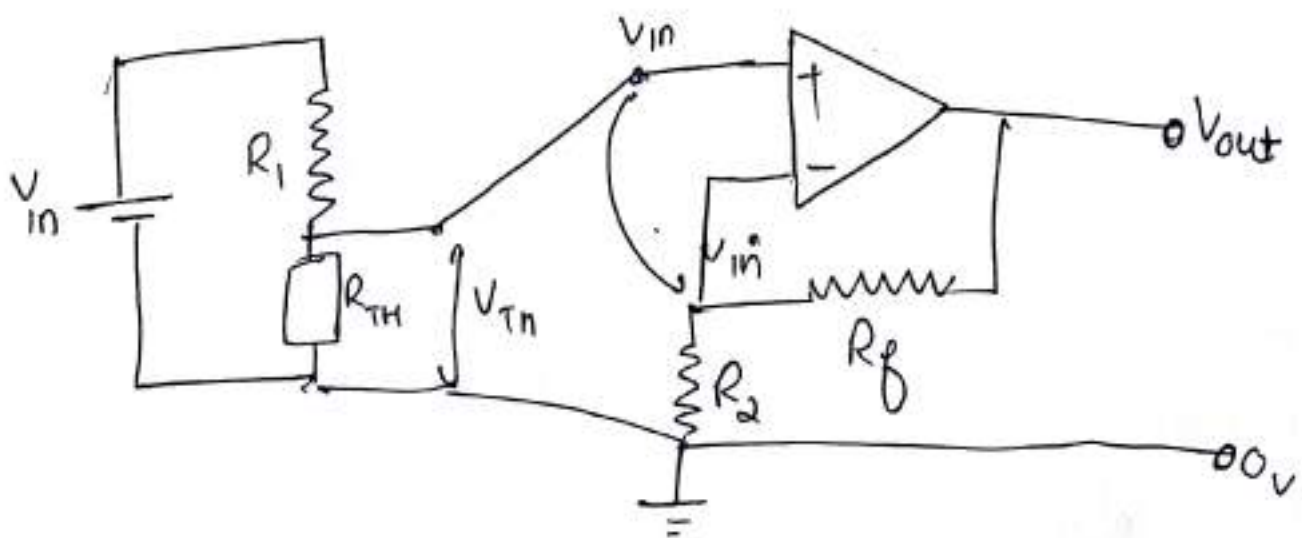
Signal conditioning

- Differential amp (with WSB voltage)
- Single-ended non inverting amplifier (with V.D.N.)

$Z_{in} \Rightarrow$ high } loading high

Filtering

low-pass filtering may be needed to reduce noise



$$V_{out} = V_{in} (\text{gain}) = V_{in} \left(1 + \frac{R_f}{R_2} \right)$$

$$V_{in} = V_{TH} = \frac{E_x (R_{TH})}{(R_{TH} + R_S)}$$

$$V_o = \frac{E_x (R_{TH})}{(R_i + R_{TH})} \left(1 + \frac{R_6}{R_3} \right)$$

$$\frac{V_{in}}{R_3} = \frac{V_{out} - V_{in}}{R_f}$$

$$\frac{R_f}{R_3} = g - 1$$

$$\boxed{\left(g - 1 + \frac{R_f}{R_3} \right)}$$

characteristics →

- High sensitivity

- Accuracy →

They are considered highly accurate temperature sensors.

- Limited temperature range →

thermistors typically have a limited operating temperature range, often within 0°C to 100°C .

- Nonlinear characteristics -- overall range

-- approx. linear relation in small span

uses

- ① Overcurrent protection → (PTC)

- ② Low - temp. measurement → Some thermistors, especially made with germanium, are used in cryogenic temp.

- ③ Digital thermostat → glyser, heater etc.

② Difference →

Thermistor
Sem

RTD

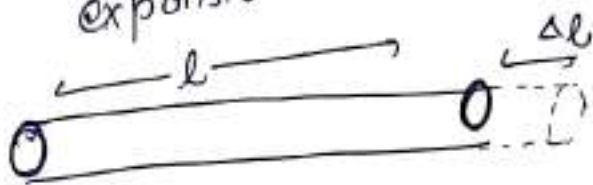
Dynamic response of thermal sensor

Q4 } Bimetallic thermometer

↳ (Mechanical sensor)

→ based on thermal expansion of metals

→ k.E ↑ → intramolecular spacing ↑ → linear expansion



$$\Delta l = (l_0 \alpha \Delta T)$$

$$\boxed{\Delta T = T - T_0}$$

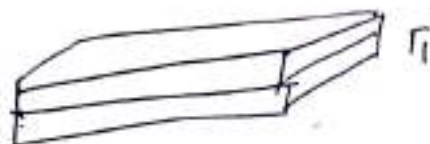
$$\boxed{l = l_0 (1 + \alpha (T - T_0))}$$

② Working principle

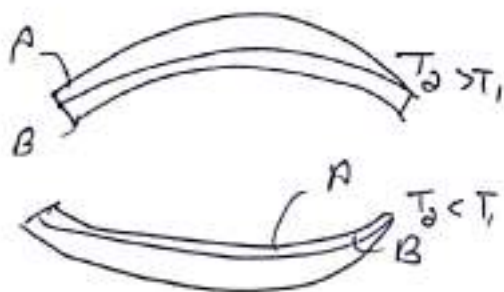
A 
 B 

$\alpha_A > \alpha_B$
 A is active $\rightarrow$ higher coen
 B is passive - lower coeff

Bonded



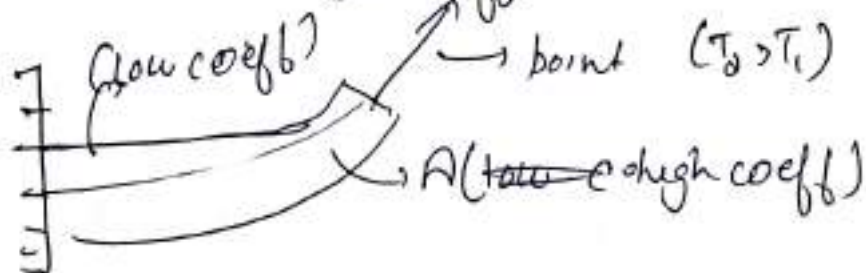
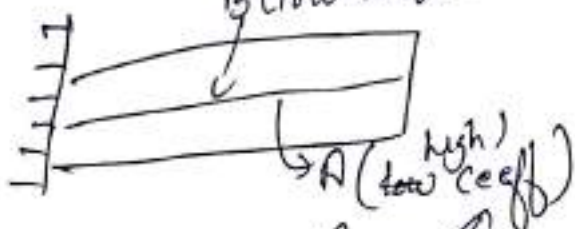
Temperature change $\rightarrow$



② Construction

1) Cantilever B's strip

At temp T_0
 B (low coeff)



Passive material $\rightarrow$ Invar (low coeff)
 $\rightarrow$ Invar (low coeff)
 Active material $\rightarrow$ Brass
 $\rightarrow$ Chromium

• Linear thermal coefficient of expansion $\rightarrow$ Expansivity

$$= \frac{\Delta l}{l_0 \Delta t^{\circ}C}$$

* Radius of curvature of the bimetallic strip
free at both ends

(Higher coeff material)
 α_2, E_2, t_2

α_1, E_1, t_1
 (low coeff material)

$$x = t \times \frac{[3(1+m)^2 + (1+mn)(m^2 + 1/mn)]}{6(\alpha_2 - \alpha_1)(T - T_0)(1+m)^2}$$

$$(t = t_1 + t_2)$$

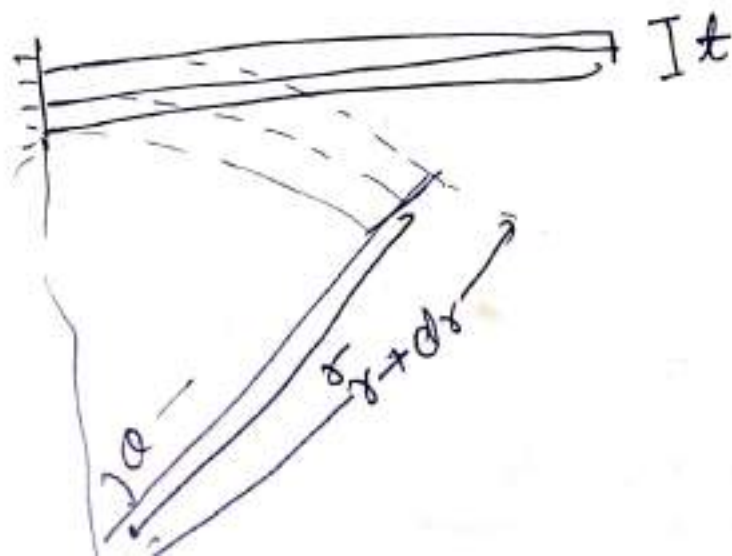
$$m = \frac{t_1}{t_2} = \frac{\text{low coeff material thic.}}{\text{high coeff material thic.}}$$

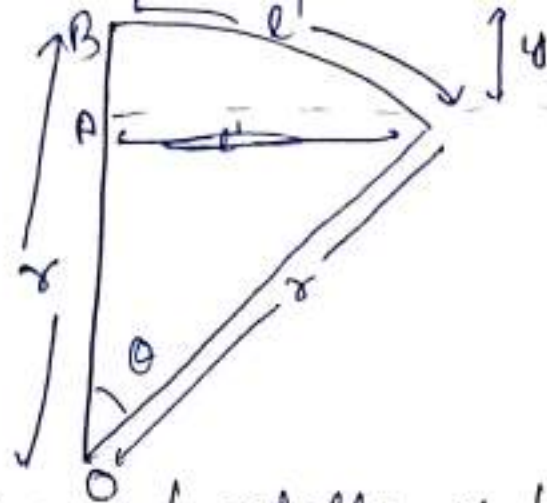
$$n = \frac{E_1}{E_2}$$

if $t_1 = t_2$ and if the materials are so chosen
 that $E_1 \approx E_2$

$$x = \frac{2t}{3(\alpha_2 - \alpha_1)(T - T_0)}$$

• To find the deflection of free tip of bimetallic cantilever strip in response to temperature





- Consider that a bimetallic strip in the form of cantilever is subject to temp T from initial temp T_0 . This results in the bending of the strip due to differential linear expansion.

$$\frac{l + \Delta l}{r} = \frac{\text{expanded length of strip having higher } (\alpha_2)}{\text{expanded length of strip having higher } (\alpha_1)}$$

$$\frac{l + \Delta l}{r} = \frac{l (1 + \alpha_2 (T - T_0))}{l (1 + \alpha_1 (T - T_0))}$$

Assume $\alpha_1 \ll \alpha_2$ i.e. θ

$$\text{and } \Delta l = \frac{t}{2}$$

$$\frac{l + \frac{t}{2}}{r} = (1 + \alpha_2 (T - T_0))$$

$$1 + \frac{t}{2r} = 1 + \alpha_2 (T - T_0)$$

$$\frac{t}{2r} = \alpha_2 (T - T_0)$$

$$r = \frac{t}{2\alpha_2 (T - T_0)}$$

$$\theta = \frac{l'}{r}$$

$$y = OB - OA$$

$$y = r - r \cos \theta$$

$$y = r - r \cos\left(\frac{l'}{r}\right) \quad \text{--- (B)}$$

$$y = r - r \cos\left(\frac{l(1 + \alpha_2(T_2 - T_0))}{r}\right) \quad \text{--- (C)}$$

Q] A bimetal strip of nickel-chrome-iron alloy
thickness 1mm
($T_0 = 25^\circ\text{C}$)

i) radius of curvature at 200°C

ii) movement of free end in perpendicular direction

$$\alpha_1 = 1.7 \times 10^{-6} / ^\circ\text{C} \quad E_1 = 150 \text{ GPa/m} \quad (\text{Invar})$$

$$\alpha_2 = 12.5 \times 10^{-6} / ^\circ\text{C} \quad E_2 = 220 \text{ GPa/m} \quad (\text{nickel-chrome-iron})$$

$$t = (t_1 + t_2) = (1 + 1) = 2 \text{ mm}$$

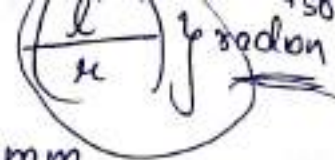
$$m = \frac{1}{1} = 1$$

$$n = \frac{150}{220} = 0.682$$

$$r = \frac{2 \times \left(\frac{1}{3}(1+m)^2 + (1+mn) \left(m^2 + \frac{1}{mn} \right) \right)}{6(\alpha_2 - \alpha_1)(T - T_0)(1+m^2)}$$

$$r = \frac{3 \times 4 + (0.682)(1 + 1.466)}{3 \times 10^{-6} \times (12.5 - 1.7) \times 175 \times 4} = 712 \text{ mm}$$

$$\begin{aligned}
 \text{ii) } y &= \kappa (1 - \cos \theta) \\
 &= 712 (1 - \cos 4) = 1.78 \text{ mm}
 \end{aligned}$$



free end height

c) Vertical displacement in case of cantilever

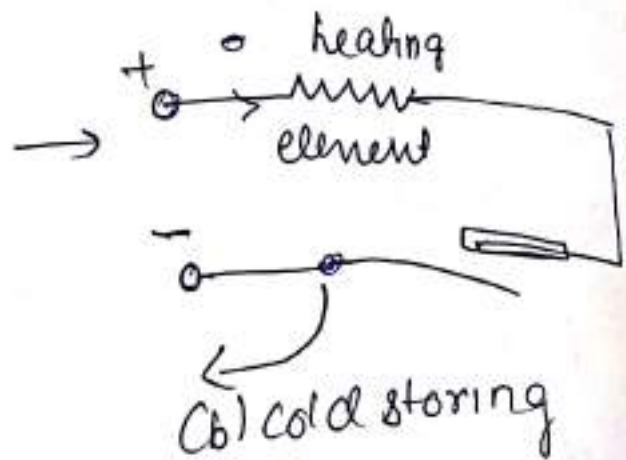
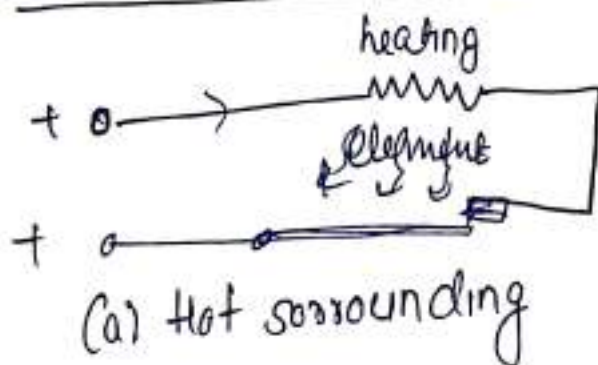
$$\kappa = \frac{l}{2(\Gamma - \Gamma_0) \alpha_0} = \frac{2}{2 \times 175 \times 12.5 \times 10^{-6}} = 457.14 \text{ mm}$$

$$\theta = \frac{l}{\kappa} = \frac{50}{457.14} = 6.27^\circ$$

$$\frac{6}{160} \times \frac{180}{\pi}$$

$$\underline{y = \kappa (1 - \cos \theta)}$$

B.M.S as a switch

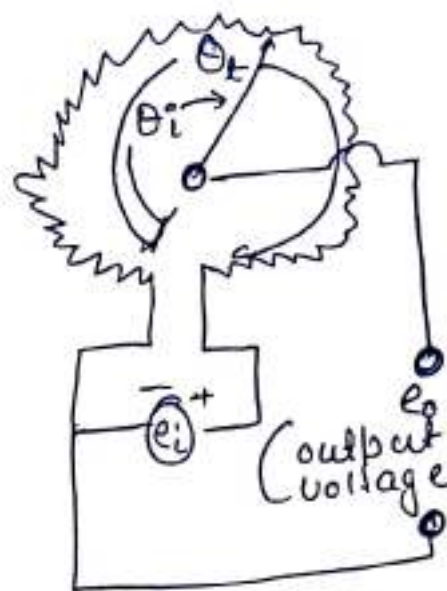
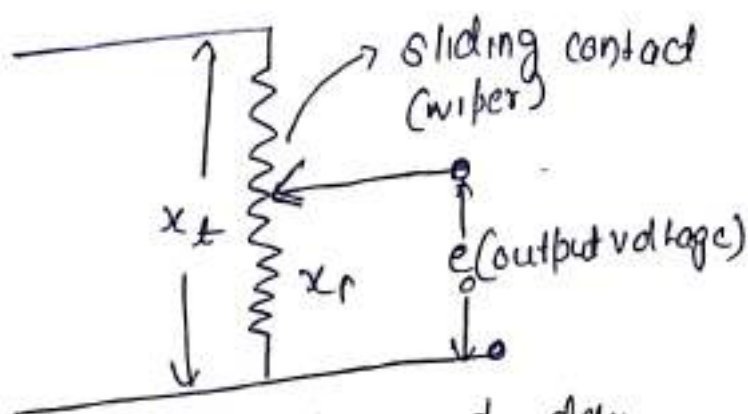


#

Q_{nt} - 2Resistive transducer - Potentiometer

→ a resistive passive transducer that measures displacement (linear, angular)

(a) Translation



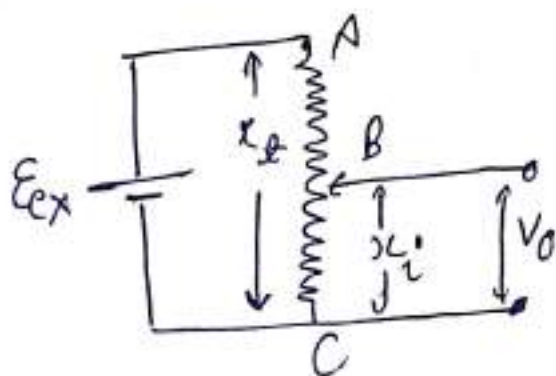
⊙ Use as a voltage divider to vary voltage or current

⊙ $\rightarrow$ voltage (wale) -ve to wiper

Voltage measurement $\rightarrow$ { wiper and ground }

Deviations for linear potentiometer

(i) Output voltage under unloaded conditions (ideal case) ($R_m \sim \text{high}$)



$$V_o = E_{ex} \frac{R_{BC}}{R_{AB} + R_{BC}} \quad \text{if } x_t \text{ is } x_i C$$

⊙ Assume $\frac{R_{AC}}{x_t} = C$

if x_t is $x_i C$

$$V_o = \frac{E_{ex} x_i \ell}{x_t \ell} = \frac{E_{ex} x_i}{x_t}$$

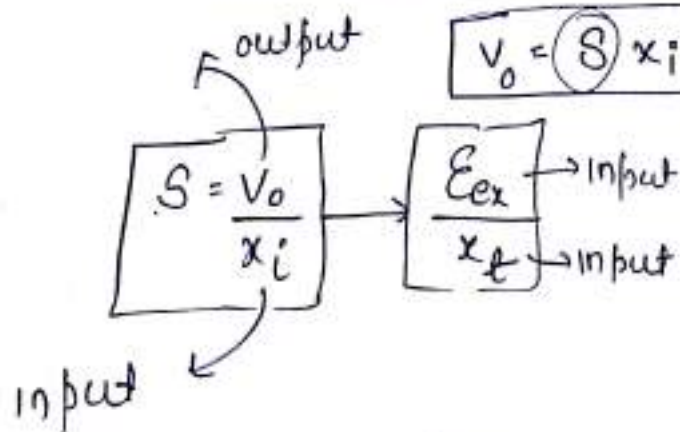
unloaded condition and $\frac{R_p}{x_t} = C$

C = resistance per unit length

$$E_{ex} \times V_o = K E_{ex}$$

$$K = \frac{x_i}{x_t}$$

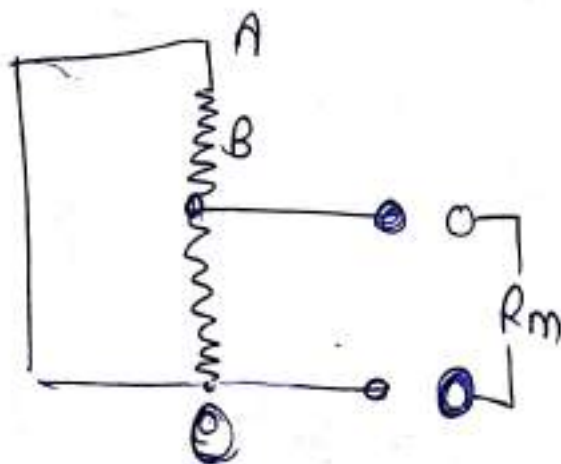
$$V_o = (S) x_i$$



(ii) Loaded condition

$$V_L = \frac{V_o}{1 + \frac{R_s}{R_m}}$$

$R_s \Rightarrow$ thevenin resistance



$$R_s = R_{AB} \parallel R_{BC}$$

$$= \frac{(1-K)R_p \times R_p K}{(1-K)R_p + R_p K}$$

$$R_s = (1-K) K R_p$$

$$K = \frac{x_i}{x_t}$$

$$V_L = \frac{K E_{ex}}{1 + \frac{(1-K)K R_p}{R_m}}$$

$$\boxed{\text{Max \% error } E_{\max} = \frac{15 R_p}{R_m}}$$

(error) $V_L = E_L = V_L - V_0$

$$= \left(\frac{V_0}{1 + \frac{(1-K)K R_p}{R_m}} - V_0 \right)$$

$$= V_0 \left(\frac{R_m}{R_m + K(1-K)R_p} - 1 \right)$$

$$\boxed{E_{\text{error}} = V_0 \frac{K(1-K)R_p}{R_m + K(1-K)R_p}}$$

$$\% E_L = \frac{K(1-K) \times 100}{K(1-K) + R_m/R_p}$$

Displacement Transducer

→ A displacement transducer is an electromechanical device used to convert motion of an object into electrical signal.

1) Mechanical Input

2) Conversion

3) Electrical Output

4) Data Interpretation

* Type of Displacement Transducers

↳ Linear → measure linear motion

↳ Rotary → measure rotational motion

Examples

-) Inductive Displacement Transducer
-) Potentiometer Transducer
-) Piezoelectric
-) Capacitive

* Potentiometer →

important parameters of potentiometer

• → like Resistance, Power.

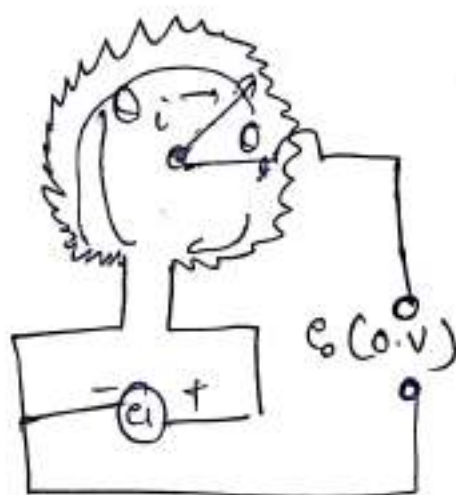
-) Resistive Element
-) Sliding Contact
-) Displacement $\uparrow$ / $\downarrow$
- Voltage $\uparrow$ / $\downarrow$

2) Angular potentiometer →

i) Ideal Case →

$$V_o = \frac{E_x \theta_i}{\theta_t}$$

$$\boxed{S = \frac{E_x}{\theta_t}}$$



3) Helical (accuracy $\uparrow$, resolution $\uparrow$)

Based on construction and material

- i) Wire wound - wire wound on a former
- ii) Non wire wound - made of a thin film of different material that forms a resistive element with a variable contact

Resolution = $\frac{\text{length}}{\text{number of turns}}$

① Classification based on material

i) Wire wound POT

- Resolution is 0.05mm
- Good linearity
- Relatively inexpensive
- Displacement range (a few cm)
- Can withstand higher currents and temperatures

② Single turn POT

- rotary position sensor
- allow to rotate shaft just once 370 degrees
- Where it works?

- Volume or brightness controls
- Motor speed setting
- Simple control knobs.

→ Non-wire wound P.O.

i) Cermet Potentiometer →

- Ceramic + precious metal
- Precision high, exc. durability, wide temp. range
- Suitable for industrial and precision applications
- High cost.

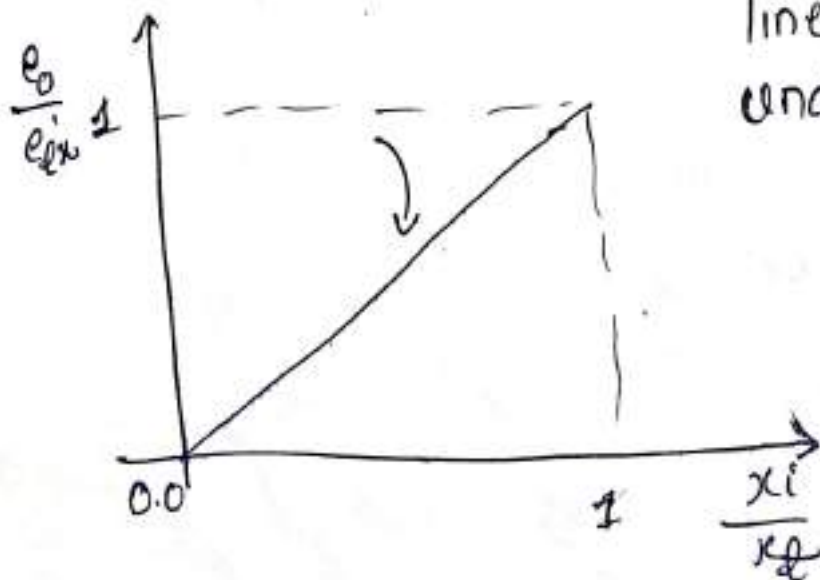
ii) Carbon film →

- Low precision applications
- Offers infinite resolution
- Uses moulded carbon resistive element

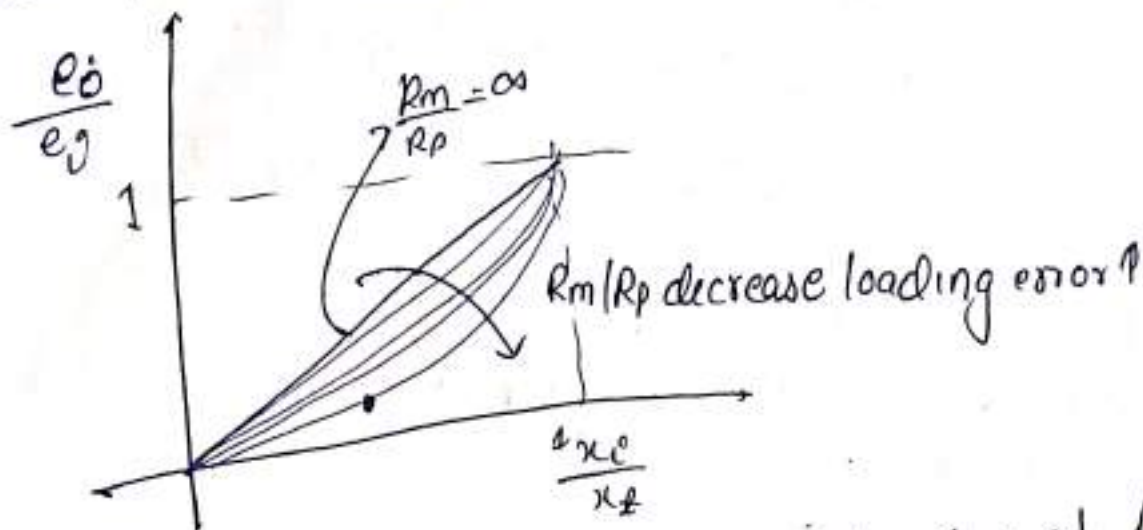
iii) Metal film →

- Thin film of metal
- Provides high precision, stability
- High accuracy
- Used on a.c.

O/P - I/P characteristics of linear Potentiometer under ideal case



loaded potentiometer



① Factors affecting important characteristics of Potentiometer

i) Power rating, input excitation and R_p

$$(E_o)_{max} = \sqrt{P R_p}$$

→ In order to keep the power dissipation at a low level, input voltage should be small and the resistance of the potentiometer should be high.

ii) Resolution

→ Smallest

→ discrete

full length of

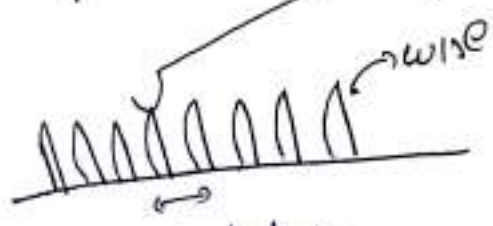
increment

steps

of wiper

wire

the resistive element.



$$\text{resolution} = \frac{\text{total resistance}}{\text{total length of resistive element}}$$

$$\text{Resolution} = \frac{\text{Total displacement range}}{\text{Number of terms.}}$$

① Factors affecting Resolution
 # Wire Diameter (wire resistance) and turns (construction)
 More turns $\rightarrow$ Small Diameter $\rightarrow$ high resolution.

Wiper design

A well designed wiper with multiple contact points ensure better and more consistent resolving by maintaining contact.

Manufacturing Focus $\rightarrow$

iii) Linear

O/P-I/P relationship is ideally linear

$\rightarrow \frac{R_m}{R_p} \uparrow \rightarrow$ To reduce R_p , reduces resolution
 Also as R_p decreases, power diss increases,
 $\rightarrow$ Hence Resolution and linearity are conflicting parameters

iv) Sensitivity

$$E_o = \frac{E_{ex} x_i}{x_d}$$

$$\boxed{\frac{E_o}{x_i} = S = \frac{E_{ex}}{x_d}}$$

② Factors affecting

$\rightarrow E_{ex} \uparrow \rightarrow S \uparrow$ as long as heating effect doesn't

take place

→ To keep power dissipation $\downarrow \rightarrow E_{ex} \downarrow \rightarrow R_p \uparrow$

→ high sensivity $\rightarrow E_{ex} \uparrow \rightarrow$ so it demands for high $R_p \uparrow$
but it will increase non linearity

● Features

- ① inexpensive
- ② simple to operate
- ③ large amplitude of displacement
- ④ Electrical efficiency is very high.
- ⑤ Resolution - mm to micrometer
- ⑥ Almost linear o/p

⑥ The disadvantages are

- i) large force to move wiper.
- ii) Sliding contact can wear out
- iii) Prone to loading error.

Q] Consider potentiometer

$L = 5\text{cm}$ $R_t = 1000 \Omega$ $R.P = 0.5\text{W}$ P_o of turn 200

a) displacement resolution

$$= \frac{5}{200} = 0.025\text{cm}$$

b) permissible $E_{ex} = \sqrt{R \times P}$

$$= \sqrt{500} = 22.36\text{V}$$

c) Voltage gradient = $\frac{22.36}{5} = 4.472\text{V/cm}$

$$d) V_o(5cm) = \frac{22.36}{50} = 2.236V$$

$$K = \frac{5}{50} = 0.1$$

$$e) (R_{eq} \text{ at } 5mm) = \frac{1000 \times 0.5}{8} = 100\Omega \rightarrow K(1-K)R_p = \frac{0.1 \times 9000}{1 - 0.1} = 90\Omega$$

$$f) V_L \text{ at } (100K\Omega R_m) = \frac{0.1 \times 22.36}{1 + \frac{0.9 \times 90 \times 100 \times 10^3}{1000 \times 10^3}}$$

$$= \frac{2.236}{1 + \frac{9000}{1000}} = 2.236V$$

$$= \frac{2.236}{1 + \frac{9}{1000 \times 10^3}} = \frac{2.236}{1 + 9 \times 10^{-6}}$$

g) Percentage

non linearity error at 5mm

$$= \% \left(\frac{2.233 - 2.236}{2.236} \right) \times 100$$

$$= 0.0013\%$$

$$= \frac{2.236}{1 + 9 \times 10^{-6}} = 2.233V$$

(h) What should be the input impedance of the meter so that max loading error is 1 percent

$$0.01 = 15\% \quad 1 = 15 \times \frac{R_p}{R_m}$$

$$R_m = 15K\Omega$$

Q] A potentiometer has a resistance of 5000Ω and is rated at $3W$. What is the maximum allowable voltage. Calculate the value of sensitivity and resolution if the $x_0 = 0.1m$ and $N = 200$.

• Calculate the loading error at 0.67 of the travel in
 $R_m = 5000 \Omega$

$$\rightarrow E_x = \sqrt{R \times P} \\ = \sqrt{15000} = 122.47 \text{ V}$$

$$\rightarrow S = \frac{E_x}{x_t} = \frac{122.47}{0.1} = 1224.7 \text{ V/mm}$$

$$\rightarrow R = \frac{0.1}{200} = 0.5 \text{ mm}$$

$$\rightarrow \% E_L = \frac{k^2(1-k) \times 100}{k(1-k) + \frac{R_m}{R_p}} = \frac{(0.67)^2(1-0.67) \times 100}{0.67(1-0.67) + 1} \\ = 12\%$$

Q] A linear potentiometer is 50 mm long, $R_p = 10000 \Omega$,
 slider is at the centre. Find linear displacement
 of the potentiometer if R (w.s.B)
 i) 3850Ω and ii) 7560Ω

Are two displacement in same direction?

ii) It is possible to measure a minimum value of
 10Ω resistance with the above arrange find
 R_L of the potentiometer in mm.

$$\rightarrow \text{Resistance at normal} = (10000/2) = 5000 \Omega$$

$$\text{i) Resistance per unit length} = \frac{10000}{50} = 200 \Omega/\text{mm}$$

$$\text{ii) Change in } R \text{ from normal position} = 5000 - 3850 = 1150 \Omega$$

$$\therefore \text{Displacement} = \frac{1150}{200} = 5.75 \text{ mm}$$

$$\text{iii) Change in } R = 7560 - 5000 = 2560 \Omega$$

$$\therefore D = \frac{2560}{200} = 12.8 \text{ mm}$$

→ Since on displacement → increasing R → and on x decreasing R both are in opposite direction

$$\text{Resolution} = \frac{10}{200} = 0.05 \text{ mm}$$

→ minimum measurable value of resistance $\frac{\text{mm}}{\text{cm}}$

$$10 \times \frac{1}{200} = 0.05 \text{ mm.}$$

Q] A variable potential divider has a total resistance of $2 \text{ k}\Omega$ and is fed from a 10 V supply. The output is connected across a load resistance of $5 \text{ k}\Omega$. Determine the loading error for $k = 0, 0.25, 0.5, 0.75$ and 1.0 . Use result to plot graph % loading error at x_0/x_t

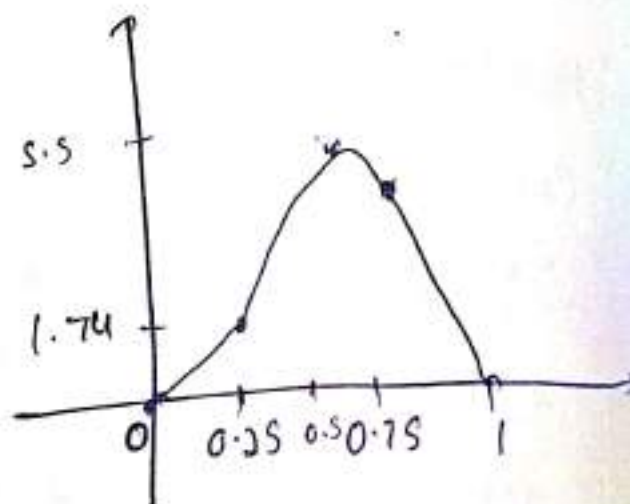
i) $\% (0) = 0$

ii) $\% (0.25) = \frac{(0.25)^2 \times 0.75 \times 100}{0.25 \times 0.75 + 2.5} = 1.74\%$

iii) $\% (0.75) = \frac{(0.75)^2 \times 0.25 \times 100}{0.75 \times 0.25 + 2.5} = 5.23\%$

iv) $\% (0.5) = 5.5\%$

v) $\% (1) = 0$



LVDT

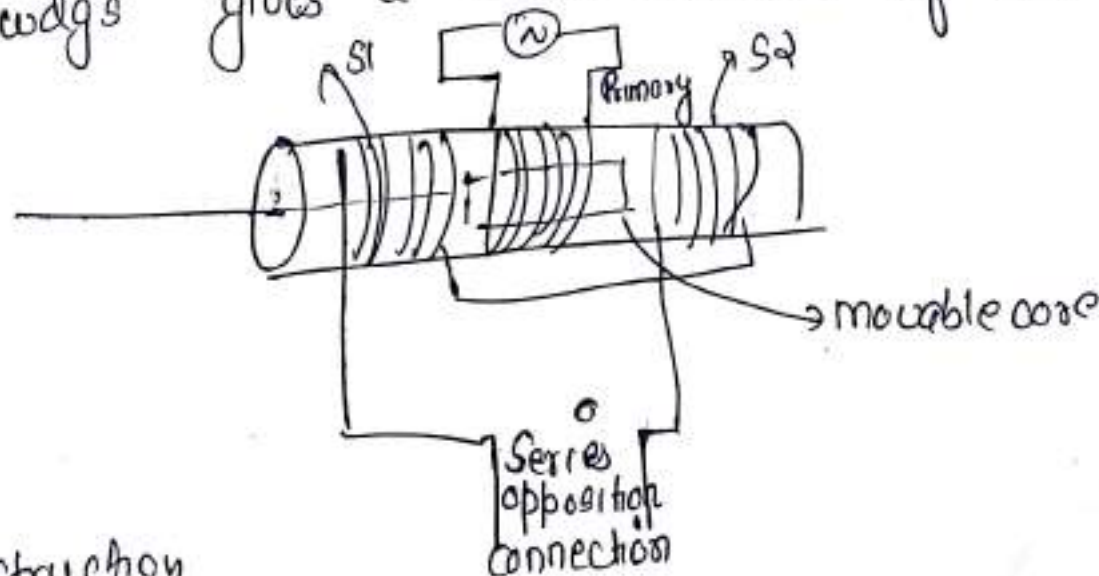
(Linear Variable Differential Transformer)

→ a passive inductive transducer used for linear displacement

↳ Works on principle of transformer action based upon electromagnetic induction

→ An a.c. through PC created magnetic field that links with two S.G

- The rate of change of M.F by displacement of the core between primary and secondary changes the induced voltage in two secondary wdg's.
- The differential output voltage across the secondary wdg's gives a linear measures of the displacement



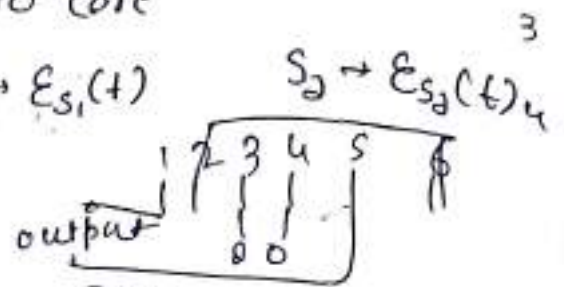
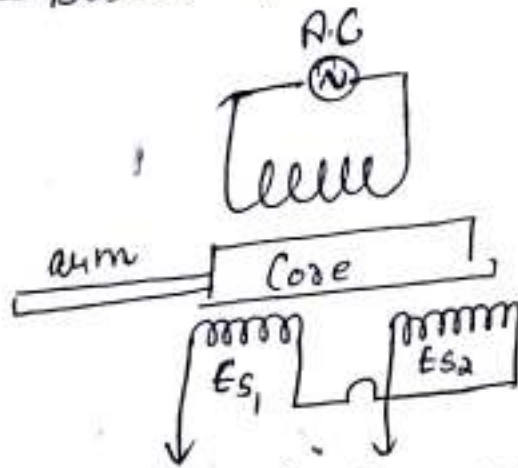
* Construction

→ Consist of single primary winding P₁ and two secondary wdg's S₁ and S₂ on a former

→ the cylinder of which wound is non-magnetic

→ S₁ = S₂ equal number of turns.

- P.S → alternator
- A movable soft iron core is placed inside former
- The linear displacement to be measured to be measured is applied to core
- Based upon E.H.I



Differential output
 $= E_{S1} - E_{S2}$

(b) Basic circuit

$$V_o(t) = E_{S1}(t) - E_{S2}(t) = V_o + Kx$$

disp (from null point)

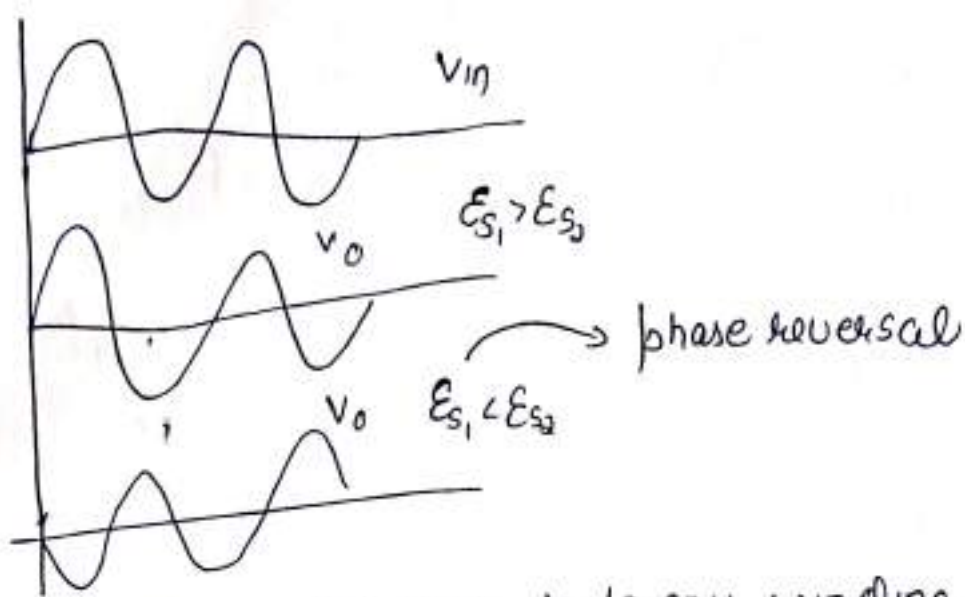
sensitivity (V/mm)

residual voltage

$V_o = E_{S1} - E_{S2}$ → amplitude and phase will vary

if core is near $S_1 \rightarrow E_{S1} > E_{S2}$ and if core is near $S_2 \rightarrow E_{S1} < E_{S2}$

phase reversal



but the closer the core is to any winding the higher will be the magnitude of V_o .

* Effect of displacement of core on output voltage of LVDT.

i) Null Position →

- When the core is at centre, $E_{s1} = E_{s2}$.

$$\therefore V_o = 0$$

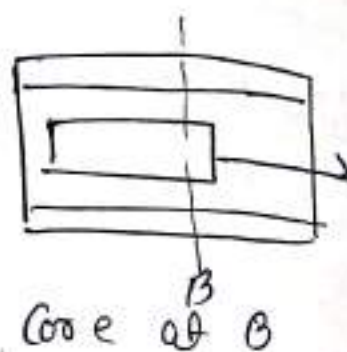
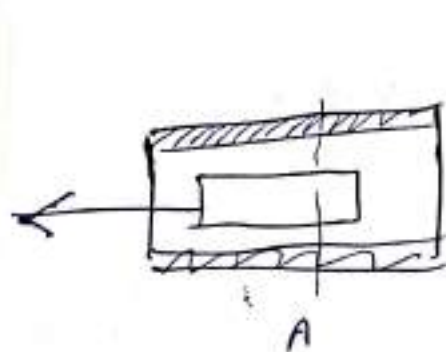
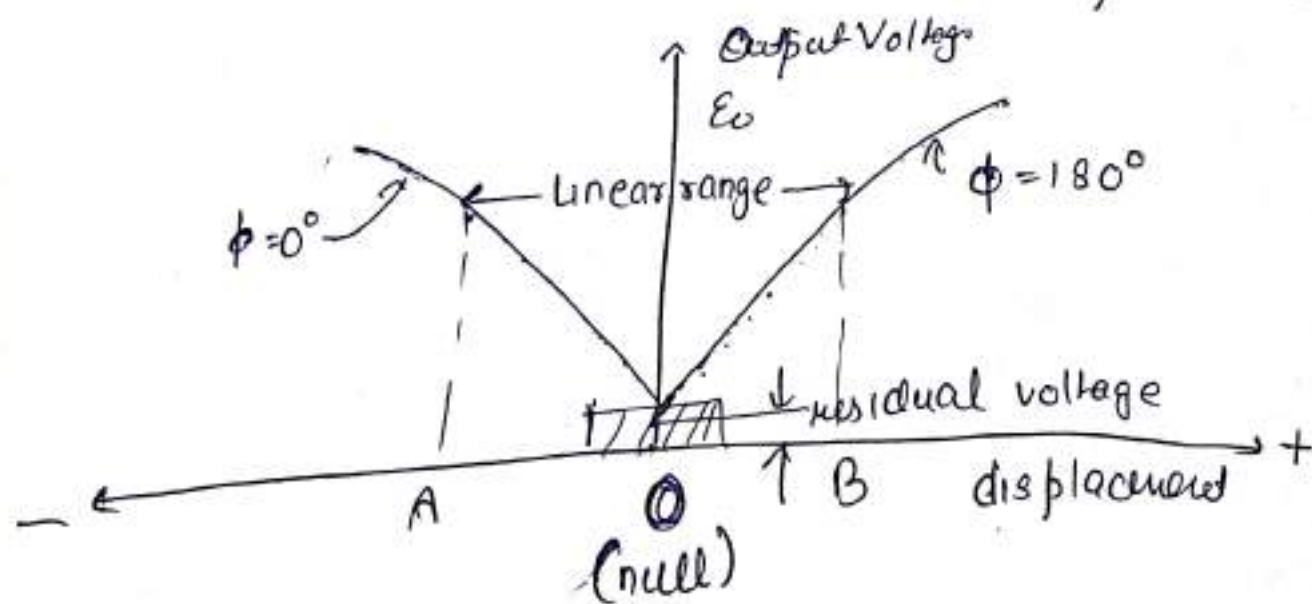
ii) Displaced Position →

→ When input displacement is given to core, null position core change

→ The core moves away, the magnetic flux with one secondary coil increases, while the other decreases

→ This creates unequal induced voltages, Magnitude increase as displacement of the core on either side increases, and phase reverses on a change in direction of movement

Core displacement	$\epsilon_{s1}, \epsilon_{s2}$		Remarks
Null position	$\epsilon_{s1} = \epsilon_{s2}$	$V_o = 0$	Ref position of LVDT
Displaced position from null (right direction)	$\epsilon_{s2} > \epsilon_{s1}$	$V_o = -ve$	mgf with phase is 0°
left direction	$\epsilon_{s1} > \epsilon_{s2}$	$V_o = +ve$	mg ↑ phase is 180°



① Numerical Problem

Q]. The output voltage of LVDT is 1.5V at maximum displacement. At a load of $0.5 \text{ M}\Omega$, the deviation from linearity is maximum and it is $\pm 0.003 \text{ V}$ find non linearity.

$$\% \text{ of linearity} = \frac{\pm 0.003}{1.5} \times 100 = \pm 0.2\%$$

Q] The output of an LVDT is connected to a 5V voltmeter through an amplifier whose $\alpha = 250$. An output of 2mV appears across the terminals moved through a distance of 0.5mm. Calculate the sensitivity of LVDT and that of setup. The milli-voltmeter has 100 divisions. The scale can be read to 1/5 of a division. Calculate the resolution of instrument in mm.

$$\rightarrow S(\text{LVDT}) = \frac{2 \times 10^{-3}}{0.5} = 4 \text{ mV/mm}$$

$$S(\text{instrument}) = \frac{4 \times 10^{-3} \times 250}{0.5 \times 10^{-3}} = 1000 \text{ mV/mm}$$

$$1 \text{ scale division} = \frac{5}{100} = 50 \text{ mV}$$

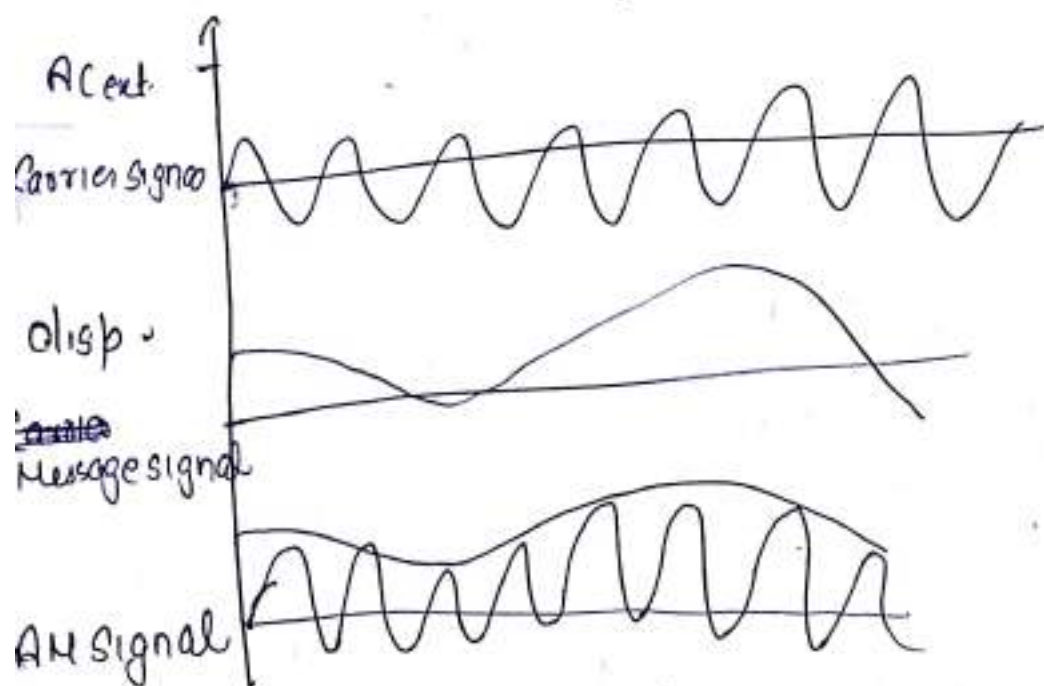
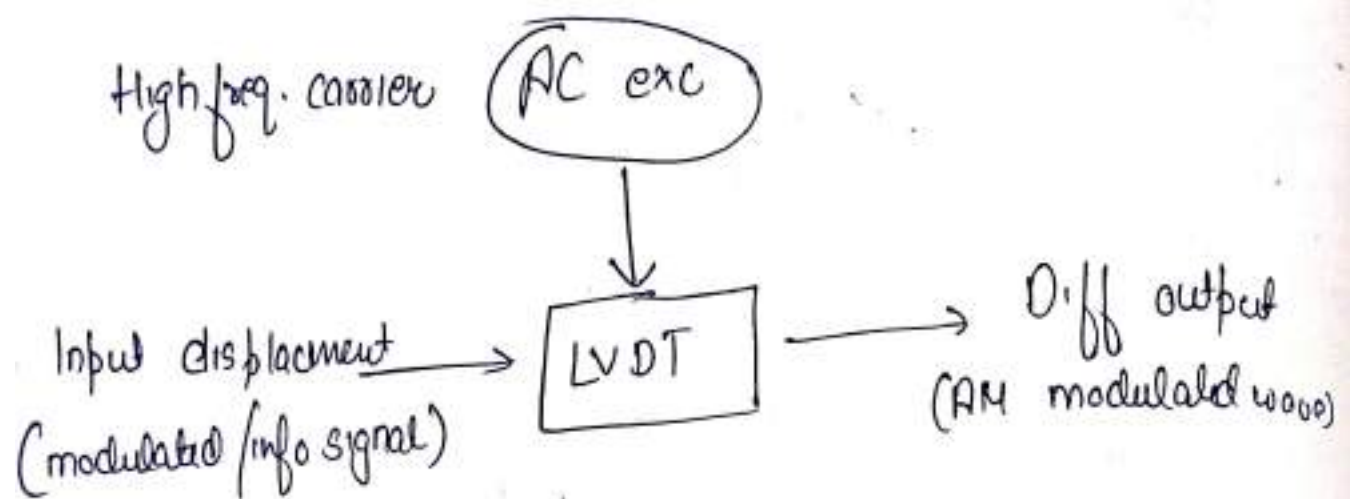
$$\text{Minimum voltage that can be read} = \frac{50}{5} = 10 \text{ mV}$$

$$\therefore \text{Resolution of instrument} = \frac{1 \times 1}{1000} = 1 \times 10^{-3} \text{ mm}$$

Q] An LVDT has an output of 6V r.m.s when the displacement is $0.4 \times 10^{-3} \text{ mm}$. Determine sensitivity of instrument in V/mm. A 10V voltmeter with 100 S.P is used to read the output. Two-tenths of a division can be estimated with ease. Determine the resolution

of the voltmeter. And the resolution with which displacement can be measured.

- The raw differential output of LVDT is an AC-modulated waveform
- An AM modulated wave in on LVDT is the output voltage which is being modulated by the displacement (info. signal) but frequency will be prop. to the carrier signal



A.M $\rightarrow$ Amplitude modulated

$\rightarrow$ The magnitude of output AC represents the amount of the core displacement, and its phase angle indicate the direction of movement. Changes in the magnitude of the output voltage depends how fast displacement magnitude changes

$$\rightarrow V_o = Kx(t) \sin(\omega_n t + \phi)$$

$\rightarrow x(t)$ is time varying displacement

$\rightarrow K$ is sensitivity

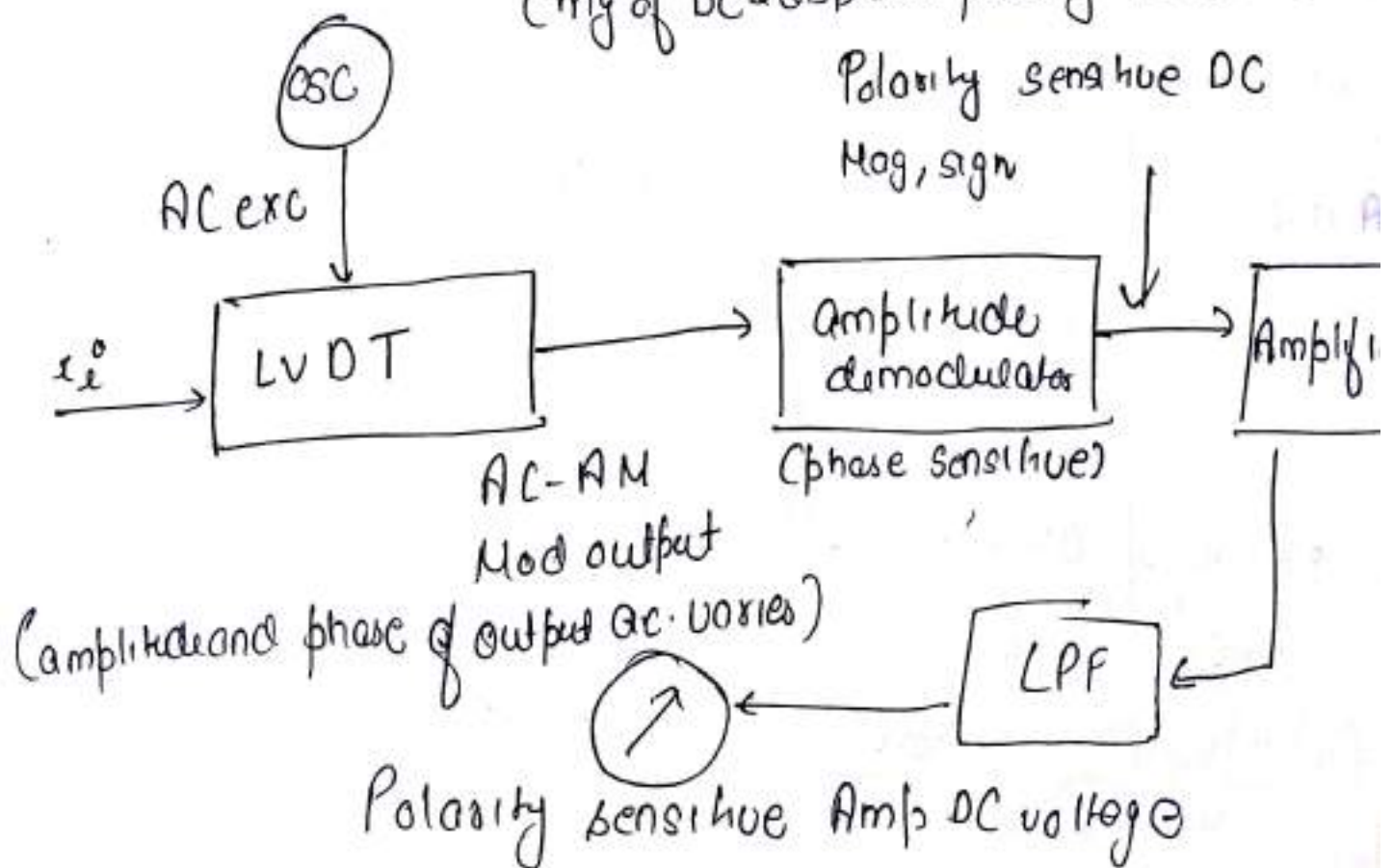
$\rightarrow F$ is freq of ex signal (carrier)

⑥ Signal Conditioning of LVDT

Comp of DC & disband polarity indicate direct

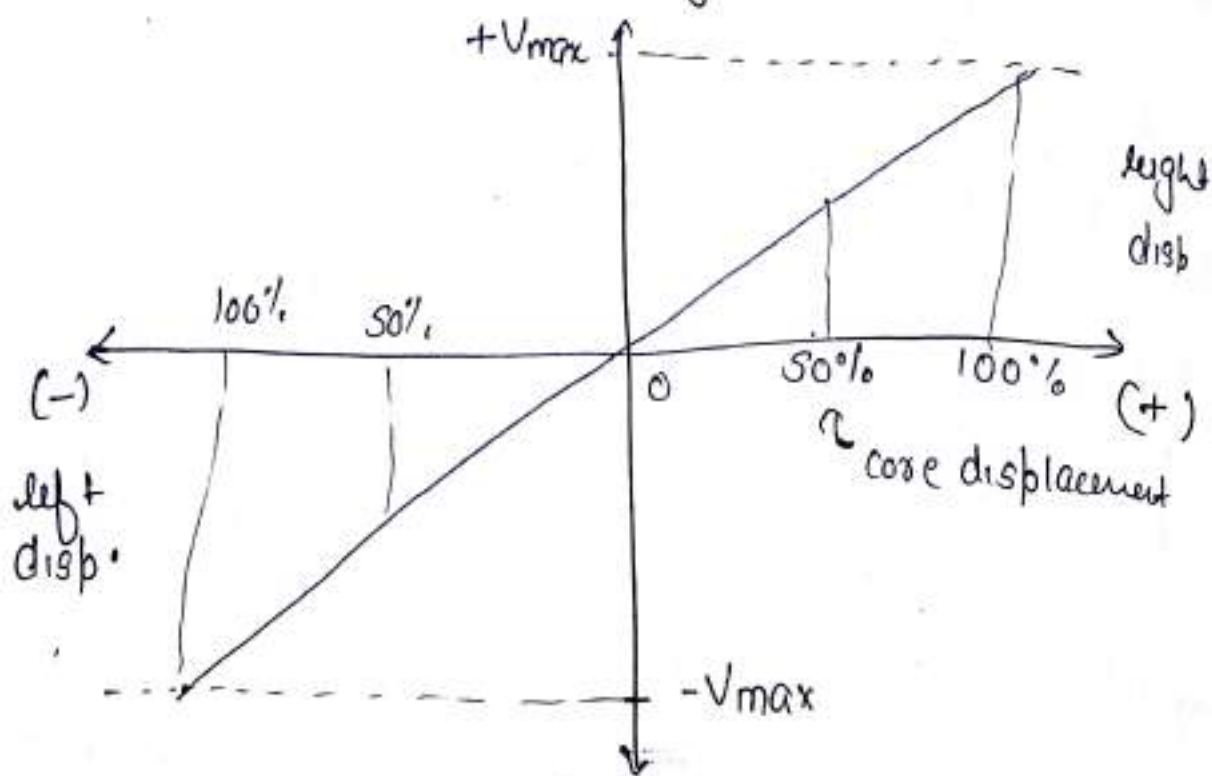
Polarity sensitive DC

Mag, sign



OSC frequency should be at least 10 times more than the displacement frequency.

② DC output voltage of LVDT, mV is linearly prop to the distance of either side of null, indicates the direction of displacement. polarity



Features of LVDT

(i) High sensitivity - sensitivity is a measure of its responsiveness to linear disp. (V/mm)
 $\sim 0.5 \text{ V/mm}$

• Ratio of output voltage to input displacement per unit excitation

(ii) infinite resolution,

of the first of January

at the first of January
the first of January
the first of January
the first of January
the first of January

12) Frontend theory

(Volatility curve is left skewed)

- a) Suggests : run mostly towards high degree of stock intervention
- b) Low liquidity

iii) Excellent linearity

(Nonlinearity error is less $\pm 0.05\%$)

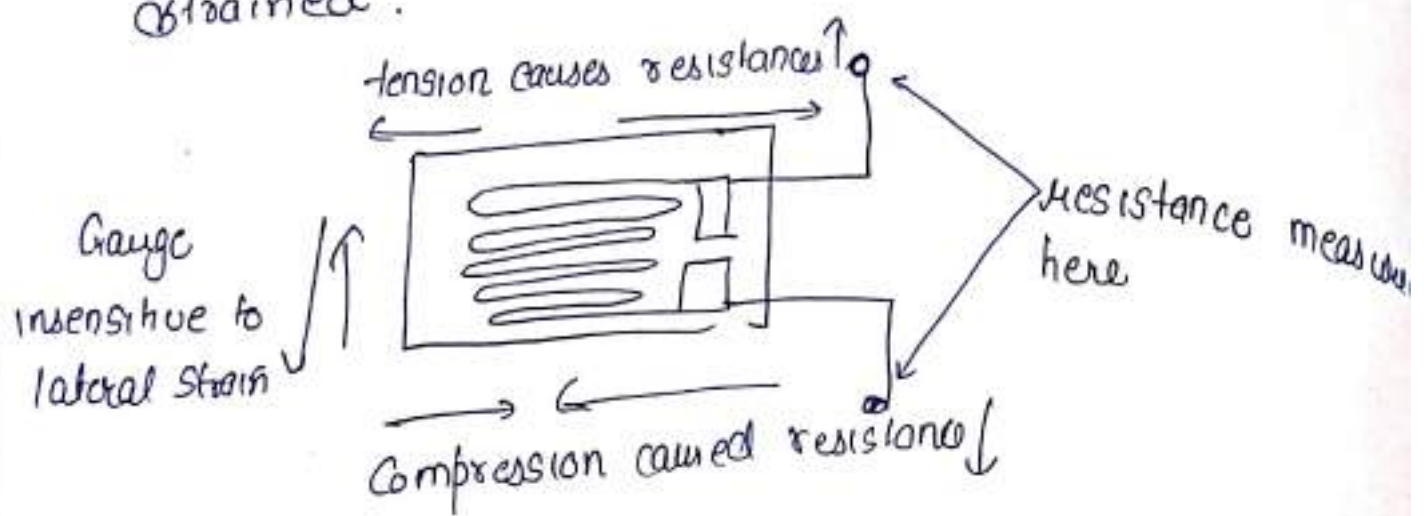
iv) Ruggedness - can usually tolerate high degree of shock and vibration

v) Low hysteresis

Strain gauge

→ A strain gauge is a resistive transducer that converts mechanical strain into a measurable signal

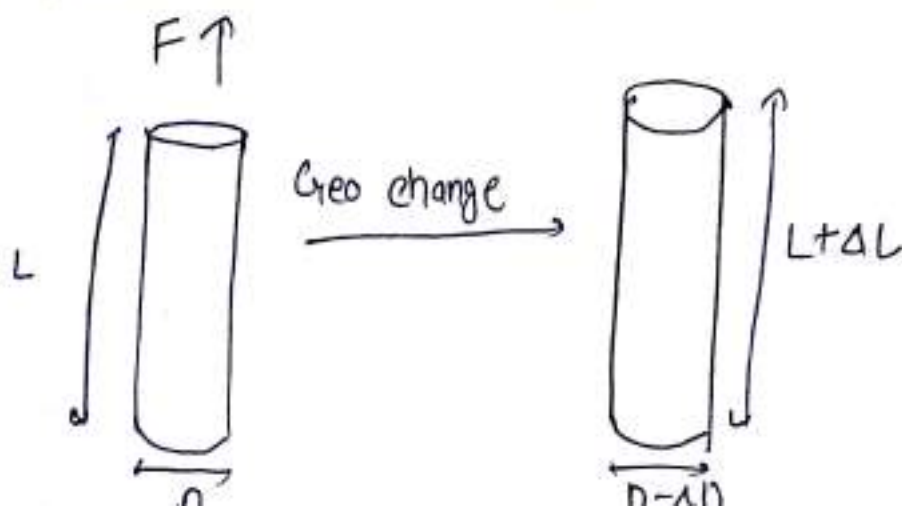
→ consists of a thin, zigzag-patterned metallic foil. As the object deforms under an external force, the SG stretches or compresses or is strained.



→ Strain of SG

• Longitudinal strain $\epsilon_L = +\Delta L/L$

• Transverse strain (lateral) $\epsilon_T = -\Delta d/d$



Tensile force $\rightarrow +ve$ strain $\rightarrow R \uparrow$ $\Delta R = P \frac{L}{A \Delta}$

$\hookrightarrow L \uparrow$ $D \downarrow$

Compressive strain $\rightarrow -ve$ strain $\rightarrow R \downarrow$

$\hookrightarrow L \downarrow$ $D \uparrow$

$$\Delta R = P \frac{L \Delta}{A \uparrow}$$

① Relationship between transverse and longitudinal strain

$$\text{Poisson's ratio } (\mu) = -\frac{\epsilon_T}{\epsilon_L}$$

Poisson effect $\rightarrow$ ① When material pulled
② it elongates longitudinally and contracts laterally

Strain unit $\rightarrow$ unitless but sometimes like 1 microstrain
 $\rightarrow 10^{-6} \text{ m/m}$ strain $1 \mu\text{m/m}$

$\rightarrow$ all the forces should be under elastic limit

$$\sigma = \frac{\text{stress}}{\text{strain}} \quad \left\{ \begin{array}{l} \text{Hook's law} \\ \downarrow \\ \text{Linear stress-strain curve} \end{array} \right.$$

$\rightarrow$ the change in dimension is reversible

① Fraction change in resistance $\propto$ strain

$$\frac{\Delta R}{R} \propto \epsilon \quad \rightarrow \text{Gauge factor}$$

$$\frac{\Delta R}{R} = (G.F.) \epsilon$$

$$G.F. = \frac{\Delta R}{R \cdot \epsilon}$$

$\downarrow$ directly tells the sensitivity
long strain

a.c.f. defines the sensitivity of S.G. The higher the gauge factor of S.G., more is the change in resistance

Change in R →

$$R = \rho \frac{l}{A}$$

metal → l and A change only

Semi-conduction → l, A change and also ρ

piezo-resistive effect

changes in mobility of electron and holes

Material	G.F	T.C.R	Stability
Metallic alloys:	1-4	Low	⊗ Excellent suited for 10^{-6} strain
Nichrome	50-100	Very high	fragile
Piezoresistor			limited to 10,000 microstrain
Foil type/Thick-film	2-40	Moderate	very good

⊗ 3 types

i) Wire wound S.G

ii) foil type

iii) Semiconductor (piezoresistive)

② Derive C.F in terms of material parameters

$$R = p \left(\frac{l}{A} \right) \dots (1)$$

unstrained R

when s is applied on R,

$$\frac{dR}{ds} = \frac{\partial p}{\partial s} \left(\frac{l}{A} \right) + \frac{\partial l}{\partial s} \left(\frac{p}{A} \right) - \frac{pl}{A^2} \frac{\partial A}{\partial s} \dots (2)$$

dividing by R

$$\frac{dR}{R ds} = \frac{\partial p}{\partial s} \left(\frac{l}{pA} \right) + \frac{\partial l}{\partial s} \left(\frac{p}{pA} \right) - \frac{pl}{pA^2} \frac{\partial A}{\partial s} \dots (3)$$

$$\frac{1}{R} \left(\frac{dR}{ds} \right) = \frac{\partial p}{\partial s} \left(\frac{1}{p} \right) + \frac{\partial l}{\partial s} \left(\frac{1}{A} \right) - \left(\frac{1}{A} \right) \frac{\partial A}{\partial s} \dots (4)$$

Considering cross-section area of shape 'circular' or 'rectangular' term $\left(\frac{\partial A}{\partial s} \right)$ is evaluated.

$$\text{Consider } A = \left(\frac{\pi}{4} \right) D^2$$

$$= \left(\frac{1}{A} \right) \frac{\partial A}{\partial s}$$

$$= \left(\frac{1}{A} \right) \frac{\pi}{4} \frac{\partial D^2}{\partial s}$$

$$= \left(\frac{2}{D} \right) \frac{\partial D}{\partial s} \dots (5)$$

Substitute (5) in (4)

$$\left[\frac{1}{R} \right] \frac{dR}{ds} = \frac{1}{p} \frac{\partial p}{\partial s} + \frac{1}{l} \frac{\partial l}{\partial s} - \frac{2}{D} \frac{\partial D}{\partial s} \dots (6)$$

$$\frac{\Delta R}{R} = \frac{\Delta p}{p} + \frac{\Delta l}{l} - 2\left(\frac{\Delta d}{d}\right) \quad (7)$$

$$\mu = -\frac{\mu_T}{\mu_L} \Rightarrow \mu \epsilon_L = -\epsilon_T$$

μ_L $\downarrow$ μ_T
 long $\uparrow$ transverse

$$\frac{\Delta d}{d} = -\mu \epsilon_L \quad (8)$$

$$\frac{\Delta R}{R} = \frac{\Delta p}{p} + \frac{\Delta l}{l} - 2(-\mu \epsilon_L)$$

$$\frac{\Delta R}{\epsilon_L R} = \frac{\Delta p}{\epsilon_L p} + \frac{\Delta l}{\epsilon_L l} (1 + 2\mu)$$

$$G.F = \left(\frac{\Delta p}{\epsilon_L p} \right) + 1 + \frac{2\mu}{\epsilon_L}$$

piezoresistivity
 metal alloys

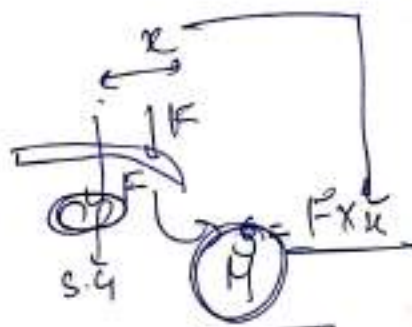
for metal wires
 $\left(\frac{\Delta p}{\epsilon_L p} \right)$ can be neglected

Contoller of beam
 Moment of cantilever = $\frac{bt^3}{12}$

$$\text{Deflection } x = \frac{Fl^3}{3EI}$$

$$S = \frac{x}{p}$$

deflection



$$S = \frac{M \cdot l}{I \cdot x^2}$$

QPF =

Measurement of resistance of Strain Gauge

i) Strain Gauge bridge

① Important points

i) Exc source $\rightarrow$ AC or DC, acc the output

ii) Types of bridge $\rightarrow$ - balanced or unbalanced

Balanced - when strain is applied, bridge-unb. and the bridge resistor is change to balance the change in r gives a measure of input strain

Unbalanced - when strain is applied, bridge unb. and we will calculate the value of voltage produced.

iii) bridge config.

-- Quarter bridge (1 SG, other 3 simple resistors)
or (with/without dummy SG)

no strain only for temp comp

-- Half bridge (2 SGs, $\pm$ -strain; other two simple)

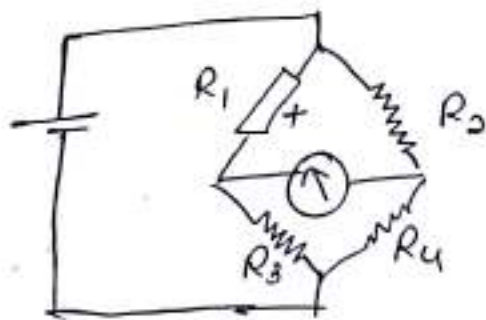
-- Full bridge (4 SGs, $2+$, $2-$ strain).

iv) Temp compⁿ - SG is temp sensitive, so to minimize

cross-sensitivity arrangement is made on the setup to use more active SGs in typical configuration.

S.G on structure elements and corresponding bridge configuration

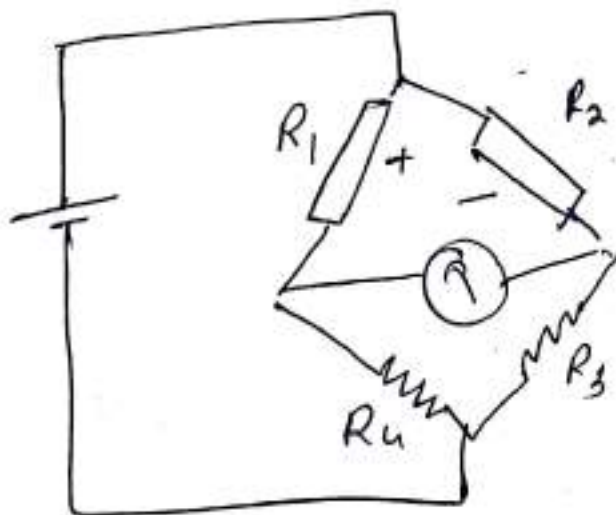
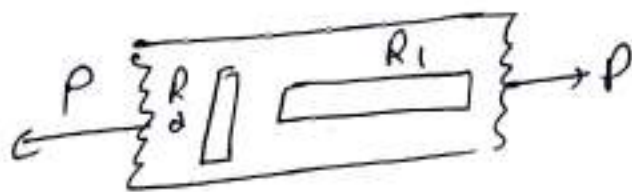
$R = R_1 = R_2 = R_3 = R_4$
unstrained value



- Lowest sensitivity
- Lowest temp. compensation

$$V_o = \frac{E_{ex} G_i F_{strain}}{4}$$

2 SA (perpendicular to each other) bonded to a structure undergoing tensile force (Poisson's arrangement)



- R_1 changes due to axial strain $+ \epsilon$
- R_2 changes due to transverse strain $- \epsilon \nu$

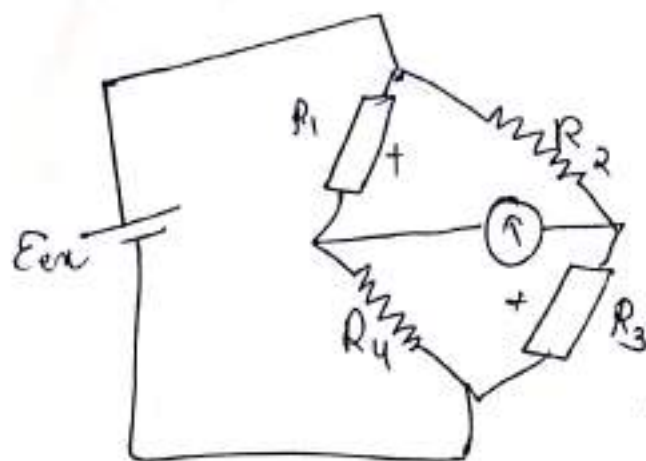
Sensitivity increase $(1 + N)$

$$V_o = \frac{E_{ex} G_i F_o \epsilon (1 + N)}{4}$$

→ The different output of bridge cancels the temp effect.

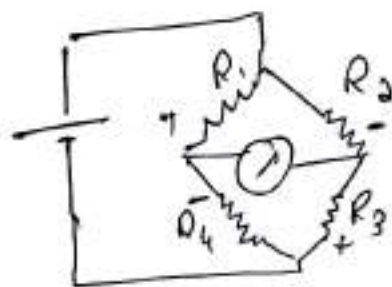
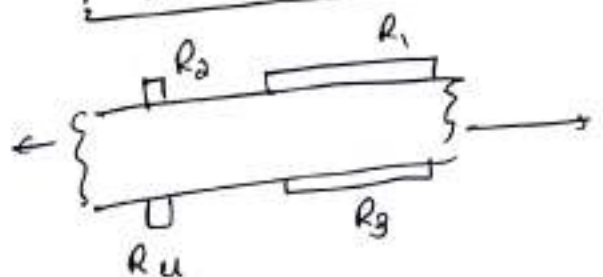
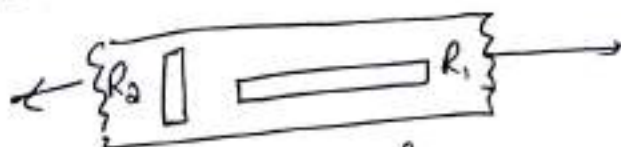
again $R = R_1 = R_2 = R_3 = R_4$

ii)



- 2 at top and bottom of structure
- R_1 and R_3 changes due to tensile strain $+\epsilon$
- Sensitivity inc of 2
- $V_{out} = \frac{E_{ex} G.F. strain}{2}$

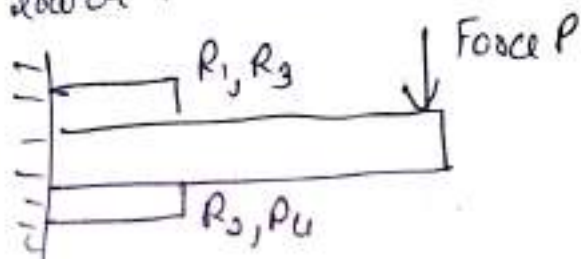
③ 4 S.G (right angle to each other) →



- R_1, R_3 change axial strain $+\epsilon$
- R_2, R_4 changes due to transverse strain $-\epsilon\mu$
- Sensitivity increase by $2(1+\mu)$
- o/p voltage $= \frac{E_{ex} G.F. \cdot \epsilon(1+\mu)}{2}$

④ use (2 at top, 2 at bottom)

upper → expansion
lower → compression

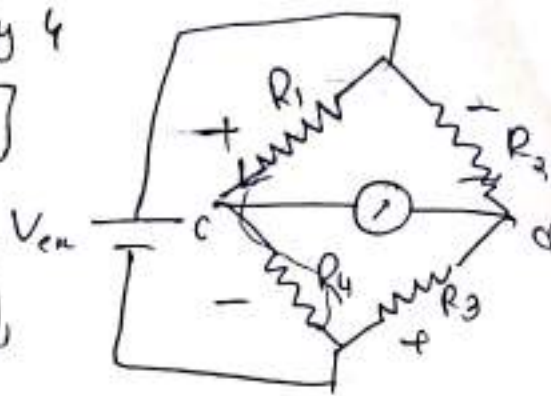


- all 4 are measuring longitudinal strain
- $R_1, R_3 \rightarrow +\epsilon$
- $R_2, R_4 \rightarrow -\epsilon$

- a. • Sensitivity increases by 4

$$V_o = \epsilon_z \cdot G_f \cdot \text{strain}$$

• most sensitive bridge



* Derivations

$$R_1 = R_2 = R_3 = R_4 = R$$

under balance condition $V_o = 0$

When strain is applied $\rightarrow$

$$\left. \begin{array}{l} R_1 = R + \Delta R \\ R_3 = R + \Delta R \\ R_2 = R - \Delta R \\ R_4 = R - \Delta R \end{array} \right\} \begin{array}{l} \text{undergo} \\ \text{longitudinal } \tau \\ \text{undergo} \\ \text{longitudinal } \epsilon \end{array}$$

$$V = V_c - V_d$$

$$V = \frac{V_{ex} R_4}{(R_1 + R_4)} - \frac{V_{ex} R_3}{(R_2 + R_3)}$$

$$V = \frac{V_{ex} (R - \Delta R)}{(R + \Delta R + R - \Delta R)} - \frac{V_{ex} (R + \Delta R)}{(2R)}$$

$$V = V_{ex} \left(\frac{R - \Delta R - R - \Delta R}{2R} \right)$$

$$V = V_{ex} \frac{-\Delta R}{R}$$

$$V = V_{ex} G_f \epsilon$$

Q) a strain gauge bridge of two fixed resistors of value 120Ω , one is active and one is unstrained temp comp. gauge. Q.1 & 2.2 Find bridge output of $3V$. $\epsilon = 600 \text{ microstrain}$

Q.2) A single electric strain gauge of $R = 120 \Omega$ having G.F. of 2 is bonded to steel elastic limit of 400 MN/m^2 and modulus of $E = 200 \text{ GN/m}^2$. Calculate the change in resistance

(a) due to a change in stress equal to $1/10$ of elastic limit and b) due to change of 20°C $\alpha = 20 \times 10^{-6}$

→ change of stress $S = 40 \text{ MN/m}^2$

$$\text{change of strain } \epsilon = \frac{S}{E} = \frac{40 \times 10^6}{200 \times 10^9}$$

c) $\Delta T = 20$ steel $\alpha = 12 \times 10^{-6}$ allow $\alpha = 16 \times 10^{-6}$

$$\epsilon = (12 - 16) \times 10^{-6} \times 20 = -80 \times 10^{-6} = -80 \text{ microstrain}$$

$$\Delta R = G.F. \epsilon R = 2 \times 80 \times 10^{-6} \times 120 = -0.0192$$

, strain due to differential thermal expansion of gauge metal and steel

$$\begin{aligned} \text{Q.3) i) } \Delta R &= R \epsilon \text{ G.F.} \\ &= 10^3 \times 0.001 \times 2 \\ &= 202 \Omega \end{aligned}$$

iii) Temp change due $\Delta T = 20$

$$\Delta R = 10^3 \times 10 \times 10^{-6} \times 20$$

$$\Delta R = 0.2 \Omega$$

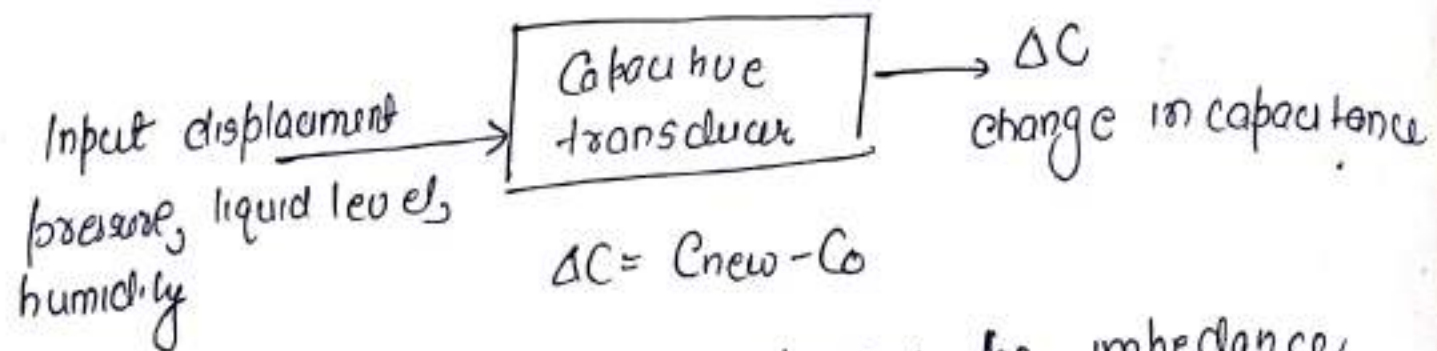
$$\begin{aligned} \text{ii) } V_0 &= \frac{10(10^3 + 2)}{(10^3 + 102)} - \frac{10 \times 100}{200} \\ &= \frac{1064}{1102} - 5 \quad \boxed{= 4.07 \text{ V}} \end{aligned}$$

$$= \frac{10 \times 100}{1000 + 1000} = \frac{1000}{2000} = 0.5$$

$$= -4.09$$

Capacitive transducer

A class of transducer in which the capacitance of the sensor element changes with the measurand due to the changes in distance, dielectric, or area between the capacitive plates.

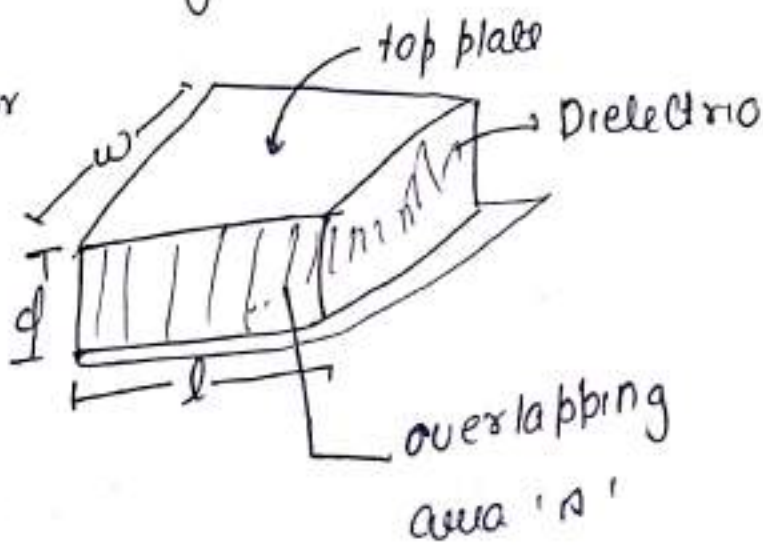


The change in capacitance changes the impedance (very high) of sensor, which is effectively measured by a DC capacitive bridge to provide V_0 . The V_0 is related to the change in the input measurand.

parallel plate capacitor

$$C = \frac{\epsilon_0 \epsilon_r A}{d}$$

$$C = \frac{q}{V}$$



② $A = \text{overlapping area of the plates (L} \times \text{w)}$

$$\epsilon_0 \epsilon_r = \epsilon$$

permittive of free space

$$8.85 \times 10^{-12} \text{ F/m}$$

$$(\epsilon_r \geq 1)$$

Dielectric constant

permittive of material

Classification

Based on design

- ① parallel plate capacitor
- ② Cylindrical & plate capacitor
- ③ Angular (semi-circular) plate capacitor
- ④ Multi plate parallel capacitors
- ⑤ Three plate dual capacitors

Based on method

- of variation
 - ↳ variable dist
 - ↳ variable area
 - ↳ variable dielectric

⑥ Based on ~~the~~ measurement

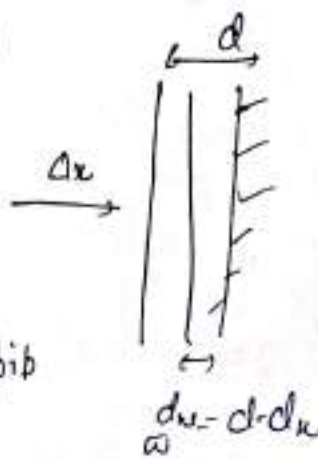
- displacement, pressure, RH, sound pressure, liquid level.

1) Variable d type parallel plate

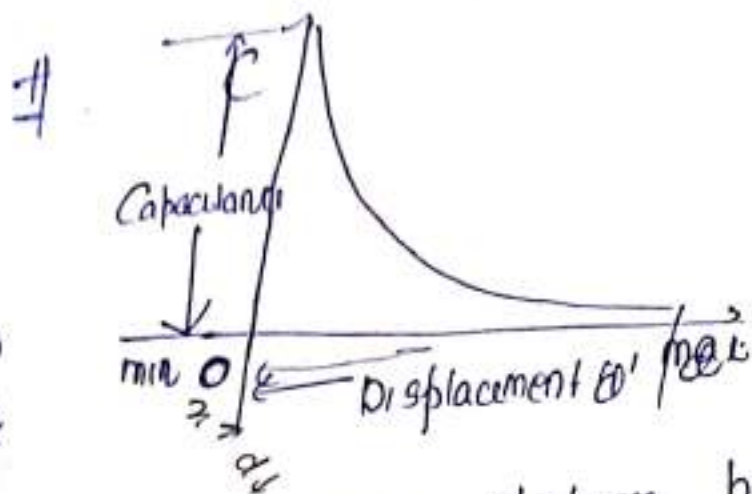
$$C = \frac{\epsilon_0 \epsilon_r A}{d} \text{ (nominal capacitance)}$$

$$C_{\text{new}} = \frac{\epsilon_0 \epsilon_r A}{d - \Delta x}$$

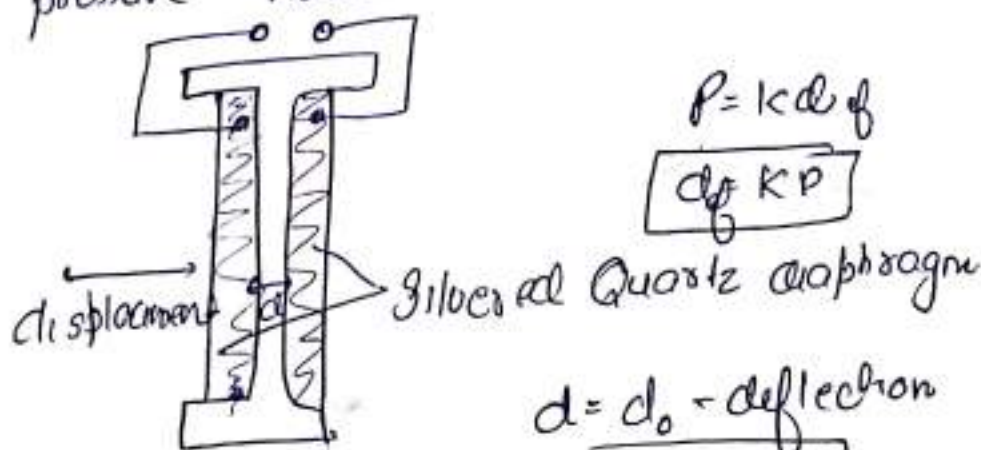
$$S = \frac{\partial C}{\partial d} = - \frac{\epsilon_r A}{d^2} \left\{ \begin{array}{l} \text{Op-tp relationship} \\ \text{is non linear} \end{array} \right.$$



$C \propto \frac{1}{d}$ — non linear (slightly) but inverse
 (distance between plates)



→ Variable distance based C-transducer for pressure measurement



$$P = k d$$

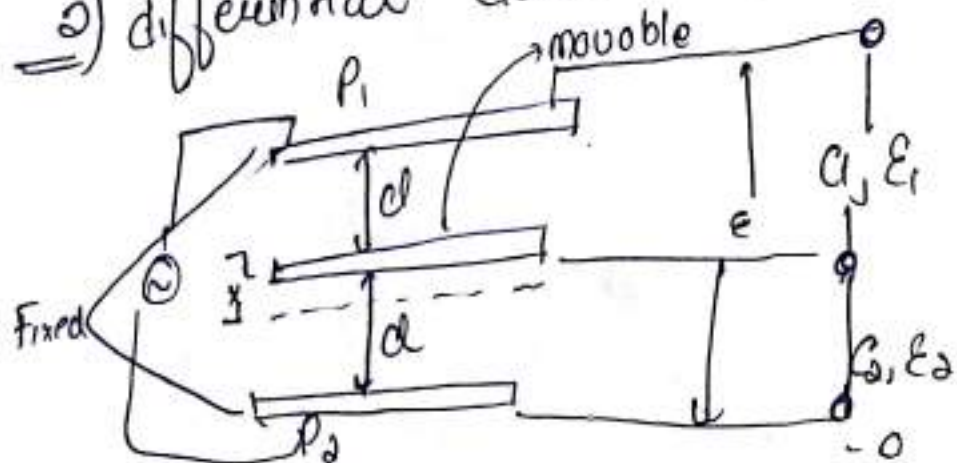
$$d = \frac{P}{K}$$

$$d = d_0 - \text{deflection}$$

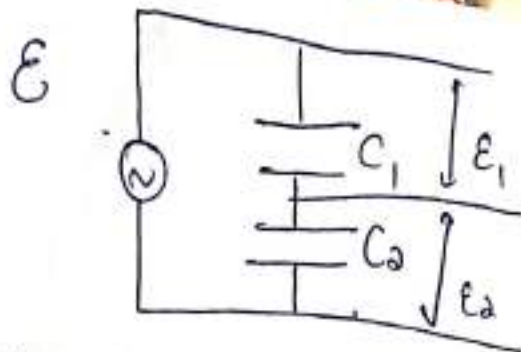
$$d = d_0 - K P$$

$$C = \frac{\epsilon_0 A}{d_0 - K P}$$

2) differential dual capacitive parallel plate CT
 $C_1 \uparrow C_2 \downarrow$



change in input capac.
 of $P_1 \uparrow P_2 \downarrow$ opposite
 effect in sensor
 → Push-pull arrangement



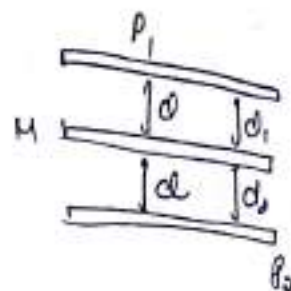
Equivalent voltage divider network of two capacitors C_1 and C_2

① derivation

Initial $C_1 = C_2$ and $E_1 = E_2$

$$C_1 = \frac{\epsilon_0 \epsilon_r A}{d}$$

$$C_2 = \frac{\epsilon_0 \epsilon_r A}{d}$$



$$E_1 = \frac{C_2 E}{C_1 + C_2} \quad \text{and} \quad E_2 = \frac{C_1 E}{C_1 + C_2}$$

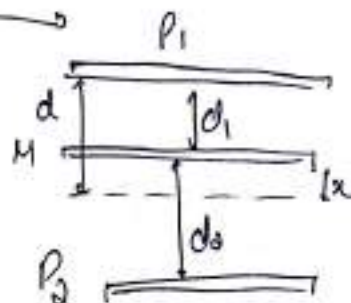
differential output voltage, $E_0 = E_1 - E_2 = 0$

② When displacement (x) applied to M

$$d_1 = d - x \quad \text{and} \quad d_2 = d + x$$

$$C_1 = \frac{\epsilon_0 \epsilon_r A}{d - x}$$

$$C_2 = \frac{\epsilon_0 \epsilon_r A}{d + x}$$



$$E_1 = \frac{C_2 E}{C_1 + C_2}$$

$$E_2 = \frac{C_1 E}{C_1 + C_2}$$

$$E_1 = \frac{\frac{\epsilon_0 \epsilon_r A}{d + x} E}{\frac{\epsilon_0 \epsilon_r A}{d - x} + \frac{\epsilon_0 \epsilon_r A}{d + x}}$$

$$E_1 = \frac{E}{\frac{1}{d - x} + \frac{1}{d + x}}$$

$$E_1 = \frac{d - x}{2d} E$$

$$E_2 = \left(\frac{d + x}{2d} \right) E$$

$$E_0 = E_1 - E_2$$

$$= \frac{x}{d} E$$

Output voltage is proportion to displacement
sign indicates direction of displacement

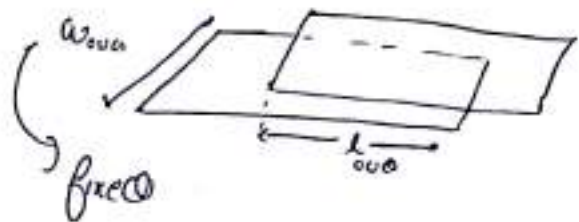
$$\text{Sensitivity} = \frac{\epsilon_0}{d} = \frac{\epsilon}{d} = \text{constant}$$

2) variable area type parallel plate

⊙ overlapping length changes the overlapping area that changes the capacitance

$$C = \frac{\epsilon (\ell_{\text{over}} \times w_{\text{over}})}{d}$$

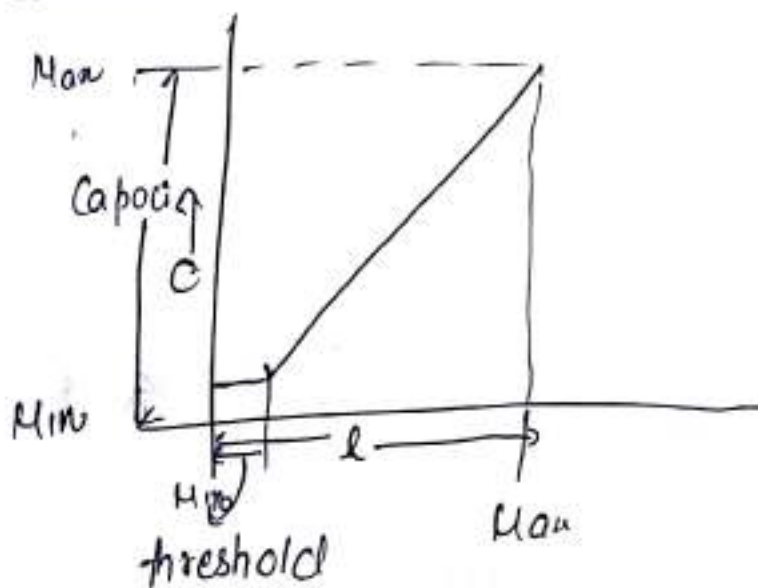
$$C \propto \ell_{\text{over}}$$



$$\text{Sensitivity } S = \frac{\partial C}{\partial \ell} = \frac{\epsilon w}{d}$$

⊙ Sensitivity is constant and high, linear output over a displacement range

⊙ Suitable for measuring both linear and angular displacement

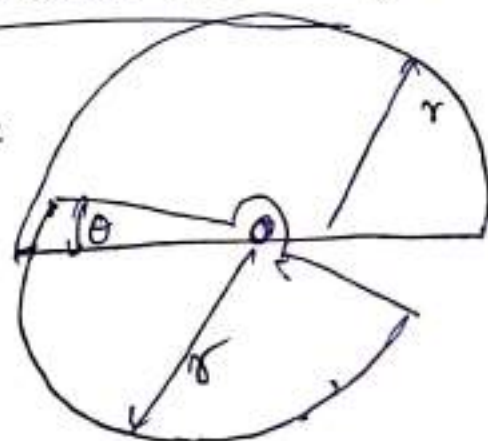


3) Variable area type parallel plate (angular)

Overlapping area for deflection

$$A_0 = \frac{A \cdot \theta}{2\pi}$$

max area



Where, A is area of circular of radius r

$$A = \frac{\pi r^2}{2}$$

$$C = \frac{\epsilon_0 r^2 \theta}{2d}$$

$$S = \frac{dC}{d\theta} = \frac{\epsilon_0 r^2}{2d}$$

$$A_\theta = \frac{\theta \pi r^2}{2} = \frac{r^2 \theta}{2}$$

$$C \propto \theta$$

C_p - r/p relationship is linear
Displacement range = 0° to 180°

5) Multi-parallel plate - S plate CT - Variable area type (COM type capacitor)



Capacitance of each sensor

$$C' = \frac{\epsilon_0 (l-x)w}{d}$$

where w = width of each plate

$$\text{Capacitor of transducer } C = 4 \frac{\epsilon_0 (l-x)w}{d}$$

$$\text{Sensitivity of transducer } \left[\frac{dC}{dx} = -\frac{4 \epsilon_0 w}{d} \right]$$

6) Variable dielectric type parallel plate capacitive transducer

- d = distance between plate
- ϵ = permittivity of movable slab
- A = area of plate
- w = width of plate
- L = length of plate = $l_2 + l_1$

dielectric
air

$$C = C_1 + C_2$$

$$C = \frac{\epsilon_0 w L_1}{d} + \frac{\epsilon_0 \epsilon_r w L_2}{d}$$



:

$$C_{new} = C_1' + C_2'$$

$$= \frac{\epsilon_0 w (L_1 - x)}{d} + \frac{\epsilon_0 \epsilon_r w (L_2 + x)}{d}$$

$$= \frac{\epsilon_0 w L_1 - \epsilon_0 w x + \epsilon_0 \epsilon_r w L_2 + \epsilon_0 \epsilon_r w x}{d}$$

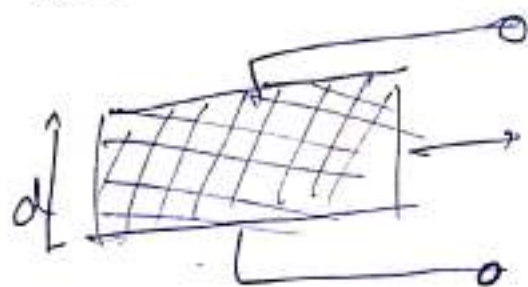
$$C_{new} = \underbrace{\frac{\epsilon_0 w L_1 + \epsilon_0 \epsilon_r w L_2}{d}}_{C_i} + \underbrace{\frac{-\epsilon_0 w x + \epsilon_0 \epsilon_r w x}{d}}_{\Delta C}$$

$$C_{new} = C_i + \frac{\epsilon_0 w x (\epsilon_r - 1)}{d}$$

$$C_{new} - C_i = \frac{\epsilon_0 w x (\epsilon_r - 1)}{d}$$

$$\Delta C = \frac{\epsilon_0 w x (\epsilon_r - 1)}{d}$$

7) ~~Variable~~ Capacitor due to change in dielectric due to ambient condition

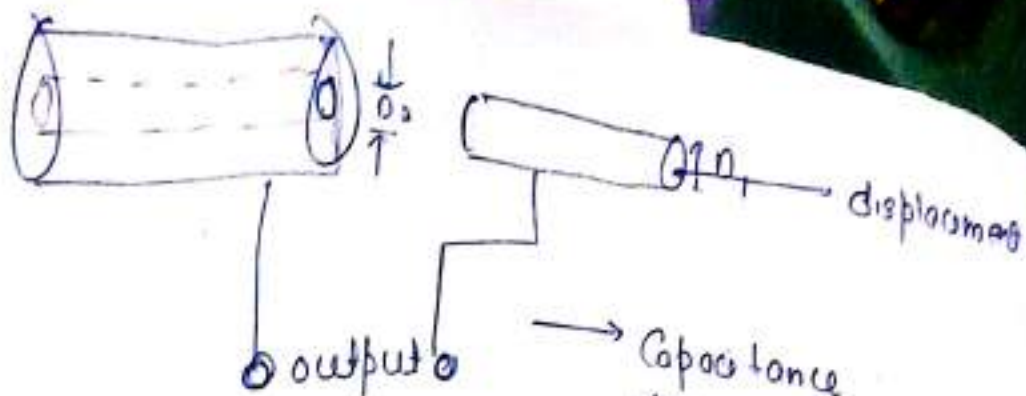


change

$$C = \frac{\epsilon_0 \epsilon_r A}{d}$$

$$\frac{\partial C}{\partial \epsilon_r} = \frac{\epsilon_0 A}{d} \rightarrow \text{const.}$$

8)



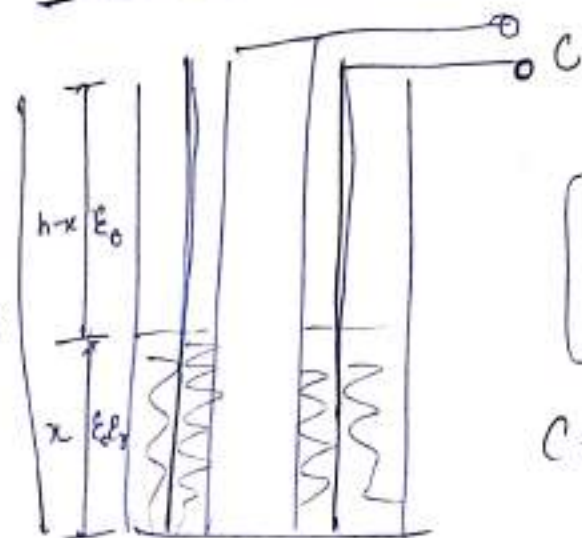
$$C_0 = \frac{2\pi \epsilon L_0}{\log_e \left(\frac{r_2}{r_1} \right)}$$

$l \rightarrow$ overlapping length

$$S = \frac{\partial C}{\partial l} = \frac{2\pi \epsilon}{\log_e (r_2/r_1)}$$

change in the capacitance, if displacement x is given to the capacitor inside towards outer cylinder $\Delta C = \frac{2\pi \epsilon (x)}{\log_e (r_2/r_1)}$

9] Liquid level measurement using cylindrical capacitor



$$C = C_{\text{liquid}} + C_{\text{air}}$$

$$C_{\text{liquid}} = \frac{2\pi \epsilon x}{\log_e \left(\frac{r_2}{r_1} \right)} \quad C_{\text{air}} = \frac{2\pi \epsilon_0 (h-x)}{\log_e \left(\frac{r_2}{r_1} \right)}$$

$$C = \frac{2\pi \epsilon}{\log_e \left(\frac{r_2}{r_1} \right)} (x + h - x)$$

$$C = \frac{2\pi \epsilon h}{\log_e \left(\frac{r_2}{r_1} \right)}$$

$$= \frac{\partial n}{\log e \left(\frac{r_2}{r_1} \right)} ((\epsilon_1 - \epsilon_2)x + \epsilon_2 h)$$

$$C_{Total} = \frac{\partial n (\epsilon_1 - \epsilon_2)x}{\log e \left(\frac{r_2}{r_1} \right)} + \frac{\partial n \epsilon_2 h}{\log e \left(\frac{r_2}{r_1} \right)} \quad \begin{matrix} \text{liquid} & \text{air} \\ & C_{\text{due to air}} \end{matrix}$$

:

$$C_{Total} = mx + c$$

m : Advantage

Sc

- Sensitivity high
- resolution high $\rightarrow 2.5 \times 10^{-3}$
- Transducer uses less power
- Minimum input loading
- accuracy = 0.0005%
- Immune to Magnetic field

Limitations

Q] A pressure measuring instrument uses a C.T having space of 4mm between its diaphragms. A pressure of 600kN/m² produces an average deflection of 0.3mm of the diaphragm. $G = 3000 \text{ N}$ frequency = 100kN₂ determine change free

$$\rightarrow f = \frac{1}{2.2 \pi \sqrt{LC}}$$

$$\frac{f_1}{f_2} = \sqrt{\frac{C_2}{C_1}} = \frac{f_1}{f_2} = \sqrt{\frac{d_1}{d_2}} = \frac{100}{f_2} \cdot \sqrt{\frac{4}{3.7}}$$

$$f_0 = 100 \times \sqrt{\frac{3.7}{4}}$$

$$f_0 = 86.17 \text{ KHz}$$

$$\Delta f = 3.83 \text{ KHz}$$

2. Miscellaneous Sensors

1. Light intensity measurement

• Energy a light wave carries per unit area

∴ an E.M.W can be specified by $\begin{matrix} \rightarrow \text{Intensity} \\ \rightarrow \lambda \\ \downarrow \\ f \end{matrix}$

2. Ways of L.I measurement:

✓
Radiometry

Photometry

L $\rightarrow$ Source

D $\rightarrow$ Sensor (detector)

(i) Radiometry unit

① measure light in terms of energy associated with light source

(ii) Photometric unit

① measure light as perceived by human eye, weighing only the visible portion.

→ two lights can have the same radiometric power but different photometric power (different colours)

3. Radiometric units

(i) Radiant flux (W or J/s)

• light energy per unit time emitted, transmitted, reflect.

→ doesn't depend on λ , used to measure any electromagnetic radiation

(ii) Spectral flux (W/nm)

• radiant flux specific to a wavelength interval

• how much power is received, transmitted or emitted per wavelength of light.

• Unit is Watts/nm.

(iii) Irradiance

• Received power of light per unit area

• W/m^2

• Most measurements are made by a detector with a finite area, so irradiance is practical convenient level of light intensity.

(iv) Radiance

→ radiance per unit of solid angle

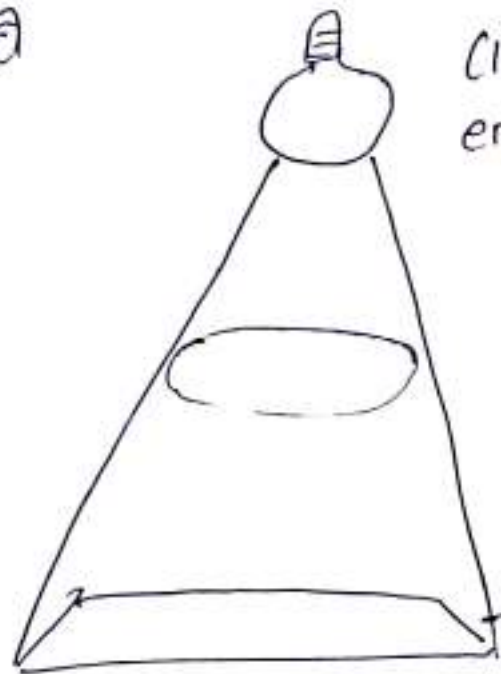
→ irradiance → power striking a specific surface from all angles.

→ radiance → specific area from a specific angle

Photometric unit

① lumen $\rightarrow$ light emitted by a radiation source in all directions.

②



(lm) is the radiant power emitted by a lamp in the sr

(cd) is the luminous flux related to a solid angle

(lx) is the luminous-flux incident on a reference area

③ luminous intensity $[cd] = \frac{\text{luminous flux } [lm]}{\text{solid angle } [sr]}$

$\rightarrow$ radiation of light emitted in a certain direction

④ illuminance (lx) $\rightarrow$ measure of brightness with which an area is illuminated. Lux indicates how much luminous flux (lumen) of a light arrives per unit area

$$\text{Lux } [lx] = \frac{\text{luminous flux } [lm]}{\text{area } [m^2]}$$

II → Light Sensor (non-contact type)

① photoelectric device converts light energy detected to electrical energy.

→ Classification →

(a) Active / Passive

(b) Based upon WP

- Photoresistors (LDR)
- Photodiode
- Phototransistor
- Photovoltaic

: actually generates voltage in response to L

(c) Wavelength based

- Visible LS

② Cadmium sulfide (CdS), photoresistors (LDR), used for brightness control, monitoring, switching operation of street light controller

- Infrared

→ ③ detect infrared radiation, photodiodes are highly sensitive to IR

- Ultraviolet

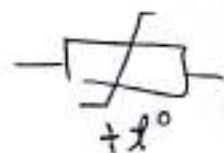
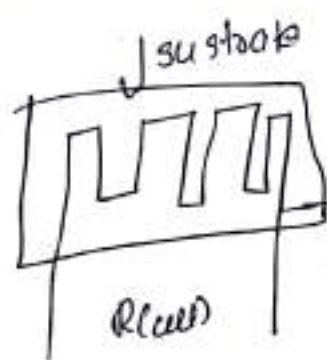
- used for monitoring UV index in industrial process

- Full-spectrum sensors

② # LDR (non linear / inverse)

consist of CdS in which the resistance change with LI.

- When no light fall R - highest $\rightarrow$ dark resistance
- When light fall $\rightarrow e^-$ - p pair generate $\rightarrow \sigma \uparrow$



③ Working principle

③ Energy gap between V.B and C.B

③ When a photon having $E > E_g$ photon is absorbed
 e^- V.B $\rightarrow$ C.B

③ As many e^- are excited $\rightarrow \sigma \uparrow \rightarrow R \downarrow$

③ $R \rightarrow$ inverse function of radiation intensity

$$E_{\text{photon}} = \frac{hc}{\lambda_{\text{max}}} = \Delta W \Rightarrow \text{Bond Gap}$$

minimum required

$$\lambda_{\text{max}} = \frac{hc}{\Delta W}$$

Relationship between R_{LDR} and intensity $\rightarrow$

$$R = R_{\text{dark}} L^{-\beta}$$

L : light intensity

β : A constant that depends on the LDR's material

Taking log in both sides

$$\log(R) = \log(R_{\text{dark}}) - \beta \log(L) \Rightarrow \text{linear relation in dark}$$

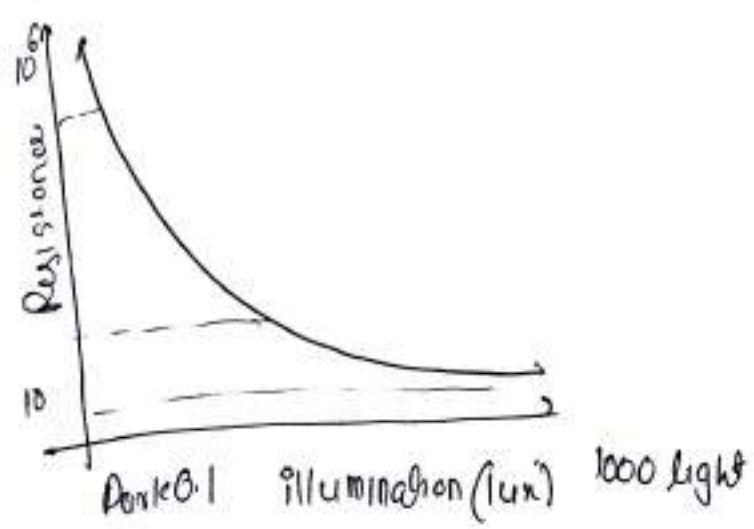
$$y = c - mx$$

~~$\frac{dR}{dL} = R_{\text{dark}}$~~

$$\frac{dR}{dL} = \beta R_0 L^{\beta-1}$$

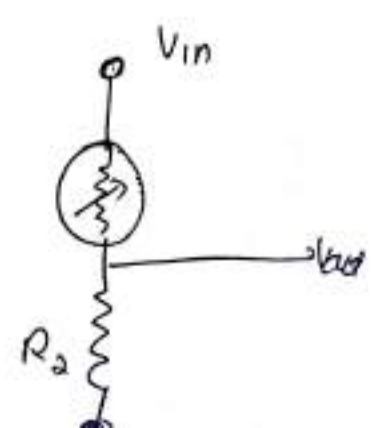
$$\frac{dR}{dL} = \frac{\beta R_0 L^{\beta}}{L}$$

$$\frac{dR}{R dL} = -\beta$$



Resistance measurement

$$V_{\text{out}} = V_{\text{in}} \times \frac{R_2}{R_{\text{LDR}} + R_2}$$



As $L1$ increases, R_{LO2} decreases and voltage
output increases, non-linearly over entire span.

For small range relationship is linear

R_{LO2} can vary from 100Ω in sun light, to $10M\Omega$ in
absolute darkness

e span.

10MΩ to

...Notes...

T.S.C

Date: _____



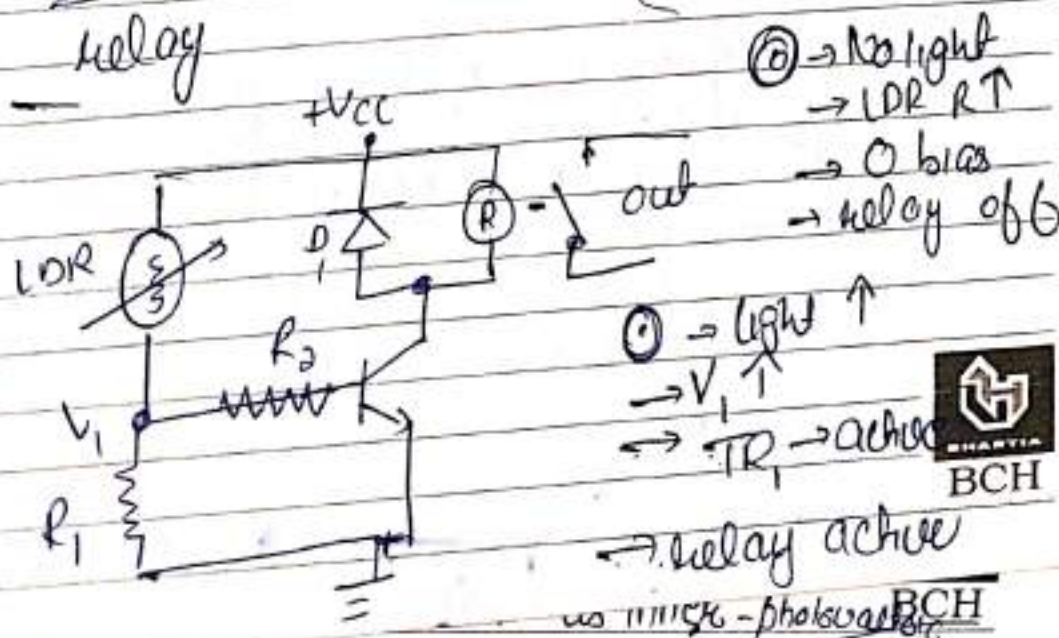
Characteristics of CdS LDR

Dark resistance $\rightarrow 10\text{M}\Omega$
fully illuminated $\rightarrow 100\Omega$

Features

- $\rightarrow$ used in auto dimming, auto brightness controls
- $\rightarrow$ darkness or twilight detector for turning the street lights "ON" and "OFF"
- $\rightarrow$ Indirectly used for distance measurement

Light activated switch - The basic light sensor circuit is used to drive a relay



as photoresistor

...Notes...

+5V



Date:

hocus

slr layers and p-n

...Notes...

Fabrication

(i) P-type junction

(ii) depletion

(iii) or or

(iv) or

(v) or

(vi) or

f

...Notes...

T.S.C

Date:

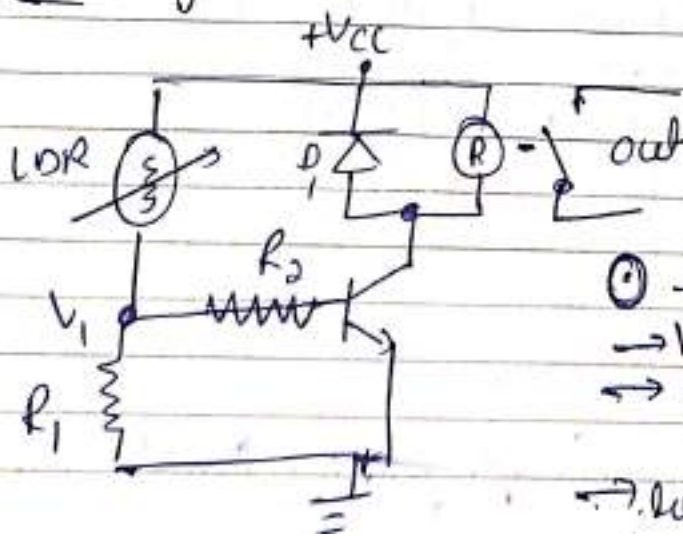
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- Darkness or twilight detection for turning the street lights "ON" and "OFF"
- Indirectly used for distance measurement

Light activated switch -- The basic light sensor circuit is used to drive a relay



@ $\rightarrow$ No light
 $\rightarrow$ LDR RT
 $\rightarrow$ 0 bias
 $\rightarrow$ relay off

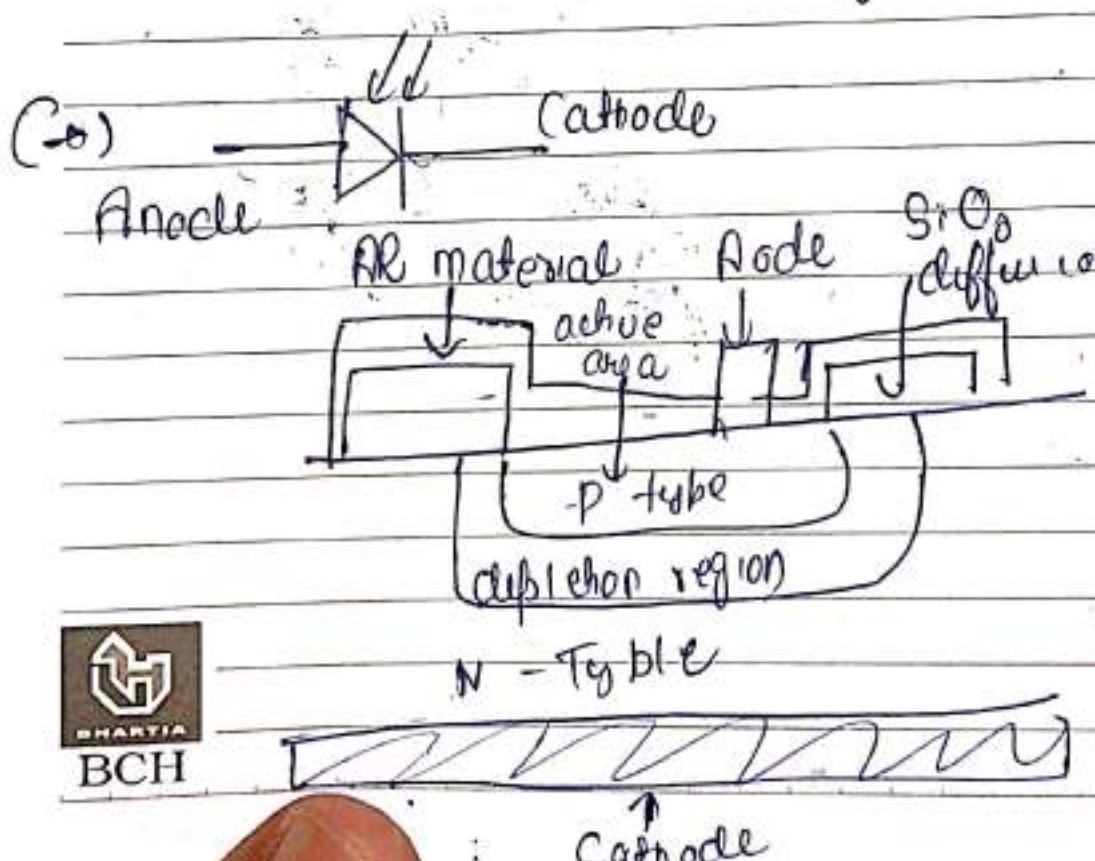
@ $\rightarrow$ light $\uparrow$
 $\rightarrow$ $V_1 \uparrow$
 $\rightarrow$ $TR_1 \rightarrow$ active
 $\rightarrow$ relay active



Photodiode

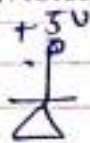
- ① Converts light into an electric current
- ② Light photons with sufficient energy strike the photodiode junction which is reverse biased they create e^-p pairs.
- ③ These charge carriers produce photocurrent.

(*) Materials → Si (190-1100nm) UV to infra
 → Germanium (400-1700) infra
 → Indium gallium arsenide
 (800-2600) → fiber optic communication



... Notes ...

Date: _____



$$I \propto \sqrt{V}$$

... Notes ...

Date: _____

Fabrication process

(i) P-type and n-type S/C layers and p-n junction

(ii) depletion region:

(iii) active region $\rightarrow$ Exposed to light (light enters and strikes depletion region)

(iv) Non-active surface $\rightarrow$

(v) Metal contact $\rightarrow$

Operating mode and working

Photoconductive mode

$\rightarrow$ in this mode, photodiode is reverse biased, a voltage is applied across the p-n junction in the reverse direction

$\rightarrow$ reverse bias significantly widens the depletion region, increasing photodiode sensitivity to light

$\rightarrow h\nu > E_g$ e^- v.b. to C.B.
hole in the v.b.

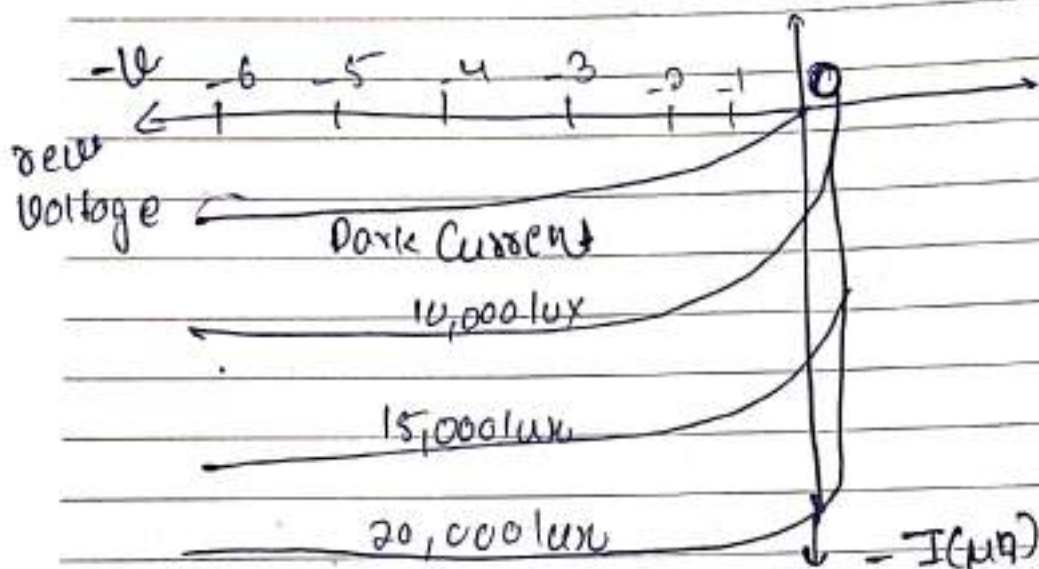
They form an e-h pair, this phenomenon is known as internal photoconductive effect



BCH

e^- pull toward n-side, h^+ to p-side
 → The separation of charge carriers creates a current called photocurrent in the external circuit

→ The magnitude of this photocurrent is directly proportional to ~~the~~ intensity of light

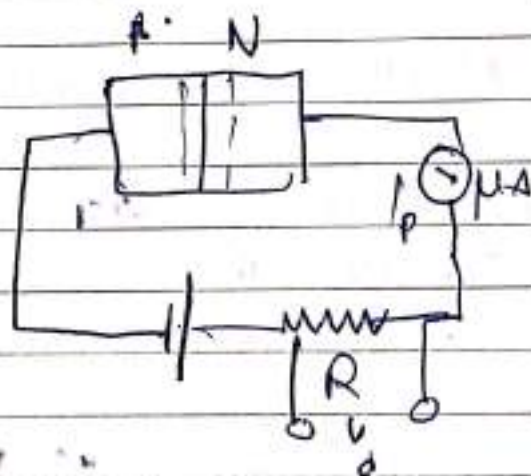


$$V_o = R I_p$$

Amp V_o

$$V_o' = V_o G$$

I_p is Sensitivity $\times P$



$$V_o' = R \times S \times P \times G$$

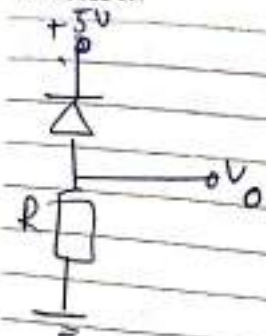
$$V_o' = I_o \times S \times P$$

→ lower input
 (light intensity)

Cathode

...Notes...

Date: _____



$$V_o \propto I_p$$

No light $\rightarrow$ No V_o

Photodiode current equation

$$I = I_0 \left(e^{\frac{qV}{nkT}} - 1 \right) - I_p$$

total photodiode current current due to light intensity

$\eta = 1$ for Si
Ge

$$k = 1.38 \times 10^{-23}$$

$$T = (K)$$

I_0 = diode reverse saturation or "dark current"

• Under reverse bias $e^{\frac{qV}{nkT}} \approx 0$

$$\text{so } I_T = -I_0$$

• $I_p \propto P$ (intensity of light)

$$I_p = R \cdot P$$

$$\Rightarrow I = -I_0 - I_p = -I_0 - RP$$

$\rightarrow$ Total response



- At no light intensity, photocurrent (depends on temp)
- As L^i increases photocurrent increases linearly

→ Key characteristics

High Speed & Responsivity

Linearity

Wide Spectral Response

Low Noise

Mechanical Ruggedness & Lifespan

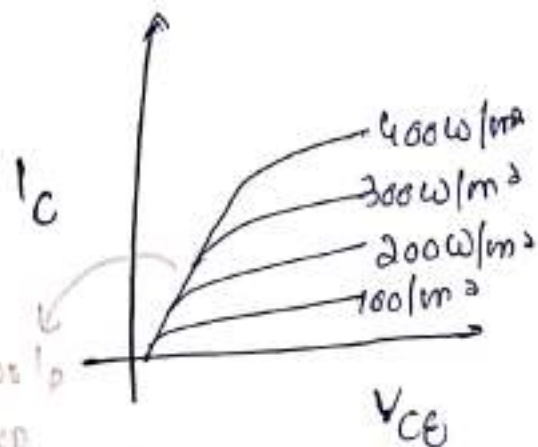
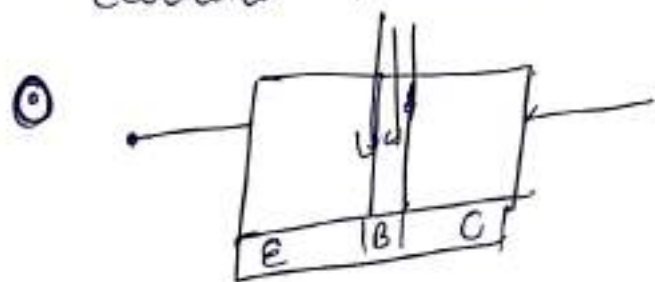
Phototransistor

① Semiconductor device with a light-sensitive base region.

② The phototransistor is basically a photodiode amplifier in the form of BJT, which has its collector-base exposed to light the I_B get amplified by β .

③ Combines photoelectric effect of a photodiode with current amplification

④ When light strike e^- - h^+ pair, which acts a base current.



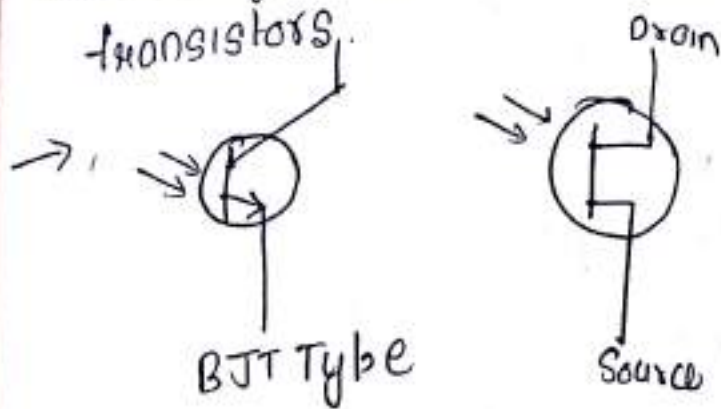
- WP
- light absorb
- e^- hole pair
- flow e^- due to internal field causing's
- current amp
- output voltage (across emitter/collector load)

Construction

• 3 terminals (E, B, C)

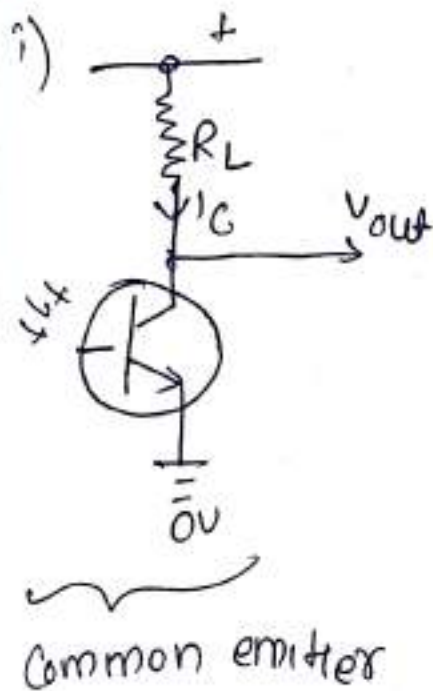
or two terminals and have light sensitive base region.

→ have larger collector and base region than regular transistors.

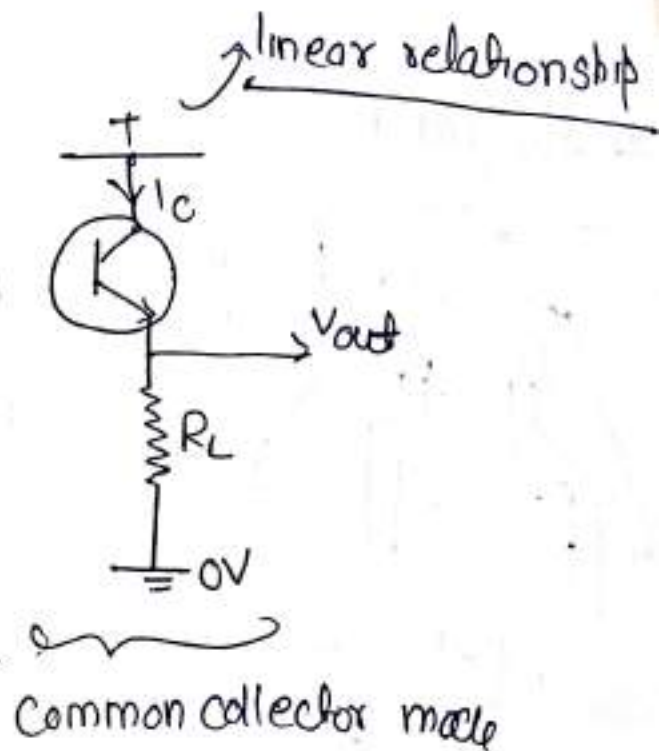


smaller lead → emitter
longer lead → collector

Common collector mode



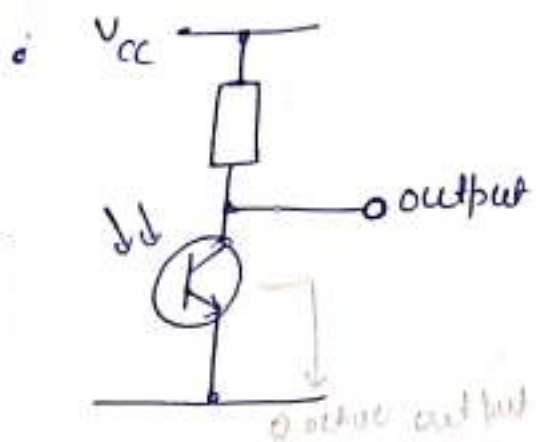
ii)



output of the phototransistor depends on various factors like

- λ of incident
- Area of light-exposed J_{BC}
- β

Common emitter configuration



(inverse relation)

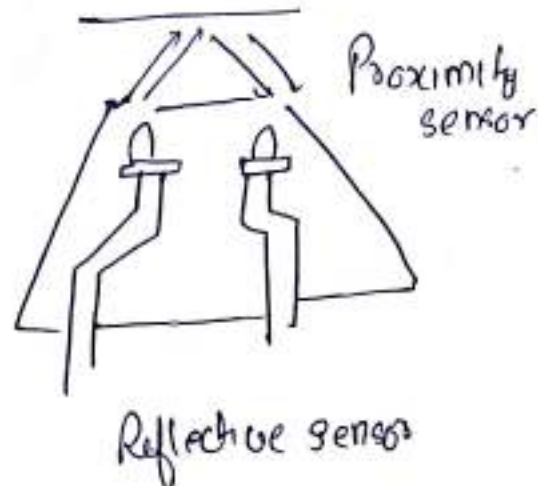
No light $\rightarrow$ output is high
no I_C $V_o = V_{CC}$

$\rightarrow$ When light is detected $\rightarrow I_C \uparrow$

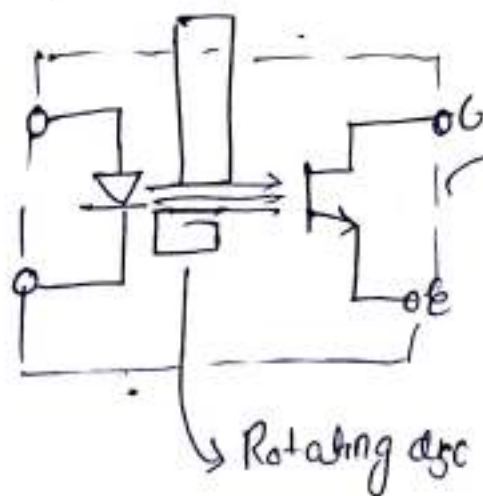
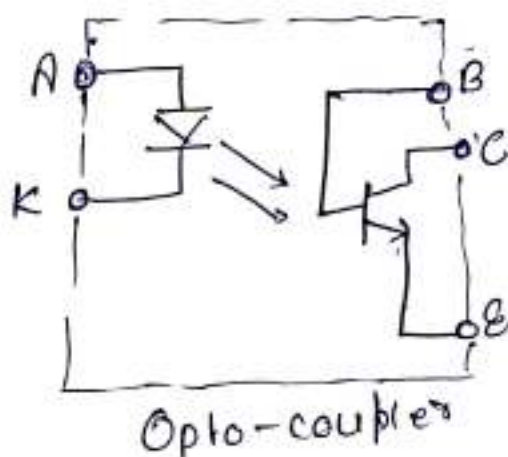
$V_o \downarrow$
 $\rightarrow$ device provides a high-voltage state to a low voltage state when light is detected.

2/ of

application



Reflective sensor



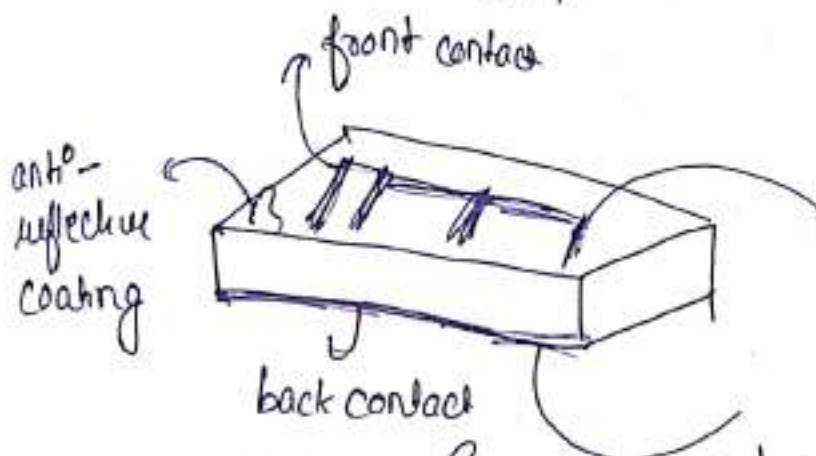
output connected to counter

P-d	P-n
Two semiconductor region	Three
Sensitivity less	high sensitivity because of P
response time is less	more

Photovoltaic cell

Si based homo-junction PV cell:

- Top of cell diffused with an "n" dopant
- base of slab: Small amount of "p" dopant
- Conducting layer are diffused at top and bottom of PV to allow current to flow to external circuit.
- Anti reflecting coating on top layer to allow max absorption.



- light strike PV cell → e^- hole pair → diffused across junction - get generated
- $E_{\text{photon}} > E_{\text{bandgap}}$

$$E_{\text{photons}} = \frac{hc}{\lambda}$$

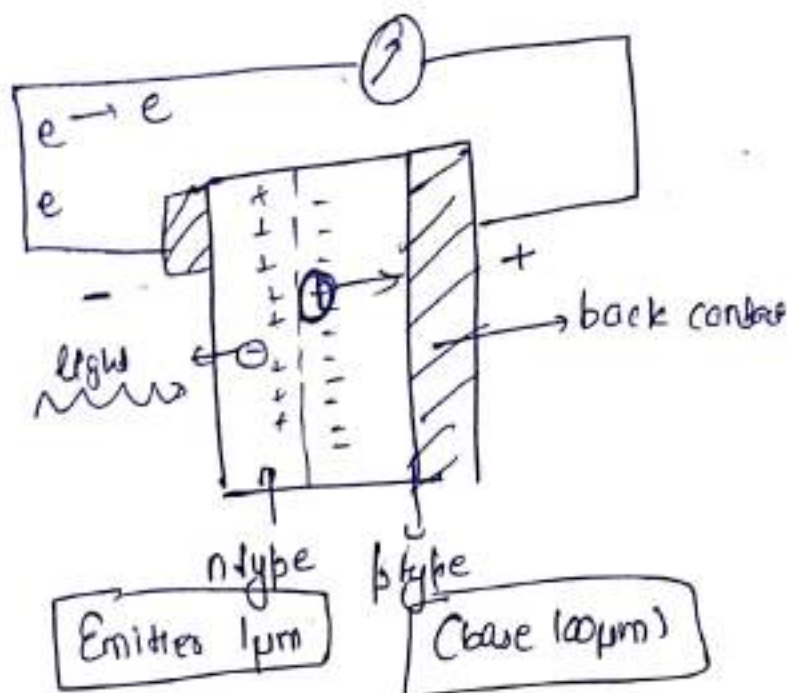
for solar cell $E_{\text{photons}} = E_{\text{band gap}}$

$$E_{bg} = \frac{hc}{\lambda}$$

$$h = 6.626 \times 10^{-34}$$

$$c = 3 \times 10^8 \text{ m/s}$$

λ = wavelength of radiation



Solar cell Efficiency

% of solar energy to which the cell is exposed
to that is converted to electrical energy

$$\eta = P_m / G A_c$$

Specified $\left\{ \begin{array}{l} P_m = \text{maximum power in W} \\ G = \text{irradiance (input light in W/m}^2) = 1000 \text{ W/m}^2 \\ A_c = \text{surface area of the solar cell in m}^2 \end{array} \right.$

Q.1 maximum wavelength of light radiation that can cause photovoltaic effect in Si based PV cell

$$E_g = 1.45 \text{ eV}$$

$$\frac{hc}{\lambda} = E_g$$

$$\lambda = \frac{hc}{E_g} = \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{1.45 \times 1.6 \times 10^{-19}}$$

$$\lambda = 857 \text{ nm}$$

Piezoelectric Sensor

(actual)

→ When mechanic stress (pressure, vibration) applied to a material, causing it to generate electric charge due to movement of ions.

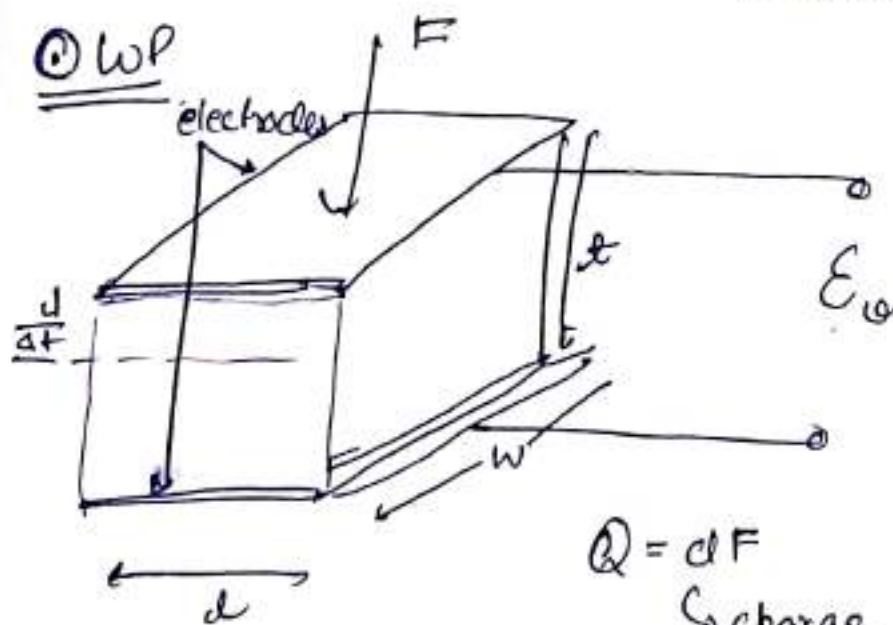
charge & stress

The generated charge can be converted to measurable electrical voltage.

① Relationship between charge sensitivity of the piezoelectric material

$$d = g \epsilon_0 \epsilon_r$$

charge sensitivity = Voltage sensitivity $\times$ permittivity of material.



$$Q = dF$$

↳ charge sensitivity

$F =$ applied force

$1 \times w \times l$
because crystal is dielectric
↑
value of capacitance

$$\epsilon_0 = \frac{Q}{C_p}$$

$$\epsilon_0 = \frac{dF}{C_p}$$

when ... within elastic limit

$$\gamma = \frac{\text{Stress}}{\text{Strain}}$$

$$\text{Stress} = \frac{F}{A} \quad \text{Strain} = \Delta l / l$$

$$\gamma = \frac{F \times l}{A \times \Delta l}$$

$$\gamma = \frac{Q \times l}{d \times A \times \Delta l}$$

$$\star \boxed{Q = \frac{\gamma \times d \times A \times \Delta l}{l}} \quad (\text{Charge generated due to strain})$$

$$\epsilon_0 = \frac{Q}{C_p} = \frac{d \cdot \gamma \cdot A \times \Delta l}{l \times C_p}$$

$$\boxed{C_p = \frac{A \times \epsilon_0 \times \epsilon_r}{d} = \frac{A \times \epsilon_c}{l}}$$

$$\epsilon_0 = \frac{d \times F}{\frac{\epsilon_c \times A}{d}} = \frac{d \times l}{\epsilon_c} \times \left(\frac{F}{A} \right) \rightarrow P$$

$$\boxed{\epsilon_0 = \frac{d}{\epsilon_r \epsilon_0} l \times P}$$

$$\epsilon_0 = g \cdot l \cdot P$$

$$\boxed{g = \frac{d}{\epsilon_r \epsilon_0}}$$

$$\boxed{g = \frac{\epsilon_0}{l \cdot P}}$$

$$\boxed{g = \frac{\epsilon_0 \cdot l}{P}}$$

$$\star \boxed{\epsilon_0 = g \cdot l \cdot P}$$

$g = \frac{\text{electric field}}{\text{stress}}$

Q1 $t = 2\text{mm}$ V. sensitivity $g = 0.055$ $P = 1.5\text{MN/m}^2$
 calculate d

$E_0 = ?$ $E_c = 40.6 \times 10^{-12}$

$$E_0 = g \times t \times P$$

$$= 0.055 \times 2 \times 10^{-3} \times 1.5 \times 10^6$$

$$\boxed{E_0 = 0.165\text{V}}$$

Q2 $d = g \times E_c$

$$d = 0.055 \times 40.6 \times 10^{-12}$$

$$\boxed{d = 2.23\text{pC/W}}$$

Q3 A piezo electric dimension $5 \times 5 \times 1.5\text{mm}^3$
 $g = 0.055$ $P = ?$ $V = 100\text{V}$

$$E = g \times t \times P$$

$$E = g \times t \times \frac{F}{A}$$

$$F = \frac{E \times A}{g \times t} = \frac{100 \times 25 \times 10^{-6}}{0.055 \times 5 \times 10^{-3}}$$

$$\boxed{= 30\text{N}}$$

Q4 dimension $5 \times 5 \times 1.2\text{mm}^3$ $F = 5\text{N}$ $d = 150 \times 10^{-10}$
 $E_c = 12.5 \times 10^{-9}$ $\gamma = 12 \times 10^{-6}$ strain $\rightarrow$ charge $\rightarrow$

capacitance

$$Q = d \times P$$

$$= 750\text{pC}$$

$$C_p = \frac{Q}{E_0} \rightarrow g \times t \times P$$

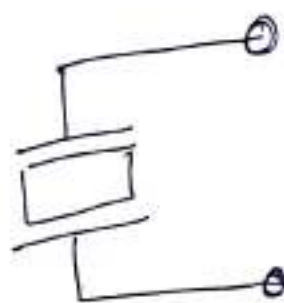
$$P = \frac{5}{25 \times 10^{-6}} = 0.2 \text{ MN/m}^2$$

$$g = \frac{d}{\epsilon_0 \epsilon_r} = \frac{150 \times 10^{-9}}{12.5 \times 10^{-9}} = 12 \times 10^{-3}$$

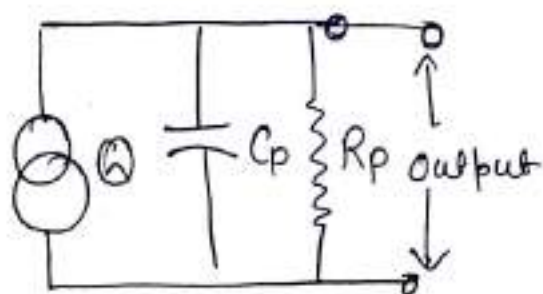
$$\epsilon_0 = g \times f \times P = 30$$

$$C = \frac{Q}{E} = \frac{750 \times 10^{-13}}{3} = 250 \text{ pF}$$

Electrical model of piezoelectric transducer



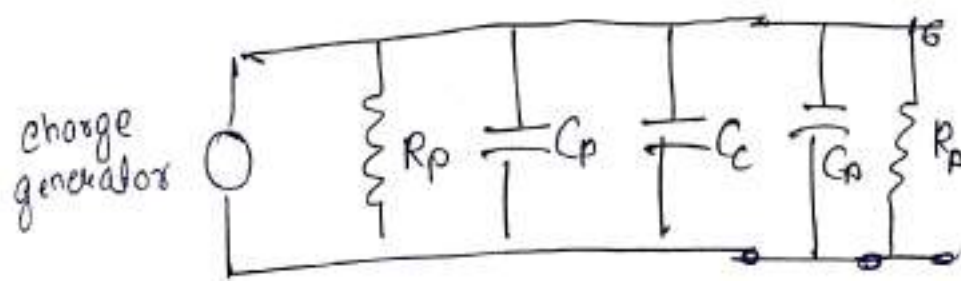
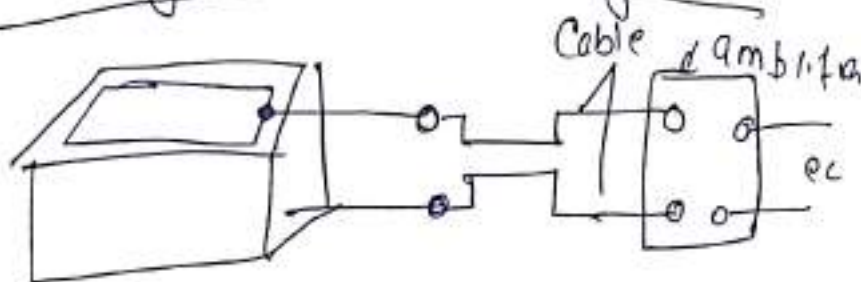
electrical
model



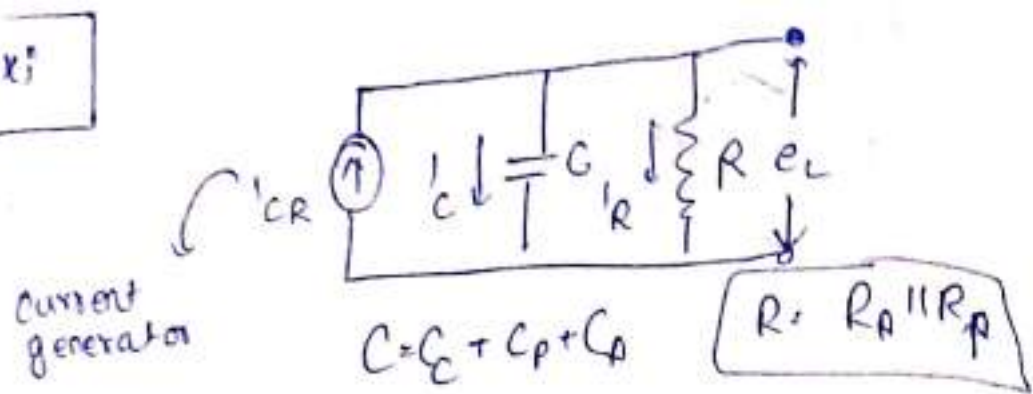
$C_p \rightarrow$ Intrinsic capacitance

$R_p =$ models the high internal resistance of transducer

Modelling the complete system



$$q = K_q x_i$$



$$i_{CR} = \left(\frac{dq}{dt} \right) = K_q \left(\frac{dx_i}{dt} \right)$$

output voltage at load $e_L = e_c = \frac{1}{C} \int i_{CR} dt = \frac{1}{C} \int (i_{CR} - i_R) dt$

$$\frac{de_L}{dt} = \frac{1}{C} (i_{CR} - i_R)$$

$$C \frac{de_L}{dt} = i_{CR} - i_R = K_q \frac{d(x_i)}{dt} - \frac{e_L}{R}$$

$$\tau \frac{d(e_L)}{dt} + e_L = \frac{K_q R}{C} \frac{d(x_i)}{dt}$$

$$\tau \frac{d(e_L)}{dt} + e_L = \frac{K_q R}{C} \frac{d(x_i)}{dt} \quad (A) \quad \frac{K_q R}{C} = K$$

Taking Laplace of A)

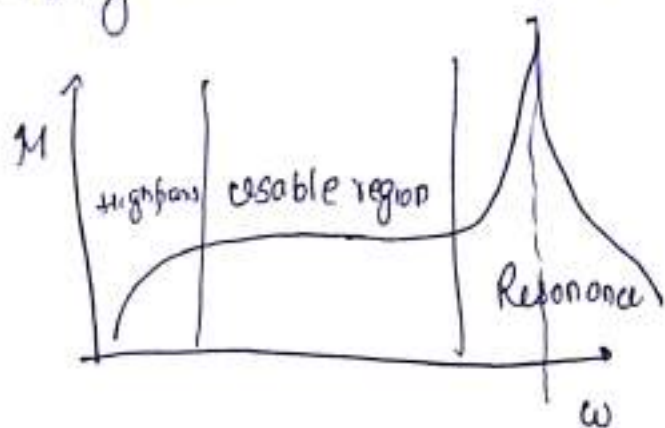
$$(Ts + 1)E_L(s) = K \cdot \tau \cdot s X_i(s)$$

$$\frac{E_L(s)}{X_i(s)} = \frac{K \cdot \tau \cdot s}{1 + \tau \cdot s}$$

Transfer function sinusoidal $\frac{E_L}{X_i}(j\omega) = \frac{j\omega K \tau}{1 + j\omega \tau}$

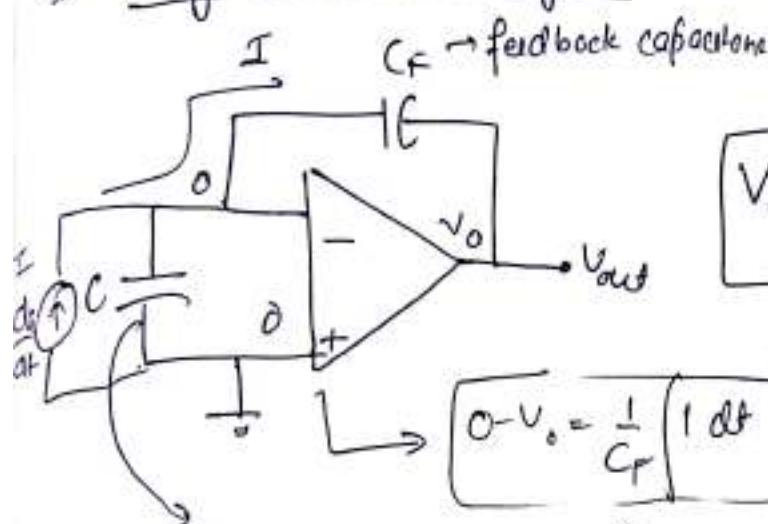
$$M = \left| \frac{E_L(j\omega)}{KX_i} \right| = \frac{\omega \tau}{\sqrt{1 + \omega^2 \tau^2}} = \frac{1}{\sqrt{1 + \left(\frac{1}{\omega \tau}\right)^2}}$$

→ only usable when frequency response is constant



$\omega \rightarrow 0$
 $M \rightarrow 0$
 not suitable for static

* Signal conditioning (charge integrator) $I = \frac{dq}{dt}$



$$V_{out} = \frac{1}{C_F} \int -I dt = -\frac{Q}{C_F}$$

$$0 - V_o = \frac{1}{C_F} \int I dt$$

internal capacitance (cable capacitance cancel out)

Application

① Used in electrical lighter in kitchen

② Automobile seat belt

③ Vibration measurements

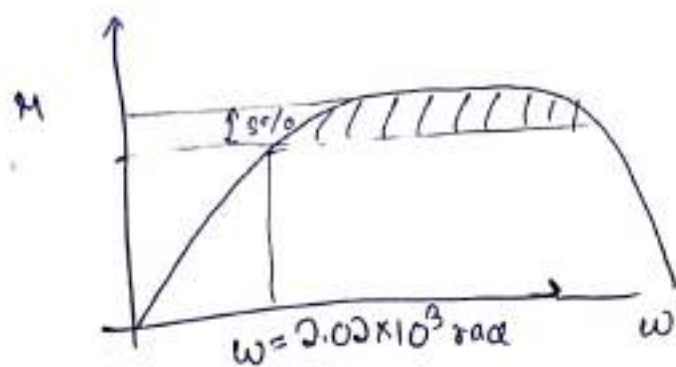
④ Ultrasonic transmitters and receivers (me)

⑤ Sound based distance measurement.

Q] In a piezoelectric transducer a flat frequency within 5% is required. Find the value of min. frequency in terms of time constant for which it can be used. If the time constant of transducer is 1.5 ms

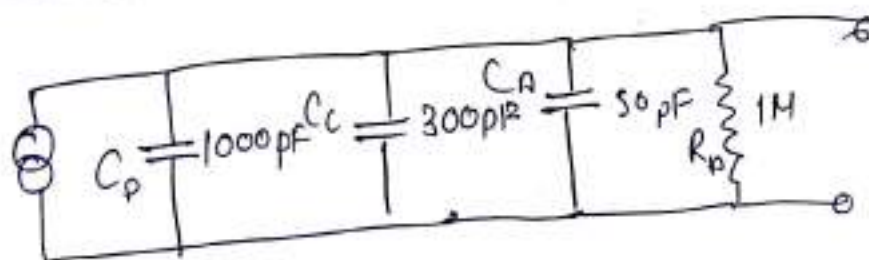
$$\rightarrow M = 1 - 0.05 = 0.95 \quad M = \frac{1}{\sqrt{1 + \left(\frac{1}{\omega\tau}\right)^2}}$$

$$\omega\tau = 3.04 \quad \omega = 2.02 \times 10^3 \text{ rad/s} \quad f = 321.5 \text{ Hz}$$



Q] $C_p = 1000 \text{ pF}$ Charge sensitivity 40×10^{-3} connecting cable
 $C_c = 300 \text{ pF}$ $R_A = 1 \text{ M}\Omega$ $C_A = 50 \text{ pF}$

→ Sensitivity of transducer?



$$C = 1350 \text{ pF} \quad R = 1 \text{ M}\Omega$$

$$\tau = 1350 \times 10^{-12} \times 10^6 = 1.35 \text{ s} \quad \boxed{1.35 \text{ ms}}$$

$$K = \frac{K_g}{C} = \frac{40 \times 10^{-3}}{1000 \times 10^{-12}} = 40 \times 10^6 \text{ V/m}$$

$$\Rightarrow 3) \text{ Time constant } \tau = R \times C$$

$$= 1.35 \text{ ms}$$

lowest frequency for 5%

$$0.95 = \frac{1}{\sqrt{1 + (1/\omega\tau)^2}} \quad \text{or } \omega\tau = 3.04$$

$$\omega = \frac{3.04}{1.35 \times 10^{-3}} = 2254 \text{ rad/s}$$

$$f = \frac{2254}{2\pi} = 358.7 \text{ Hz}$$